

The Cost of Assuming Ionisation Balance

Transient Error in Balmer-Ratio Diagnostics of
Divertor Hydrogen Plasmas

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Abstract

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Chapter 1

Reading the Light from a Divertor

A physicist sits at a console and looks at a spectrum. An optical fibre viewing the bottom of a fusion reactor has collected light along a narrow line of sight, and a spectrometer has spread it out by wavelength. Two lines stand out: a red one at 656 nm and a blue-green one at 486 nm, both emitted by hydrogen atoms. She divides the brightness of the first by the brightness of the second, looks the resulting number up in a table, and reads off an electron temperature and an electron density. That is how the hottest and least accessible part of the machine gets measured.

This thesis is about the table. It was built by solving a model of hydrogen's excited states to steady state, and it is indexed on two numbers: temperature and density. A table can be indexed on two numbers only if every other quantity the answer depends on has been eliminated by an assumption. The assumption in this case is that ionisation and recombination in the plasma have come into balance — not that they are nearly balanced, or changing slowly, but that the plasma has *finished changing*.

The plasma in question never finishes changing. It is the most violently unsteady region of the machine. So the question is whether the table still holds, and if not, by how much it is wrong and where.

That is all this chapter does: put the reader in front of the spectrum, show where the table comes from, and expose the assumption buried inside it. Section 1.1 explains what the region being measured is for and why a probe cannot be put into it. Section 1.2 builds the hydrogen atom and its spectrum from scratch, because everything downstream is a statement about two of its lines. Section 1.3 says what the instrument records and what it does not. Section 1.4 follows the chain from a measured ratio to a pair of plasma parameters. Section 1.5 is the hinge: it identifies the quantity the table has eliminated and separates two approximations that are routinely confused with one another. Section 1.6 asks whether the plasma is ever in the state the table assumes. Section 1.7 says what is already known and by whom, which is more than one might guess. Section 1.8 states the question this thesis answers and names the suspect it went looking for.

1.1 Why the divertor is hard

The question: what is the thing being measured, and why is measuring it hard enough to be worth a thesis?

Start with the machine. A tokamak is a doughnut-shaped vessel in which a gas of hydrogen isotopes is heated until its electrons separate from its nuclei — the resulting mixture of free electrons and bare nuclei is a plasma — and then held away from the walls

by a magnetic field wound around the doughnut. If the plasma gets hot enough and stays confined long enough, deuterium and tritium nuclei fuse and release energy. ITER, under construction in southern France, is the machine built to demonstrate this at reactor scale.

Confinement is never perfect, and this is the useful part. Particles and heat leak steadily outward across the magnetic field, and once they cross the last magnetic surface that closes on itself they are no longer trapped: they stream along field lines instead. The layer outside that surface is called the scrape-off layer (SOL), and in a modern tokamak the field lines in it are not allowed to strike the wall at random. They are steered magnetically to a dedicated set of water-cooled plates at the bottom of the vessel. Those plates, together with the volume of plasma and gas in front of them, are the *divertor*.

A chemical engineer already owns the right picture for this. The divertor is the quench-and-separation section bolted onto the outlet of a very hot reactor. It receives the reactor's waste heat and its ash — helium, the product of fusion — cools the stream, and pumps the ash away, and it must do so without being destroyed by the duty.

Being destroyed by the duty is the entire difficulty. The exhaust power does not spread itself over the plate; it arrives concentrated in a narrow strip where the field lines land, and the resulting surface heat flux would be far above what any solid can survive. [TODO: The ITER divertor target steady-state heat-flux design limit and the unmitigated parallel heat flux are deliberately left unstated. The design-basis reference is Pitts *et al.* (2019) [1]; its full text could not be obtained for this draft, and the figures in circulation could not be traced to a source that was actually read. Pull that paper through the institutional subscription and insert both numbers with a page reference, or delete this sentence's quantitative claim.]

The way out is not a better plate material. It is to stop the plasma before it arrives. If enough neutral hydrogen gas is allowed to build up in front of the target, the plasma ions collide with it: in a charge-exchange collision an ion takes an electron from a neutral atom and becomes a fast neutral itself, carrying its momentum out of the plasma stream, and the energy that remains is radiated away by excited atoms rather than deposited on the plate. Over the last few centimetres the plasma pressure and the heat flux both collapse. The plasma is then said to be *detached* from the target: the region where ionisation happens lifts off the plate and sits in the volume in front of it. Detachment is not a fault to be avoided. It is the intended operating regime of a reactor divertor.

Detachment is also what makes the atomic physics of hydrogen unavoidable. A detached divertor plasma is cold — of order a few electronvolts, where one electronvolt corresponds to about 11 600 K — and at a few electronvolts hydrogen is only partly ionised. Neutral atoms and ions coexist in comparable numbers, recombination competes with ionisation on equal terms, and the light emitted by neutral hydrogen becomes the brightest thing in the visible spectrum. The plasma one most needs to characterise is therefore exactly the plasma whose light carries the most information.

What conditions are those, in numbers? Edge modelling of the ITER baseline burning plasma gives target electron temperatures spanning roughly one to a few tens of electronvolts across the design database, with the large majority of cases below 15 eV [2, 2], and falling below 1 eV across half a metre of each target in the most strongly detached solutions obtained so far [3]. The near-target electron density under partial detachment is expected to reach about $10^{15}/\text{cm}^3$ [4] — and the same work makes the point this thesis depends on, that at such densities radiative decay is no longer fast compared with electron-impact processes, so the excited states must be treated with a full collisional–radiative calculation rather than a simple equilibrium formula.

Every calculation in this thesis is performed on a fixed grid spanning $T_e = 1\text{--}10\text{ eV}$ in electron temperature and $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$ in electron density. The upper density matches the expected near-target value cited above and the temperature range covers the cold, detached end of the modelled database, which is where the atomic physics of neutral hydrogen governs the emission. It does not cover the hotter attached cases, and it is a modelling choice rather than a prediction: it should be read as the range over which the model was exercised, not as a claim about what any particular machine will do. [TODO: The lower end of the density range is not sourced. Every $10^{13}/\text{cm}^3$ -scale figure that could be verified is an *upstream* separatrix or X-point density, not a divertor or near-target density, and no source read for this chapter states a divertor density that low in prose. Either pull Pitts *et al.* (2019) [1] or Loarte *et al.* (2007) [5] through the institutional subscription and read the number directly, or narrow the claim. Do not assert the lower bound without doing one or the other.]

It would be simpler to put an instrument into the plasma, and ITER will do exactly that. A Langmuir probe is a small electrode exposed to the plasma and biased to a range of voltages; the current it draws as a function of bias gives the local electron temperature and density. ITER carries a designed divertor Langmuir probe system whose stated purpose is to measure plasma parameters at the target plates, and whose thermomechanical analysis concludes that the sensors survive the environment [6]. It is worth being accurate about this, because the claim that probes cannot survive in a reactor divertor is often made and is not correct.

The limitation is not survival but what a probe can see. It is mounted flush in the target surface, so it reports conditions in the thin charged layer at the plate rather than in the volume in front of it — and the volume in front of it is precisely where a detached plasma does its recombining. It perturbs what it measures. Its interpretation relies on sheath theory that is least trustworthy in exactly the cold, strongly recycling conditions of interest. And the coverage is necessarily sparse: a set of sensors embedded in a handful of divertor cassettes samples points on a surface, not a volume.

Light has neither limitation, and one decisive advantage: it leaves the plasma by itself. A window, a fibre, a spectrometer and a detector make a measurement that touches nothing, perturbs nothing, and samples the emitting volume rather than the surface. Spectroscopy is therefore complementary to probe measurement rather than a replacement for it, and it is the only one of the two that reports on the recombining volume where a detached plasma does its interesting work. This is why hydrogen line emission is a workhorse divertor diagnostic — and why the chain of reasoning that converts photons into T_e and n_e carries so much weight. The next section builds the atom that emits them.

1.2 Light from a hydrogen atom

The question: where do the two lines in the opening paragraph come from, and what would make their brightnesses differ?

A hydrogen atom is one electron bound to one proton. The electron cannot take any energy it likes: the bound states form a discrete ladder, labelled by an integer $n = 1, 2, 3, \dots$ called the principal quantum number, with energies

$$E_n = -\frac{13.6\text{ eV}}{n^2}. \quad (1.1)$$

Here E_n is the energy of the level in electronvolts, measured from the free-electron state

at $E = 0$, so all bound levels are negative and $n = 1$ — the ground state — is the deepest at -13.6 eV. What Eq. (1.1) does physically is fix the whole spectrum of hydrogen with one constant: the levels crowd together as n grows, and 13.6 eV is the energy needed to remove the electron entirely from the ground state, which is called the ionisation energy.

Each level n is not a single state. The electron's orbital angular momentum is also quantised, by an integer $\ell = 0, 1, \dots, n-1$, and states of different ℓ have historically been labelled by letters: $\ell = 0$ is written S, $\ell = 1$ is P, $\ell = 2$ is D, $\ell = 3$ is F. So the $n = 2$ level contains a $2s$ state and a $2p$ state, the $n = 3$ level contains $3s$, $3p$ and $3d$, and so on. In an isolated hydrogen atom all the ℓ states of a given n have almost exactly the same energy, which is why they can often be lumped together — and Chapter 2 explains the one place in this problem where lumping them destroys the physics. The number of independent quantum states in a level is its statistical weight g , equal to $2n^2$ for hydrogen when electron spin is counted; a level with a larger g has more ways of being occupied and is correspondingly favoured in thermal equilibrium.

An atom in an excited level does not stay there. It can fall to a lower level and emit the energy difference as a single photon. This is spontaneous emission, and it is a first-order rate process exactly like a unimolecular decomposition: the probability per unit time that an atom in level q decays to a lower level p is a constant A_{qp} , in 1/s, called the Einstein coefficient for spontaneous emission. The number of such decays per unit volume per unit time is therefore $A_{qp}n_q$, where n_q is the population of level q in $1/\text{cm}^3$. Not every pair of levels is connected: quantum mechanics permits a strong (electric-dipole) transition only when ℓ changes by exactly one, a restriction called a selection rule, and Chapter 2 shows that the levels which *cannot* decay this way are the ones that dominate this problem.

Because the energy released is fixed by Eq. (1.1), every transition emits at one wavelength, and the emission appears in a spectrograph as a narrow *spectral line*. Its brightness is proportional to the population of the upper level. That single sentence is why spectroscopy is a diagnostic at all: the light tells you how the atoms are distributed over their levels, and the distribution is set by the plasma conditions.

The transitions that end on $n = 2$ fall in the visible, which is convenient for windows, fibres and detectors, and they are collectively called the Balmer series. The two brightest are the subject of this thesis:

$$H_\alpha : \quad n = 3 \rightarrow 2, \quad \Delta E = 13.6 \text{ eV} \left(\frac{1}{4} - \frac{1}{9} \right) = 1.89 \text{ eV} \Rightarrow 656 \text{ nm}, \quad (1.2)$$

$$H_\beta : \quad n = 4 \rightarrow 2, \quad \Delta E = 13.6 \text{ eV} \left(\frac{1}{4} - \frac{1}{16} \right) = 2.55 \text{ eV} \Rightarrow 486 \text{ nm}. \quad (1.3)$$

The arithmetic is worth doing once by hand, because it shows that the red line and the blue-green line of the opening paragraph are not empirical facts to be looked up: they follow from Eq. (1.1) and the relation $\lambda = hc/\Delta E$, with $hc = 1240$ eV nm.

What these two lines have in common is their lower level. Both end on $n = 2$, so the ratio of their brightnesses does not depend on anything about $n = 2$: it depends only on how the atoms are shared between the $n = 3$ and $n = 4$ levels. The whole of this thesis is a statement about that one ratio, and Section 1.3 explains why it, and not the brightness of either line by itself, is what a divertor spectroscopist can actually use.

1.3 What a spectrometer records

The question: given that a spectrometer measures brightness, what can and cannot be extracted from that measurement?

Picture the geometry first. The instrument looks along a straight line through the plasma and collects the light emitted anywhere along it, from a small solid angle. It cannot tell where along the line a photon came from. Whatever it reports is a sum over the whole chord.

Formally, if each decay from level q to level p releases a photon of energy $h\nu_{qp}$ in joules and the decays occur at $A_{qp}n_q$ per unit volume per unit time, the detector records the brightness

$$I_{qp} = \frac{h\nu_{qp}}{4\pi} \int_{\text{chord}} A_{qp} n_q(s) ds, \quad (1.4)$$

in $\text{W}/(\text{m}^2 \text{sr})$, where s is distance along the line of sight in cm and the factor 4π accounts for the fact that an atom radiates into all directions equally, so only that fraction of the emission reaches the detector. Equation (1.4) says the brightness is the emission rate per unit volume, integrated over the length of plasma the instrument happens to be looking through.

It also assumes that a photon, once emitted, escapes without being reabsorbed by another atom on its way out. A plasma for which that is true is called *optically thin* at that wavelength. For the Balmer lines in a divertor this is usually a fair assumption, because the lower level $n = 2$ holds very few atoms to absorb them; for transitions that end on the ground state it is often badly false, since the ground state holds almost all of them. Chapter 6 takes that up, quantifies it for this model, and it is the single most important restriction on what follows.

Now the discouraging part. Equation (1.4) has one measured number on the left and, on the right, the population we want multiplied by a path length nobody knows, integrated over an emitting region whose extent nobody knows either. Worse, n_q is itself proportional to the amount of neutral hydrogen in the viewing volume, and that is set by how much gas recycles off the target and how it moves — neither of which the spectrometer sees. An absolutely calibrated brightness of a single line therefore determines a product of unknowns and none of them individually.

Take the ratio of two lines and most of this cancels. Define the observable used throughout this thesis:

$$R \equiv \frac{I_{H\alpha}}{I_{H\beta}} = \frac{\nu_{32} A_{32}}{\nu_{42} A_{42}} \cdot \frac{\int n_3 ds}{\int n_4 ds}, \quad (1.5)$$

where n_3 and n_4 are the populations of the $n = 3$ and $n = 4$ levels in $1/\text{cm}^3$ and R is dimensionless. The prefactor is a ratio of atomic constants, known to high accuracy. What Eq. (1.5) does is remove the geometry: provided the two lines are emitted by the same volume — the substantive assumption here, and a good one for two lines of the same series from the same species — the path length divides out, the absolute neutral density divides out, and the absolute calibration of the instrument divides out with them. Only the *relative* sensitivity of the detector at two nearby visible wavelengths is needed, which is a far easier measurement to make and to maintain.

What survives is a pure ratio of two excited-state populations. That is the whole appeal of a line ratio, and it is why R , rather than any single brightness, is the quantity a divertor spectroscopist works with. It is also where the burden of interpretation now sits. A line ratio contains no geometry and no absolute density, so every piece of physics in it is compressed into one question: *what sets the ratio of the $n = 3$ and $n = 4$ populations?* The next section answers that with a model, and the section after it examines what the model had to assume.

1.4 From a ratio to a temperature

The question: what stands between the number the spectrometer produces and the (T_e, n_e) the physicist writes down?

An atom in a plasma is pulled in several directions at once. Free electrons collide with it and knock its bound electron up the ladder of Eq. (1.1), or right off it, which is *ionisation*. Ions capture free electrons back into bound levels, which is *recombination*; this happens either with the excess energy carried off by a photon (radiative recombination) or with it carried off by a second electron that happens to be present (three-body recombination, which therefore needs two electrons and becomes important at high density). Meanwhile every excited atom is decaying radiatively on its own clock, and every decay from a high level cascades down through the intermediate ones. No single one of these processes dominates in a divertor. All of them run at once, at comparable rates.

The object that keeps track of all of it is a *collisional–radiative* (CR) model. It is nothing more exotic than a population balance written for every level of the atom: accumulation equals everything arriving minus everything leaving, with one term per atomic process and a measured or computed rate coefficient attached to each. The framework was set out for hydrogenic ions by Bates, Kingston and McWhirter in 1962 [7], whose central point was that neither of the two classical limits describes a laboratory plasma: not coronal equilibrium, where the density is so low that every collisional excitation is followed by a radiative decay before anything else happens, and not local thermodynamic equilibrium (LTE), where collisions are so frequent that the level populations take their thermal equilibrium values. Real plasmas sit between the two, and the populations there have to be computed.

A chemical engineer will recognise this object on sight. It is a reaction network in a well-mixed vessel: one species per energy level, one reaction per atomic process, a feed stream, and a set of coupled first-order rate equations with a rate-constant matrix. Chapter 3 builds it explicitly and pushes that correspondence considerably further than a metaphor would go.

Solve the CR model at a fixed T_e and n_e and it returns the population of every level, hence n_3 , hence n_4 , hence R through Eq. (1.5). Do that on a grid of (T_e, n_e) and the result is a table — equivalently, a family of contours in the (T_e, n_e) plane along which R is constant. Inverting a measurement means locating the measured value of R on that map. One ratio is one constraint on two unknowns, so practice supplies a second: another line ratio, the Stark broadening of a line by the plasma’s own microscopic electric fields, continuum emission, or an independent estimate of one of the two parameters. The intersection of two contours fixes the point.

This is not a hypothetical procedure. Balmer-spectrum analysis against a CR model has been used to infer divertor conditions since at least the Alcator C-Mod measurements of Pigarov, Terry and Lipschultz, who matched calculated and measured Balmer spectra — populations of the excited states supplied by a collisional–radiative model — to determine the plasma parameters in the recombining region of a divertor at electron temperatures near one electronvolt [8]. The modern versions of the technique are more elaborate but the chain is the same one.

In practice the table is rarely built by hand. The Atomic Data and Analysis Structure (ADAS) distributes precomputed coefficients for exactly this purpose, tabulated over (T_e, n_e) [9, 10]. What matters for the argument of this thesis is a detail of how ADAS stores them: for each line it supplies *two* emissivity coefficients, one for the emission

driven by excitation out of the neutral ground state and one for the emission driven by recombination from the ion. The database keeps the two apart deliberately. The next section is about what happens when they are combined.

1.5 The assumption inside the table

The question: a table has two axes; how many quantities does the answer actually depend on?

Take the physical picture before any algebra. An atom sitting in the $n = 3$ level got there by one of two routes. Either it started as a neutral atom in the ground state and an electron collision kicked it up — fed *from below* — or it started as an ion and captured a free electron into a high level, from which it cascaded down — fed *from above*. Those are two different supply lines drawing on two different reservoirs: the population of ground-state neutral atoms, n_{1s} , and the population of ions, n_{ion} . Every excited level is filled by some mixture of the two.

Because the population balances are linear in the populations, the mixture is exact rather than approximate. For every level p ,

$$n_p = \underbrace{a_p n_{1s}}_{\text{fed from below}} + \underbrace{c_p n_{\text{ion}}}_{\text{fed from above}}, \quad (1.6)$$

where n_p , n_{1s} and n_{ion} are in $1/\text{cm}^3$, the dimensionless coefficient a_p is the number of atoms in level p supported per ground-state atom, and c_p is the number supported per ion. What Eq. (1.6) does is separate the atomic physics from the plasma composition: a_p and c_p depend on T_e and n_e and on nothing else, while n_{1s} and n_{ion} describe how much of each reservoir is present. This split is as old as the CR model itself — it is the population-coefficient decomposition of Bates, Kingston and McWhirter, written by Fujimoto and McWhirter in the normalised form Chapter 3 adopts [7, 11, 12]. Its two limits have names that will recur: a plasma in which the first term dominates is called *ionising*, one in which the second dominates is called *recombining*, and a divertor passes from the first into the second as it detaches.

Now put Eq. (1.6) into the observable. For the two Balmer levels,

$$R \propto \frac{a_3 n_{1s} + c_3 n_{\text{ion}}}{a_4 n_{1s} + c_4 n_{\text{ion}}} = \frac{a_3 (n_{1s}/n_{\text{ion}}) + c_3}{a_4 (n_{1s}/n_{\text{ion}}) + c_4}. \quad (1.7)$$

The second form is the important one: dividing through by n_{ion} shows that the observable depends on the atomic coefficients, which are functions of (T_e, n_e) , *and* on the ratio n_{1s}/n_{ion} , which is not. That ratio is a third, independent quantity. It says how much neutral hydrogen is present per ion, and it is fixed by the plasma's recent history and by transport, not by the local temperature and density.

A table with two axes must therefore dispose of the third quantity, and there are only two honest ways to do it. One is to work in a region of the operating range where a single supply dominates *both* levels, so that the third argument cancels between numerator and denominator of Eq. (1.7) and the table is right whatever the neutral fraction does. The other is to adopt a closure — a rule that fixes n_{1s}/n_{ion} in terms of T_e and n_e . The rule used in practice is *equilibrium ionisation balance*: solve the CR system all the way to steady state, and take the neutral-to-ion ratio it predicts for those local conditions. The state it predicts is called collisional–radiative equilibrium (CRE), and it is worth being

clear that this is a much weaker condition than full thermodynamic equilibrium: it means only that the CR rate equations have stopped changing, not that the populations are thermal.

That closure is an assumption about the plasma, not a piece of atomic physics.

It is important to say who is and is not being criticised, because the databases are not at fault. ADAS keeps the excitation-driven and recombination-driven coefficients separate precisely so that a user can weight them by the n_{1s} and n_{ion} appropriate to the plasma in hand [9]. The most careful divertor analyses do exactly that, and treat the neutral content as an unknown rather than a given: Verhaegh and co-workers, analysing Balmer emission from the divertor of the TCV tokamak, take the neutral fraction n_{1s}/n_e to be a quantity that is estimated rather than measured, assign it a broad prior spanning more than an order of magnitude, and propagate that uncertainty through the whole inference by Monte Carlo so that every inferred quantity emerges as a probability distribution rather than a number [13]. In that community, the dependence written in Eq. (1.7) is known and is handled, and it should be said clearly that this thesis does not discover it.

The same paper is worth one more sentence, because it draws a boundary this thesis has to respect. Verhaegh and co-workers restrict their analysis to the higher Balmer lines, $n \geq 5$, on the grounds that the lower ones — H_α above all — are contaminated by emission from molecular processes that their atomic treatment does not model [13]. This thesis studies H_α and H_β , which is to say precisely the two lines that careful practice avoids in detached conditions, and it does so with a model containing no molecules. Section 1.7 returns to this and Chapter 6 quantifies what it costs; it is stated here so that no reader reaches those chapters feeling it was kept back. The assumption enters at the moment a measured ratio is inverted against a two-parameter lookup, because a two-parameter lookup returns exactly one value of R per (T_e, n_e) pair and therefore requires the two supplies to have been combined in some fixed proportion.

There are now two approximations in play, and keeping them apart is the single most important thing this thesis does.

The quasi-steady-state (QSS) approximation. That the *excited* levels have already reached steady state with whatever reservoirs are present at that instant. This is the standard shortcut of CR theory, and it is what makes a_p and c_p in Eq. (1.6) functions of (T_e, n_e) alone. **The approximation being tested in this thesis is the Bodenstein steady-state approximation, applied to atomic level populations instead of radical concentrations.** It is the same approximation the reader has already met for reactive intermediates, with the same justification — the intermediates turn over far faster than anything else in the system — and, in principle, the same failure mode.

The ionisation-balance assumption. That the *reservoirs themselves*, n_{1s} and n_{ion} , stand in the proportion that CRE would give for the local T_e and n_e . This is what removes the third argument from Eq. (1.7) and allows the table to have two axes.

These are statements about different populations relaxing at different speeds, and only the first is what the literature normally means by “the validity of QSS”. A two-parameter table needs both to hold. Chapter 3 makes the distinction exact, defines a separate error measure for each, ε^{QSS} and ε^{CRE} , and shows why conflating them makes it possible to measure the second and report it as the first. Whether either holds in a real divertor is the subject of the next section.

1.6 A plasma that never finishes changing

The question: how long does the plasma get between disturbances, and is that long enough?

The ionisation-balance assumption would be harmless if divertor plasmas sat still. They do not, and they are not meant to.

The largest regular disturbance is the edge-localised mode (ELM). The confined plasma maintains a steep pressure gradient at its edge; periodically that gradient becomes unstable, the edge relaxes, and a burst of particles and energy is expelled into the SOL and flushed down the field lines onto the divertor.

The numbers that matter here are the ones describing a single event, and they are short. In present-day machines an ELM heat pulse at the target lasts between about 0.75 ms and 3 ms, and the scaling to ITER gives a deposition rise time of 250 μ s with a decay roughly twice that, the pulse being well approximated by a triangular waveform; each event carries away between two and ten per cent of the energy stored in the plasma edge [14]. So the conditions in the divertor are reset over a few hundred microseconds, repeatedly.

How often they are reset is a more delicate question, and it is worth being careful because the usual figure is weaker than it looks. The expected natural ELM frequency in ITER is of order 1 Hz at the full plasma current of 15 MA, rising to a few hertz at reduced current and scaling roughly as the inverse square of the current [15]; present-day machines, being smaller, run at tens to hundreds of hertz. The frequently quoted range of one to two hertz for ITER traces, as far as could be established for this chapter, only to a conference paper that was not peer reviewed, and it is not repeated here for that reason. [TODO: If a refereed derivation of the ITER natural ELM frequency exists, cite it here in place of Kirk *et al.* (2013), whose statement of the number sits in an introduction rather than in a result.]

The distinction matters for the argument. An ELM is not a gentle modulation that the plasma tracks quasi-statically; it is a step change delivered in a few hundred microseconds, after which the plasma is left to recover. The relevant question is therefore not how often the steps come but how long the recovery takes relative to a single step — and that is a question about the plasma’s own internal clocks.

An ELM is not the only transient. The detachment front itself moves: as the neutral pressure in the divertor is raised, the ionisation region lifts off the plate and travels back up the field lines, and where it settles is controlled by feedback on a gas valve. Operating near detachment onset means operating near a threshold, where the plasma responds to a small change in the control parameter by moving a long way. Add the current ramps at the start and end of a discharge, periodic core relaxations, and deliberate density modulation, and the picture is of a system that is repeatedly stepped rather than one that is ever allowed to settle. Whether the intervals between steps are milliseconds or seconds does not by itself decide the question; what decides it is whether the plasma reaches its new balance before the next step arrives.

Now set that against the clock the table assumes, and note carefully which clock that is. It is not the lifetime of an excited state. Those are nanoseconds — Eq. (1.4)’s Einstein coefficients are of order 10^8 1/s — and nothing a tokamak does on a millisecond disturbs them. The relevant clock is the time the *ground-state* population takes to come back into balance with the ions after conditions change, and that is a much slower thing.

One can see why it is slower without any model at all. To ionise a hydrogen atom out of the ground state an electron must bring 13.6 eV. In a plasma at $T_e = 3$ eV,

the fraction of electrons in the Maxwellian tail with at least that energy is of order $\exp(-13.6/3) \approx 1.1 \times 10^{-2}$: about one electron in a hundred. To excite the same atom to $n = 2$ costs only 10.2 eV, and to move an atom that is *already* in $n = 3$ takes a fraction of an electronvolt, which essentially every electron can supply. The ground state is therefore separated from everything else by a barrier that only the tail of the distribution can cross, while the excited levels exchange population freely among themselves. That is the physical origin of the two very different clocks, and Chapter 3 turns the argument into a calculation.

A chemical engineer will recognise this as a Damköhler-number question, and that is exactly what it is. If the characteristic time of the disturbance is short compared with the slowest chemical relaxation in the system, then the composition during the disturbance is not the equilibrium composition, and a chart that assumes equilibrium describes a state the system has not reached. Here the disturbance lasts milliseconds; the fast levels relax in nanoseconds; and the slow reservoir relaxes on a timescale Chapter 3 computes and Chapter 5 compares against the event.

So the complication, stated in full: a table built on two assumptions is being applied to a plasma that plainly violates the second one, and possibly the first. Which of the two actually fails, and what it costs, is not obvious from anything said so far — and it turns out not to be obvious at all.

1.7 What is already known

The question: how much of this is new, and how much has been known for decades?

This question has been asked before, in several forms, and it is better to be precise about what was established than to be told at an examination. The contribution of this thesis is narrower than the question it asks, and the narrowing is worth stating at the outset.

The time-dependent hydrogen problem has been solved before, and the overlap is close enough to state plainly. Sawada and Fujimoto studied exactly this system — atomic hydrogen, a CR model, and an abrupt change of plasma parameters — and their paper is titled, in part, for the validity range of the quasi-steady-state solution of the coupled rate equations [16]. What they established is a statement about *which level controls the approach to steady state*. Solving the coupled rate equations after a step, they found that in an ionising plasma the levels below Griem’s boundary — the level at which radiative and collisional depopulation rates are equal — relax at their own natural lifetimes while every level above it relaxes at the rate of the boundary level itself; in a recombining plasma the roles invert, the levels above the boundary relaxing on their own timescales and those below following the boundary. Either way the overall response of the excited manifold is set by the relaxation time of Griem’s boundary level.

That is a result about the excited levels, and it is the reason this thesis does not claim to have discovered that the excited manifold responds on a finite timescale. The difference is in what is being asked. Sawada and Fujimoto asked when the excited-level solution is valid, and answered it with a timescale internal to that manifold. The question here is what a *diagnostic inversion* costs when the ground-state reservoir underneath that solution has not settled — a different quantity, controlled by a different and far slower clock, and one that can be wrong even when the excited-level solution is perfectly

accurate. Chapter 5 returns to Griem’s boundary level, which the present model locates independently, and uses it as a check.

The general validity question was settled, as a matter of principle, twenty-five years ago. Greenland analysed the reduction of CR systems in general form, in two papers of the same year, and reached two conclusions that bear directly on this work. The first is that the conditions for a valid reduced CR model are conditions on the *eigenvectors* of the full matrix of rates, and that — in his words, and in contrast to the conventional view — the eigenvalues have secondary importance [17]. The second is that the validity criteria he derives are *not related to the equilibrium timescales* of the system [18]. The two statements come from different papers and are cited separately here because they are separately attributable. Those statements are general, they are prior, and this thesis does not improve on them as statements. What it does is quantify them for one specific diagnostic: Chapter 5 shows that the standard timescale-separation criterion does not predict the diagnostic error, measures the error it fails to predict, and maps where in the operating range that error is large. Turning a general criterion into a number a diagnostician can apply is the contribution; discovering the criterion is not.

That the ratio depends on the neutral reservoir is current best practice, not a discovery. As Section 1.5 noted, Verhaegh and co-workers already treat the neutral fraction as a free assumed parameter and propagate its uncertainty by Monte Carlo [13]. The dependence in Eq. (1.7) is therefore known. What this thesis adds is a *closed form* for the sensitivity — an exact expression for how strongly the ratio responds to a change in the reservoir, derived in Chapter 3, together with a bound showing that the response can never exceed a certain size — and a *map* of where in the (T_e, n_e) plane that sensitivity is large enough to matter. A closed form does something a Monte Carlo cannot: it says which conditions are safe without requiring the calculation to be repeated.

The classical validity criteria answer a different question. Griem asked at what principal quantum number collisions become fast enough to hold a level in thermal equilibrium with the free electrons, and defined a boundary level above which the upper part of the manifold is thermalised [19, 20]. That is a statement about the internal structure of the excited levels at one instant. It says nothing about whether the ground state has settled, which is the question here. Chapter 5 nonetheless finds Griem’s boundary level turning up where the present model says it should, which is a useful consistency check rather than a new result.

Real divertor Balmer light is not purely atomic. One line of work must be flagged now because it bounds what any conclusion of this thesis can claim. In detached conditions a substantial part of the observed Balmer emission does not come from the atomic processes modelled here at all. Hydrogen in a cold divertor is largely molecular, and molecules open additional routes to excited atomic levels — among them molecular-assisted recombination (MAR), in which a molecular ion recombines and dissociates into an excited atom. Analyses of discharges on the TCV and MAST-U tokamaks attribute a large share of the low- n Balmer light to these channels near detachment onset [21, 22]. The model built in this thesis contains no molecules at all. Chapter 6 states the consequence plainly, and it is the principal reason the results here are presented as statements about a class of CR model rather than as predictions for a particular machine.

Taken together: the system has been modelled, the general validity criterion has been stated, and the dependence on the neutral reservoir is known and handled in practice. What is missing is a quantitative, closed-form answer to the specific question of how much a two-parameter inversion of the Balmer ratio is in error when the reservoir has not settled, and a map of where that error lives. That is the gap this thesis fills.

1.8 The question, and the suspect

The question: what is the thesis actually going to compute, and what does the obvious physical argument predict it will find?

The question, stated plainly:

When a hydrogen plasma at divertor conditions is disturbed on a timescale short compared with the relaxation of its ionisation balance, how large is the error in a (T_e, n_e) inversion of the H_α/H_β ratio, what physical quantity sets that error, and where in the operating range is it largest?

Answering it needs a model that does not assume the answer. This work therefore builds a time-dependent CR model of atomic hydrogen — every level tracked as a dynamical variable, with orbital angular momentum resolved wherever that resolution changes the physics — subjects it to an abrupt change in electron temperature of the kind an ELM delivers, and follows the line ratio in time. Chapter 2 assembles the atomic data it runs on and Chapter 3 assembles the equations.

What we expected to find. Section 1.5 identified two assumptions, and the obvious suspect is the first one. The QSS approximation is the step that eliminates several dozen dynamical variables by fiat; it is the step that every review of CR modelling flags; and it is the step whose chemical-engineering analogue — the Bodenstein approximation — every undergraduate is warned can fail when the intermediates are not fast enough. In a plasma whose conditions are reset over a few hundred microseconds by every ELM, the natural expectation is that some of the excited levels will simply fail to keep up, the population distribution will lag behind the conditions, and R will be wrong for that reason. That was the expectation this work set out to test, and it is why the model tracks every level in time rather than assuming any of them are stationary.

What is at stake if that expectation is right. If the excited levels lag, then the coefficients a_p and c_p of Eq. (1.6) are not functions of (T_e, n_e) during a transient, the whole population-coefficient framework loses its footing during exactly the events one most wants to diagnose, and the repair is a genuinely time-dependent inversion.

And if it is wrong? Then the second assumption is still standing there, untested, and it is the one the table needs just as much.

Which of the two fails, by how much, and what physical quantity controls it, is the subject of Chapters 3 and 5. The answer is not the one stated above as the expectation — that much can be said here, because the shape of the argument depends on it — and the reason it is not is the reason this thesis exists.

Two things should be said in the same breath, before any of it. First, the claim this work defends is conditional: *if* the ground-state neutral population is frozen for the

duration of the disturbance, which requires that neutral transport and recycling do not renew it faster than the atomic physics does, then the inversion carries the error computed here. Second, the model is zero-dimensional, optically thin, and contains no molecules. Chapter 6 examines every one of those boundaries and quantifies what it costs, rather than listing them as caveats. A result stated without its scope is the characteristic failure of this subject, and it is not repeated here.

Everything from this point on is construction. The section below says what each of the remaining chapters contributes to it.

1.9 Outline of the thesis

The question: where does the argument go from here?

Chapter 2, *The Atoms*, establishes the inputs. It describes everything that can happen to a hydrogen atom in a plasma and how fast each process runs, sets out the levels the model tracks and the reason orbital angular momentum is resolved for the lower ones and bundled above them, shows how a measured collision cross section becomes a rate coefficient, and gives the provenance and accuracy of every dataset used. It makes a structural argument for including a process the standard treatments often omit, and is explicit about where the atomic data are least secure.

Chapter 3, *Two Clocks*, is the conceptual core. It assembles the CR rate equation from a population balance on each level, checks the assembly against particle conservation, and then diagonalises the rate matrix to expose the timescale structure the Boltzmann-factor argument of Section 1.6 anticipated: one very slow mode — the ground state — sitting far above a dense band of fast ones. It presents the QSS approximation as the Bodenstein approximation applied to that fast band, derives the exact dependence of R on the ground-state reservoir, and defines the two reference states and the two error measures the rest of the thesis keeps apart.

Chapter 4, *Does the Model Work?*, asks what a wrong model would fail, and then applies those tests: detailed balance, the coronal and LTE limits, the timescale hierarchy, and an external benchmark against independently published population coefficients. It also records the errors this project’s own audit found, what each cost, and how each was caught.

Chapter 5, *What the Model Says*, reports the result: the relaxation in time, the size of each of the two errors, the mechanism that sets the larger one, a map of where in the (T_e, n_e) plane it lives, whether it survives averaging over a transient event, and whether the standard timescale-separation criterion predicts any of it.

Chapter 6, *What This Model Cannot Say*, bounds the claim — optical thickness, molecular channels, the truncation of the level manifold, the fixed ion reservoir, and the remaining assumptions — with each consequence quantified where quantification is possible and stated as open where it is not.

Chapter 7 states the criterion in a form another modeller can apply to their own rate matrix, says where in the divertor operating range a (T_e, n_e) inversion should be distrusted, and sets out what a time-dependent inversion would require.

The argument begins with the atoms. Before anything can be said about how fast the populations change, we need to know what the processes are and how well their rates are known, which is where Chapter 2 starts.

Chapter 2

The Atoms

Chapter 1 left us with a question and no instrument to answer it. A spectroscopist measures the ratio of two hydrogen lines in a divertor, looks the answer up in a table, and reads off a temperature and a density. The table was computed on the assumption that the plasma has stopped changing; a divertor never stops changing. Everyone asked whether the table still holds worries about the same thing first: the excited states of the atom are the fast, delicate part of the system, and surely they cannot keep up with a plasma torn about on microsecond timescales. Testing that suspicion needs a model that follows every level of the atom in time, assuming in advance that none of them has settled.

This chapter builds the raw material for that model. There is no time derivative anywhere in it. It is about the *atoms*: what can happen to a hydrogen atom in a plasma, how fast, where the numbers come from, and how far they can be trusted. Chapter 3 puts them in motion.

One sentence belongs before any physics, because it is the reader's fastest route into everything that follows. **The approximation this thesis tests is the Bodenstein steady-state approximation, applied to atomic energy levels instead of radical concentrations.** Anyone who has written $d[R^\bullet]/dt \approx 0$ for a reactive intermediate has used it. The 43 energy levels below are the chemical species, the atomic processes are the elementary reactions, and this chapter measures the rate constants before that approximation is made.

2.1 What happens to a hydrogen atom in a plasma

2.1.1 The atom, from scratch

A hydrogen atom is one electron bound to one proton. Quantum mechanics allows the electron only certain energies, labelled by the *principal quantum number* $n = 1, 2, 3, \dots$, with binding energy

$$I_n = \frac{I_H}{n^2}, \quad I_H = 13.6058 \text{ eV}. \quad (2.1)$$

I_n (eV) is the energy needed to pull the electron free and leave a bare proton. Both sides are an energy, so the dimensional check is trivial; the *shape* of Eq. (2.1) is what matters. The levels crowd together as n grows: $n = 1$ lies 10.20 eV below $n = 2$, but $n = 14$ and $n = 15$ are separated by 0.0089 eV. High levels are easy to move between and easy to strip off; low levels are not. That asymmetry generates the two timescales of Chapter 3.

Within a shell the electron also carries *orbital angular momentum*, quantised by $\ell = 0, 1, \dots, n-1$. In hydrogen, and only in hydrogen, all ℓ states of a given n have the same energy: they are *degenerate*. Spectroscopy labels them $\ell = 0, 1, 2, 3, \dots \equiv \text{S, P, D, F, } \dots$, so “ $2p$ ” means $n = 2, \ell = 1$. The number of quantum states sharing a label is the *statistical weight*

$$g_{n\ell} = 2(2\ell + 1), \quad g_n = \sum_{\ell=0}^{n-1} g_{n\ell} = 2n^2, \quad (2.2)$$

the factor 2 counting electron spin. g is a dimensionless counting number and appears only where a thermodynamic argument needs to know how many ways a state can be occupied: $g = 6$ for $2p$, $g = 2$ for $2s$.

An atom dropping to a lower level can emit the energy difference as one photon. The levels are discrete, so the photon energy is too, and the light appears at one sharp wavelength: a *spectral line*. Lines ending on $n = 2$ form the Balmer series and lie in the visible, which is why a fibre-coupled spectrometer sees them; this thesis uses H_α ($n = 3 \rightarrow 2$, 656 nm) and H_β ($n = 4 \rightarrow 2$, 486 nm). A line’s brightness is proportional to the population of the level it comes from, so measuring light measures populations — which is why the populations are worth computing.

2.1.2 Six processes, and how fast they run

Six things can happen to an atom in a plasma of electron density n_e ($1/\text{cm}^3$) and temperature T_e (eV), and the model is built from these and nothing else. *Electron-impact excitation* pushes the bound electron up, and needs a free electron carrying at least the energy gap. *Electron-impact de-excitation* is the reverse and has no threshold, so it is generally the faster of the pair. *Ionisation* removes the electron altogether, and is the only process that takes an atom out of the set of levels we track. *Recombination* does the reverse, by two distinct mechanisms treated in Section 2.5. *Spontaneous emission* drops the atom to a lower level with no collision partner at all — the only process whose rate is independent of density. *Proton-impact ℓ -mixing* shuffles population between the ℓ sublevels of one shell; it costs no energy, because those sublevels are degenerate, and Section 2.7 shows it is the fastest process in the model.

The chemical-engineering reading is exact. Each level is a species and each process an elementary reaction. Excitation, de-excitation, ionisation and ℓ -mixing need a collision partner, so their rates carry a density factor exactly as a second-order rate law does. Spontaneous emission is unimolecular, like a first-order decomposition. Recombination is the odd one out, because it creates atoms out of a species we have chosen not to track, and Chapter 3 shows it therefore behaves like a constant feed stream into a stirred tank rather than a reaction inside it.

Before quoting numbers it is worth saying what we expect, so the table can disagree. An allowed optical transition should run at $\sim 10^8$ 1/s; collisions at $n_e \sim 10^{14}/\text{cm}^3$ should be slower, since a coefficient of order $10^{-9} \text{ cm}^3/\text{s}$ times that density is only 10^5 1/s; and ground-state ionisation at a few electronvolts should be slower still, needing an electron above 13.6 eV from a distribution whose mean energy is far below it. What ℓ -mixing will do is not obvious in advance.

Table 2.1 confirms the expectations and exceeds one of them by three orders of magnitude. Radiative decay of $2p$ takes 1.60 ns and ground-state ionisation 71.0 μs , both as anticipated — but ℓ -mixing empties $2s$ in under 7 ps, two hundred times faster than the fastest radiative decay in the atom. That number is why Section 2.7 exists. It is also the

Table 2.1: Representative rates for the six processes at the benchmark point $T_e = 2.947$ eV, $n_e = 1.389 \times 10^{14}/\text{cm}^3$ (grid indices [23, 5]). Rate coefficients are read from the stored arrays of Table 2.4 at temperature index 23; a rate in 1/s is the coefficient times n_e , except for spontaneous emission, which needs no partner. The recombination entry is a feed per unit proton density, not a rate an atom experiences. The spread is fourteen orders of magnitude, which is why the problem is numerically stiff.

Process	Example	Rate (1/s)	1/rate
Proton ℓ -mixing	$2s \rightarrow 2p$	1.475×10^{11}	6.78 ps
Spontaneous emission	$2p \rightarrow 1s$	6.268×10^8	1.60 ns
Ionisation	$2p \rightarrow \text{H}^+$	8.639×10^6	116 ns
Collisional de-excitation	$2p \rightarrow 1s$	8.173×10^5	1.22 μs
Collisional excitation	$1s \rightarrow 2p$	7.686×10^4	13.0 μs
Ionisation	$1s \rightarrow \text{H}^+$	1.409×10^4	71.0 μs
Recombination feed	$\text{H}^+ \rightarrow 1s$	1.095×10^1	—

first sight of the difficulty: a chemical engineer recognises a rate spread of fourteen decades as a stiff network, of exactly the kind that motivates the steady-state approximation for intermediates in the first place.

2.2 The state space

Hydrogen has infinitely many bound levels, so a model must stop somewhere, and must also decide how finely to describe each level it keeps. Too few levels and the cascade feeding the Balmer lines is cut short; too much detail at high n and the matrix grows without the extra states carrying independent information.

This model tracks **43 states**. Levels $n = 1$ –8 are resolved in ℓ (indices 0–35, ordered $1S$, $2S$, $2P$, $3S$, $\dots$, $8J$); levels $n = 9$ –15 are *bundled*, one state per shell carrying the full shell weight $g_n = 2n^2$ (indices 36–42). The ordering is fixed by `data/processed/collisions/K_exc_full/state_index.csv` and every array in the pipeline is indexed in it. The ion H^+ is deliberately *not* a 44th state: it is an external reservoir, a choice whose consequences are listed in Section 2.10.

Why ℓ must be resolved at low n . The obvious guess is that ℓ is a refinement — nice to have, not load-bearing. At $n = 2$ it is wrong, and a selection rule is the reason. Electric-dipole (E1) radiation, much the strongest emission mechanism, requires $\Delta\ell = \pm 1$. The transition $2p \rightarrow 1s$ satisfies this and runs at $A_{2p \rightarrow 1s} = 6.2684 \times 10^8/\text{s}$ (`Radiative/A_resolved.npy`, element [0, 2]). The transition $2s \rightarrow 1s$ does not: both states have $\ell = 0$, and the E1 rate is exactly zero. In this dataset the total radiative loss from $2s$ is $\gamma_{2s} = 0$ identically; the two-photon decay hydrogen really possesses, of order 10/s, is not included, and would be negligible beside the collisional rates of Table 2.1 in any case. So the two sublevels of $n = 2$ are not refinements of each other: one empties in 1.6 ns and the other cannot empty radiatively at all. Bundling them and weighting the decay statistically would hand the metastable $2s$ level a decay rate it does not have.

Why bundling is safe at high n . Above $n = 8$ the argument reverses. Proton ℓ -mixing scales roughly as n^4 (Section 2.7), so by $n \approx 9$ it redistributes population within a shell far faster than anything else happens to that shell. The sublevels then sit permanently in the ratio of their statistical weights and one number per shell holds all the information.

That is an argument, not a measurement, and it can be tested: do the ℓ sublevels of a shell the model *does* resolve come out statistical anyway? For $n = 4$ the statistical share of the $4f$ sublevel is $g_{4f}/g_4 = 14/32 = 0.4375$. Solving the model across the grid gives 0.4361–0.4375 (register entry A6) — statistical to four figures, with the largest departure, 0.3%, at $n_e = 10^{12}/\text{cm}^3$, the lowest density on the grid and exactly where ℓ -mixing is weakest. Two caveats belong in the same breath. The bundled block carries no ℓ -mixing of its own — it is *assumed* statistical rather than driven to it, and that assumption has not been tested directly. And the truncation at $n_{\text{max}} = 15$ is a real limitation whose size Chapter 6 reports as measured rather than assumed.

Where the ordering and the grids come from. A 43-state model has one correct row ordering, and every array must agree on it; getting it wrong produces no error, only plausible numbers about the wrong states. The pipeline therefore never lets a script declare the ordering for itself. `src/validation/cr_context.py` loads the grids, the matrix and the state ordering from the files the pipeline wrote, cross-checks their shapes, and raises if any is missing or inconsistent; its docstring names the failure mode it prevents — “If a verification script hardcodes them and the pipeline later changes, the script keeps running and silently mislabels every result — which is exactly the failure mode verification is supposed to catch.” Every verification result in Chapters 4 and 5 comes from a script that imports it. The grid itself is declared in `src/rates/assemble_cr_matrix.py` and written to disk so downstream code reads rather than redeclares it: 50 temperatures logarithmically spaced over $T_e = 1\text{--}10\text{ eV}$ and 8 densities over $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$, giving 400 points. The benchmark point carried through this thesis is the one nearest $T_e = 3\text{ eV}$, $n_e = 10^{14}/\text{cm}^3$: index $[23, 5]$, $T_e = 2.947\text{ eV}$, $n_e = 1.389 \times 10^{14}/\text{cm}^3$. It is an illustrative anchor, so that the reader accumulates familiarity with one place in parameter space. No conclusion rests on it; every general claim is carried by the full grid.

2.3 From a cross section to a rate coefficient

Atomic collision theory does not deliver rate constants; it delivers *cross sections*. The cross section $\sigma(E)$ (cm^2) is the effective target area an atom presents to an electron of kinetic energy E : a beam of flux Γ ($1/(\text{cm}^2\text{ s})$) drives the process $\Gamma\sigma$ times per second. For excitation σ vanishes below the threshold ΔE , rises above it, and falls again at high energy as the electron passes too quickly to interact.

A plasma is not a beam. What a kinetic model needs is one number per temperature, the *rate coefficient* $C(T_e) = \langle \sigma v \rangle$, which for a Maxwell–Boltzmann electron distribution reduces to

$$C(T_e) = \sqrt{\frac{8}{\pi m_e}} (k_B T_e)^{-3/2} \int_{\Delta E}^{\infty} \sigma(E) E e^{-E/k_B T_e} dE, \quad (2.3)$$

with m_e the electron mass (kg), $k_B T_e$ the temperature as an energy (J), E the incident energy (J), ΔE the threshold (J) and σ in m^2 ; the result is converted to cm^3/s . Dimensionally $\text{cm}^2 \times \text{cm}/\text{s} = \text{cm}^3/\text{s}$, and $C n_e n_p$ is then $1/(\text{cm}^3\text{ s})$, a number of events per unit volume per unit time: this is mass-action kinetics with C as a second-order rate constant.

What Eq. (2.3) does physically is weight the cross section by how many electrons arrive at each energy and how fast, with the lower limit enforcing that electrons below threshold contribute nothing.

One reading of it is a common trap. The prefactor carries $T_e^{-3/2}$, but $C(T_e)$ does not fall as $T_e^{-3/2}$: the integral grows as more of the distribution clears the threshold, and for a threshold process that growth wins easily. The ground-state excitation coefficient $C_{1s \rightarrow 2p}$ is $5.5316 \times 10^{-10} \text{ cm}^3/\text{s}$ at the benchmark temperature and nearly three orders of magnitude smaller at 1 eV.

The implementation is `compute_K_CCC.py`, function `maxwell_average`, and four choices in it could have gone wrong silently. The integral is trapezoidal on a 5000-point uniform grid; a 500-point grid was tested and gives $\sim 17\%$ error at 1 eV, so the grid density is measured, not defaulted. The threshold is taken from Eq. (2.1) rather than from the data file, avoiding the noisy near-threshold region. Interpolation returns zero outside the tabulated range. And the integral stops at the top of the data (100 to 968 eV) rather than at infinity; since $e^{-E/k_B T_e} = 4.5 \times 10^{-5}$ at 100 eV and $T_e = 10 \text{ eV}$, the missing tail is below 0.01 % over the range used here.

2.3.1 Detailed balance, and why de-excitation is never computed directly

Excitation and de-excitation are not independent. This is the statement a chemical engineer makes when insisting that forward and reverse rate constants of an elementary step be consistent with the equilibrium constant, and the argument has the same shape. Put the plasma in complete thermodynamic equilibrium, where every process is balanced by its reverse. For levels 1 (lower) and 2 (upper),

$$n_1 n_e C_{12} = n_2 n_e C_{21} \implies \frac{C_{12}}{C_{21}} = \frac{n_2}{n_1} \Big|_{\text{eq}} = \frac{g_2}{g_1} \exp\left(-\frac{\Delta E}{k_B T_e}\right), \quad (2.4)$$

with C_{12}, C_{21} in cm^3/s , g_1, g_2 from Eq. (2.2), and $\Delta E = I_1 - I_2 > 0$. Both sides are dimensionless. Physically: going up needs a rare fast electron and coming down does not, so de-excitation wins by the Boltzmann factor, partly offset by there being more upper states to fall out of.

The electron density cancels, because the same electron is the partner in both directions. So do the populations: C_{12} and C_{21} are Maxwellian averages of cross sections, which depend on T_e and atomic structure and nothing else. Equilibrium was only a device to fix the ratio; the ratio holds in any Maxwellian plasma, however far the populations are from equilibrium. That is the entire justification for using Eq. (2.4) in a model whose subject is departure from equilibrium. It fails only if the electrons themselves are non-Maxwellian.

Worked number. For $1s \rightarrow 2p$ at the benchmark temperature, $g_1 = 2$, $g_2 = 6$, $\Delta E = 10.20435 \text{ eV}$, so Eq. (2.4) predicts $3 \exp(-10.20435/2.947052) = 9.404800 \times 10^{-2}$. The stored arrays give $C_{1s \rightarrow 2p}/C_{2p \rightarrow 1s} = 5.53155 \times 10^{-10}/5.88163 \times 10^{-9} = 9.404800 \times 10^{-2}$, identical to seven figures. Over all 819 transition pairs at that temperature, using g and I_n from `state_index.csv`, the largest relative deviation is 2.48×10^{-9} .

That agreement checks the implementation and nothing more, and must not be reported as validating the atomic data. It is machine precision because it is true by construction: de-excitation coefficients are never Maxwell-averaged from de-excitation cross sections but

derived from the excitation coefficients through Eq. (2.4) (`compute_K_CCC.py`, function `detailed_balance`). The reason for building it that way is that it guarantees Saha–Boltzmann populations in the high-density limit, which Chapter 4 uses as a gate — and which is therefore a weaker gate than one the construction did not guarantee. The independent statement, that the underlying cross sections themselves respect microscopic reversibility, is a separate check reported in Section 2.8.

2.3.2 Two hydrogen energies, and why both are right

A reader auditing the source will find two different numbers near 13.6 eV and should not treat the difference as sloppiness. They are two distinct physical quantities.

The first is the *Rydberg energy* $R_\infty hc = 13.605\,693\,122\,994$ eV, the ionisation energy of an idealised hydrogen atom with an infinitely heavy, point-like, non-relativistic nucleus. It is a defined constant of atomic theory, and it is the unit in which collision calculations tabulate their energies. The second is the *ionisation energy of real hydrogen*, 13.598 434 599 702 eV, the measured energy needed to remove the electron from a real H atom in its ground state. The second is smaller by 0.0534 %.

The pipeline uses each where it belongs, and the rule is simple. Cross sections and thresholds supplied in Rydbergs must be converted back with the constant they were tabulated in, so every collisional quantity — the excitation thresholds of Section 2.4, the ionisation thresholds of Section 2.5, and I_H inside the ℓ -mixing formula Eq. (2.10) — carries $R_\infty hc$. Photon energies and spectroscopic level energies describe real atoms and carry the real ionisation energy. Both constants were checked against their published values and both are correct for what they are.

Two warnings go with that. First, the difference is *not* simply the reduced-mass correction: scaling $R_\infty hc$ by μ/m_e reproduces the ionisation energy for about five digits and then stops, and the residual 1.47×10^{-4} eV is the Dirac fine-structure and Lamb-shift contribution. Writing one as the other times m_e/m_H is wrong at the 10^{-5} level. Second, the Rydberg energy is stored in the source under a name — `CHI_H` — conventionally reserved for the ionisation potential. Anything that reads that symbol expecting an ionisation potential, a Saha or Boltzmann exponent in particular, is 0.0534 % too large. Nothing in the assembled matrix does so, but the name is a trap and is flagged here rather than renamed.

2.4 Electron-impact excitation

Excitation lifts atoms out of the ground state and populates everything the spectrometer sees, so its cross sections matter more than any other input. They come from the *convergent close-coupling* (CCC) method of Bray and co-workers [23, 24], in which the collision problem is solved by expanding the target in a large square-integrable basis and enlarging that basis until the answer stops moving — the “convergent” is the claim that the limit is reached rather than estimated. The dataset is the 3115 raw files supplied by Prof. Igor Bray (personal communication, 22 February 2026) in `data/raw/ccc/e-H_XSEC_LS/`, parsed by `src/parsers/parse_ccc.py`. The acquisition report for the same dataset records 3117 files; 3115 are on disk, and the difference is noted here rather than reconciled.

Three instructions came with the data and all three are load-bearing.

The filenames read backwards. The convention is `FINAL.INITIAL`, so `1S.2P` holds $2p \rightarrow 1s$ de-excitation and the excitation $1s \rightarrow 2p$ lives in `2P.1S`. Reading it the wrong way round would transpose the dataset with no obvious symptom, so it is worth saying which direction the model uses: rate coefficients are Maxwell-averaged from the *excitation*-direction files, selected in `compute_K_CCC.py` by the test $n_f > n_i$, and de-excitation follows by Eq. (2.4). The de-excitation-direction files are parsed and stored but never enter the rate tables.

Transitions with $\Delta n = 0$ were excluded, on the data provider’s instruction.

The note reads: “Transitions with $n_{\text{final}} = n_{\text{initial}}$ (e.g. $2S \leftrightarrow 2P$, $3S \leftrightarrow 3P$) have unphysically large cross sections due to non-relativistic degeneracy in the CCC calculation. All $\Delta n = 0$ transitions are excluded from the database.” The reason is that in a strictly non-relativistic treatment the ℓ sublevels of a shell are exactly degenerate, the transition energy is zero, and the long-range dipole interaction gives a divergent cross section. After the exclusion 1740 transitions remain, all with $\Delta n \neq 0$. The stated accuracy for what remains is $\sim 5\%$, which the benchmark of Section 2.8 is able to test.

That exclusion has a consequence which is easy to miss and which Section 2.7 turns into the central argument of this chapter. With $\Delta n = 0$ electron impact removed there is no electron-driven route between $2s$ and $2p$ at all. Together with the fact that $2s$ cannot radiate, it leaves the $n = 2$ manifold with almost nothing to do — unless another collision partner is admitted.

Covering the shells CCC does not reach. CCC data extend to $n = 8$ resolved in ℓ , with resolved data reaching $n = 9$ and $n = 10$. Transitions involving higher bundled shells are filled from the semi-empirical formulae of Vriens and Smeets [25], coded in `compute_K_VS.py`; the implemented expressions are their Eqs. (17)–(19) directly, not the equivalent rearrangement of Hartgers *et al.* [26] that the file’s header docstring describes. Their accuracy is $\sim 20\%$ against $\sim 5\%$ for CCC, so the merge uses CCC wherever it exists. Those formulae need one further input, the shell-to-shell absorption oscillator strength, built from the same Einstein coefficients used in Section 2.6, with a Kramers fallback where a value is missing that is recorded per transition in the metadata. Table 2.2 records which source supplies which block, because “the excitation data are from CCC” would misdescribe a table that is a quarter Vriens–Smeets.

2.5 Ionisation and recombination

2.5.1 Ionisation: the only true loss

Ionisation is the only process that removes an atom from the bound manifold altogether; everything else moves an atom from one tracked level to another. In reactor terms it is the irreversible step consuming reactant to product, and the rest is internal shuffling. The ionisation rate coefficient Q_p (cm^3/s) is obtained by Maxwell-averaging the CCC *total ionisation cross section* (TICS) with Eq. (2.3), the lower limit being the binding energy of Eq. (2.1) (`compute_K_TICS.py`). The threshold is ℓ -independent in hydrogen but the cross sections are not, so the coefficients are ℓ -resolved for $n \leq 8$.

The stored coefficients are not uniform in origin and the metadata records this state by state. The 36 resolved levels carry `CCC_TICS`; the $n = 9$ bundle carries `CCC_TICS_direct`,

Table 2.2: Which source supplies which excitation transitions, from the coverage record in `src/rates/assemble_K_exc.py` and the per-transition attribution in `K_exc_meta.csv`. “res” is an ℓ -resolved state ($n \leq 8$), “bund” a bundled shell ($n = 9\text{--}15$). The collapse rows are ℓ -resolved CCC data summed over final ℓ to give a shell total, which keeps CCC accuracy for the first two bundled shells instead of falling back to Vriens–Smeets. De-excitation coefficients are not listed because they are not independent data: each follows from its excitation partner by Eq. (2.4).

Block	Source	Pairs	Stated accuracy
res $\rightarrow$ res, $n \leq 8$	CCC	546	$\sim 5\%$
res $\rightarrow n = 9$ bundled	CCC, collapsed over ℓ	36	$\sim 5\%$
res $\rightarrow n = 10$ bundled	CCC, collapsed over ℓ	36	$\sim 5\%$
res $\rightarrow n = 11\text{--}15$	Vriens–Smeets	180	$\sim 20\%$
bund $\leftrightarrow$ bund	Vriens–Smeets	21	$\sim 20\%$
total		819	

taken from the bundled `TICS.9` file rather than assembled from sublevels; the six shells $n = 10\text{--}15$ carry Lotz1968, the semi-empirical formula of Lotz [27], with a note recorded against every one of them: “overestimates CCC by factor $\sim 4\text{--}8$; negligible impact” (`collisions/tics/K_ion_final_meta.csv`). That is an honest flag, not a resolved question, and Section 2.8 returns to it, because the overestimate does not stay confined to ionisation.

Worked number. At the benchmark point $Q_{1s} = 1.0140 \times 10^{-10} \text{ cm}^3/\text{s}$ and $Q_{2p} = 6.2171 \times 10^{-8} \text{ cm}^3/\text{s}$ (`K_ion_final.npy`, temperature index 23) — about 600 times easier to ionise an atom already in $n = 2$, which has to supply only 3.40 eV instead of 13.61 eV. This step-ladder character is why the excited levels matter to a quantity as apparently unrelated as the total ionisation rate.

2.5.2 Recombination: two mechanisms, two density dependences

Recombination runs the other way, and telling its two mechanisms apart matters because they scale differently with density. In *radiative recombination* the excess energy leaves as a photon: one electron and one proton are needed, so events go as $n_e n_{\text{ion}}$ and the coefficient α_{RR} has units cm^3/s . In *three-body recombination* a second free electron carries the energy away: three particles must meet, so events go as $n_e^2 n_{\text{ion}}$ and $\alpha_{3\text{BR}}$ has units cm^6/s . Nothing here is special to plasmas. It is the difference between a bimolecular and a termolecular elementary step, with the same consequence: the second mechanism takes over as density rises.

Radiative recombination is computed shell by shell from Johnson’s three-term Gaunt-factor expansion [28], itself a fit to the treatment of Seaton [29],

$$\alpha_{\text{RR}}(n, T_e) = D y^{3/2} e^y [g_0(n)E_1(y) + g_1(n)E_2(y) + g_2(n)E_3(y)], \quad y = \frac{I_n}{k_B T_e}, \quad (2.5)$$

with $D = 5.197 \times 10^{-14} \text{ cm}^3/\text{s}$, E_i the exponential integrals and g_0, g_1, g_2 the Gaunt polynomials of Johnson’s Table 1 (`recombination_rates.py`). y and the bracket are

dimensionless and D carries the units, so both sides are cm^3/s . Physically, capture into a shell is easiest when the electron's thermal energy matches that shell's binding energy, so α_{RR} peaks near $T_e \sim I_n$. The shell total is split over ℓ statistically, $\alpha_{\text{RR}}(n, \ell) = \alpha_{\text{RR}}(n)(2\ell + 1)/n^2$ [30], a step requiring $n_e \gtrsim 10^{10}/\text{cm}^3$ — two decades below our lowest density.

One property of Eq. (2.5) needed checking rather than assuming. $g_1(n)$ is negative and grows with n , so the three-term sum can in principle turn negative for high shells at low temperature; an earlier analysis in this project raised that objection and recommended the single-term form instead. The code keeps the three-term expansion and guards it with a clamp at zero, which would silently replace a negative value rather than report it. Reading the stored arrays directly, neither `alpha_RR_resolved.npy` nor `alpha_RR_bundled.npy` contains an exact zero anywhere on the 50-point temperature grid: the clamp never fires over the range this thesis uses, though it would need revisiting for a colder grid or a longer ladder.

Three-body recombination is not computed independently at all. It comes from ionisation by detailed balance [20], exactly as de-excitation came from excitation:

$$\alpha_{3\text{BR}}(n, \ell, T_e) = Q_{n\ell}(T_e) \frac{g_{n\ell}}{2} \left(\frac{h^2}{2\pi m_e k_B T_e} \right)^{3/2} \exp\left(\frac{I_n}{k_B T_e}\right), \quad (2.6)$$

where h is Planck's constant and the bracketed factor is the cube of the electron thermal de Broglie wavelength, a volume — so the units read $\text{cm}^3/\text{s} \times \text{cm}^3 = \text{cm}^6/\text{s}$, as required. Equation (2.6) imposes Saha equilibrium between a level and the continuum: in complete equilibrium the ionisation flux $Q_{n\ell} n_e$ and the recombination flux $\alpha_{3\text{BR}} n_e^2 n_{\text{ion}}$ must balance, and rearranging Saha for $n_e n_{\text{ion}}/n_{n\ell}$ gives it directly. Like detailed balance for de-excitation, this guarantees the high-density limit rather than testing it.

Assembled, the two mechanisms give the source used throughout the thesis, $\mathbf{S} = n_e(\alpha_{\text{RR}} + n_e \alpha_{3\text{BR}})$, whose units are $1/\text{cm}^3 \times \text{cm}^3/\text{s} = 1/\text{s}$, so that $\mathbf{S} n_{\text{ion}}$ is atoms created per unit volume per unit time. Chapter 3 explains why this is a feed stream rather than an element of the rate matrix.

Worked numbers, and a surprise. The obvious guess is that three-body recombination is a high-density correction, negligible at $n_e = 10^{12}/\text{cm}^3$. That is true only for the ground state. Forming the three-body fraction $n_e \alpha_{3\text{BR}}/(\alpha_{\text{RR}} + n_e \alpha_{3\text{BR}})$ from the stored coefficients at temperature index 23 gives, for 1s, below 10^{-4} at $n_e = 10^{12}/\text{cm}^3$ and 0.0085 at $10^{15}/\text{cm}^3$; but for $n = 15$ it gives 0.9996 at the *lower* density and above 0.9999 at the higher. Capture into the ground state is radiative essentially everywhere on the grid; capture into the top shell is three-body essentially everywhere on the grid. Equation (2.6) shows why: $\alpha_{3\text{BR}}$ inherits an ionisation coefficient that is enormous for a shell bound by 0.060 eV, and the exponential that suppresses it for the ground state is close to unity at the top of the ladder. A loosely bound electron is easy to knock off and, by the same token, easy to capture in a three-body encounter.

That has a consequence Section 2.8 takes up: the three-body coefficients for $n = 10$ –15 are built by Eq. (2.6) from Lotz ionisation coefficients the metadata itself calls too large by 4–8, and those are precisely the shells whose feed is entirely three-body.

2.6 Radiative decay

An excited atom need not be disturbed to decay. The *Einstein coefficient for spontaneous emission* A_{qp} (1/s) is the probability per unit time that an atom in upper level q drops to lower level p , and the total radiative loss from q is $\gamma_q = \sum_{p<q} A_{qp}$, also in 1/s, with $1/\gamma_q$ the radiative lifetime. This is a first-order rate constant in the ordinary chemical sense — the one process with no concentration dependence, so the only one whose relative importance rises as the plasma thins.

Not every pair of levels is connected. E1 radiation requires $\Delta\ell = \pm 1$; transitions violating it are *forbidden*, meaning slower by many orders of magnitude rather than impossible. The dataset here contains E1 transitions only, so a forbidden transition has $A = 0$ exactly. The related *oscillator strength*, a dimensionless measure of how strongly a transition couples to the radiation field, feeds the Vriens–Smeets formulae of Section 2.4.

Two consequences of the selection rule matter to this thesis. First, $2s \rightarrow 1s$ is forbidden, so $\gamma_{2s} = 0$ against $\gamma_{2p} = 6.2684 \times 10^8/\text{s}$ — the $n = 2$ shell holds one level that empties in 1.6 ns and one that cannot empty radiatively. Second, $4f \rightarrow 2p$ is forbidden because ℓ would change by two, so the $4f$ sublevel holds 43.75 % of the $n = 4$ shell by weight and contributes nothing to H_β , decaying only to $3d$. The $n = 4$ shell population and the H_β line intensity are therefore not proportional, and Chapter 4 measures how much that matters to the observable. The three channels that do make H_α are $3s \rightarrow 2p$ ($6.3172 \times 10^6/\text{s}$), $3p \rightarrow 2s$ ($2.2461 \times 10^7/\text{s}$) and $3d \rightarrow 2p$ ($6.4688 \times 10^7/\text{s}$); the three that make H_β are $4s \rightarrow 2p$ ($2.5796 \times 10^6/\text{s}$), $4p \rightarrow 2s$ ($9.6735 \times 10^6/\text{s}$) and $4d \rightarrow 2p$ ($2.0637 \times 10^7/\text{s}$).

The coefficients come from the exact hydrogenic code of Hoang-Binh [31], distributed through the Computer Physics Communications Program Library as ADUU v1.0, which evaluates the radial integrals analytically rather than fitting them [32]. Its 1015 transitions for $n \leq 15$ are stored in `Radiative/H_A_E1_LS_n1_15_physical.csv`; every entry has $|\Delta\ell| = 1$ and $n_{\text{upper}} > n_{\text{lower}}$, and `radiative_rates.py` sweeps every allowed combination on load and raises if one is missing rather than assuming completeness. For the bundled shells the A values are averaged over ℓ with statistical weights — the same assumption that justified bundling in Section 2.2, applied consistently. The pattern to notice is that γ falls steeply with n , from $6.27 \times 10^8/\text{s}$ for $2p$ to $8.16 \times 10^4/\text{s}$ for the $n = 15$ bundle: radiative decay dominates at low n and collisions at high n , and the level where they cross is a measurable feature of the model that Chapter 4 locates.

These are not the recommended values, and it matters to say so. An external check of the four Lyman-series coefficients finds them 0.056 to 0.060 % *above* the critically evaluated values recommended by Wiese and Fuhr [33] — for Lyman- α , $6.2684 \times 10^8/\text{s}$ stored against $6.2649 \times 10^8/\text{s}$ recommended. The offset is systematic, not scattered, and it has a cause. What this code computes is the *exact non-relativistic, infinite-nuclear-mass hydrogenic* transition probability, reproduced to 1–4 parts in 10^5 ; the recommended values include the finite proton mass and the small relativistic and quantum-electrodynamic corrections that the non-relativistic Schrödinger problem does not contain. Both sets are correct for what they are, and using the exact hydrogenic set throughout is internally self-consistent, which is what a rate matrix needs. **But this dataset must not be described as the recommended or evaluated set, and no evaluated compilation should be cited as its source.** It is the analytic hydrogenic solution, and that is what is written here.

The largest silent error available in a rate matrix. One number in this section can be wrong by a third without anything complaining, and the trap is worth spelling out because it is the exact mistake this chapter’s ℓ -resolution is designed to avoid. The value $A_{2p \rightarrow 1s} = 6.2684 \times 10^8/\text{s}$ is an ℓ -resolved level rate: it is the decay probability of an atom that is in $2p$. The corresponding n -resolved shell rate for $n = 2 \rightarrow n = 1$ is smaller by the fraction of the shell that can radiate at all,

$$A_{n=2 \rightarrow n=1} = \frac{g_{2p}}{g_2} A_{2p \rightarrow 1s} = \frac{6}{8} \times 6.2684 \times 10^8/\text{s} = 4.7013 \times 10^8/\text{s}, \quad (2.7)$$

the factor $3/4$ being the statistical weight of $2p$ within $n = 2$, and the missing quarter being the $2s$ sublevel that has no E1 route to the ground state at all. Both sides are in $1/\text{s}$. Placing the ℓ -resolved number on an n -resolved matrix element therefore overestimates Lyman- α emission by 33.3%, and nothing in a conservation check or a sign check would notice, because the error is in a magnitude and magnitudes cancel in column sums (Section 2.9).

This model is ℓ -resolved for $n \leq 8$, so the ℓ -resolved value is the correct one to use here — but that was audited rather than assumed. Reading the stored block directly: the element $A[1s, 2s]$ is exactly zero, the element $A[1s, 2p]$ carries $6.2684 \times 10^8/\text{s}$, $\gamma_{2s} = 0$ and $\gamma_{2p} = 6.2684 \times 10^8/\text{s}$, and across all 36×36 resolved elements there is not one non-zero entry with $|\Delta\ell| \neq 1$. The shell-averaged value of Eq. (2.7) appears nowhere in the resolved block. The bundled block $n = 9\text{--}15$ is n -resolved by construction and carries statistically weighted shell rates throughout, which is the consistent treatment for states that are always in statistical ℓ -equilibrium.

2.6.1 The photons are assumed to leave

Writing γ_q as a bare sum of Einstein coefficients contains an assumption: that every emitted photon escapes and never returns. A plasma for which this holds is *optically thin*. If it is optically thick, a photon emitted by one atom is absorbed by another, which returns to the upper level, and the effective decay rate falls. The chemical-engineering analogue is a reaction slowed because the product cannot leave and keeps re-adsorbing: the kinetics are unchanged, the effective rate constant is not.

The size of the effect is set by the *optical depth* τ_c , the number of absorption lengths a photon must cross to escape, and carried by the *population escape factor* Θ_P , defined so that the effective coefficient is $\Theta_P A$. For the resonance line the model is most exposed to, Lyman- α ($2p \rightarrow 1s$), `src/analysis/escape_factor.py` evaluates the escape factor of Holstein [34] in the formulation of Behringer [35] adopted as routine ADAS214 of the Atomic Data and Analysis Structure (ADAS) [9], by direct quadrature and with no fitting formula:

$$\Theta_P(\tau_c) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} \exp(-\tau_c e^{-x^2}) dx, \quad \tau_c = n_{1s} \sigma_0 \frac{D}{2}, \quad (2.8)$$

with x the frequency offset in Doppler widths (dimensionless), n_{1s} the ground-state density ($1/\text{cm}^3$), σ_0 the line-centre absorption cross section (cm^2) from the Ladenburg relation with oscillator strength $f_{12} = 0.4162$ and $\lambda_0 = 121.567 \text{ nm}$, and D twice the escape path length (cm). Both sides of the first expression are dimensionless. It says: weight the escape probability by the fraction of photons emitted at each frequency, and note that photons in the line wings get out when photons at the line centre cannot.

The geometry is mislabelled, and the label matters. The module documents Eq. (2.8) as applying to “a homogeneous slab of half-thickness b ”. Read directly from the ADAS214 manual, equation 3.14.14 is the *isotropic* case — the escape probability at the centre of a sphere, ADAS geometry g1 — and not the slab case, g2. The distinction is not cosmetic in the thick regime: a true slab requires the further angular average $\Theta_{\text{slab}}(\tau) = \int_0^1 \Theta_{\text{iso}}(\tau/\mu) d\mu$, which is smaller by a factor approaching two, measured as 0.529 at $\tau_c = 1$ and 0.477–0.484 at $\tau_c = 100$ –1000. Everything else in the module was checked against the primary source and is correct: the convention $\tau_c = \kappa_0 b$ with b the edge-to-centre distance is exactly ADAS’s own, the Ladenburg constant $\pi e^2/m_e c = 2.654\,008\,85 \times 10^{-2} \text{ cm}^2 \text{ Hz}$ is right, the Doppler profile normalisation and thermal speed reproduce ADAS’s own tabulated widths, and the leading Holstein constant is 1 rather than the 1.6 sometimes quoted. **This is reported, not repaired.** Where the plasma is thin — which Section 2.10 and Chapter 3 show is the whole region carrying this thesis’s quantitative results, $\Theta_P \rightarrow 1$ — the two geometries agree and nothing depends on the choice. Where the plasma is thick, at the cold edge, the escape factor computed here is roughly twice too large if a slab is the right picture, and Chapter 6 carries that caveat rather than resolving it.

Two constants in this module are wrong, and both are immaterial. An external audit of the same file found the elementary charge entered as $e = 4.803\,26 \times 10^{-10} \text{ esu}$ against the CODATA value $4.803\,204\,713 \times 10^{-10}$ [36], high by 0.001 15 % and entering σ_0 as e^2 , so 0.0023 %; and the Doppler width computed with $m = 1.672\,62 \times 10^{-24} \text{ g}$, which is the *proton* mass where the physically correct object is the neutral hydrogen *atom*, $1.673\,532\,840\,7 \times 10^{-24} \text{ g}$. A neutral atom Doppler-broadens with its own mass, so the line is 0.027 % too wide. Neither shifts any conclusion in this thesis by a detectable amount, and neither has been edited: the repository is under verification rather than development, and a constant corrected quietly is a constant nobody can find again. They are recorded here so that a reader who recomputes σ_0 and gets a slightly different number knows why.

Every rate matrix in this thesis is built with bare A coefficients, $\Theta_P \equiv 1$. Equation (2.8) is a diagnostic applied afterwards, to find where that choice is defensible. Chapter 3 carries out that evaluation across the grid and Chapter 6 states the consequences where the plasma is not thin. The question can be deferred because τ_c depends on n_{1s} , an output of the model rather than an input, and on a path length D the model does not contain at all.

2.7 Proton-impact ℓ -mixing, and why it is not optional

2.7.1 The physical picture, before any formula

Collisional–radiative treatments of hydrogen often omit proton collisions on the grounds that protons are slow and heavy and therefore ineffective. For transitions that change n that is right. For transitions that only change ℓ it is exactly backwards, and the chain is worth following link by link, because it explains a rate that would otherwise be merely a large number in a table.

Degeneracy makes the mixing free, and slowness helps. Within a shell, ns , np , nd have identical energy, so moving population among them costs nothing — there is no threshold to clear, unlike excitation $n \rightarrow n + 1$. And a proton, 1836 times heavier than an electron,

moves 43 times more slowly at the same temperature, so the encounter lasts far longer. For a process needing a sustained push rather than a sharp kick, a long gentle encounter is worth more.

The mechanism is the linear Stark effect. A passing proton is a moving charge; while nearby, the atom sits in a nearly static electric field. Because s , p and d are degenerate, hydrogen responds linearly, and the field's own eigenstates are mixtures of s , p and d . An atom in a pure np state is therefore no longer in an eigenstate once the field arrives; it evolves as a superposition, and when the proton leaves the population has been redistributed across ℓ . No photon and no dipole selection rule is involved.

Distant collisions dominate, logarithmically, and Debye screening cuts them off. The field at impact parameter b falls as $1/b^2$, so the transition probability does too; but the number of protons passing near b grows as $b db$. The product is db/b , so every decade in b contributes equally and the integral produces a logarithm — which is why the process is strong, and why it needs a cut-off. That cut-off is physical: a proton is dressed by a screening cloud, its potential falls as $e^{-R/\lambda_D}/R$, and beyond about one Debye length λ_D the atom feels nothing. So $b_{\max} \approx \lambda_D$ and the Coulomb logarithm is $\ln(\lambda_D/b_{\min})$.

Assembling those links by an impact-parameter calculation gives the scaling

$$\nu_{\ell\text{-mix}} \propto n_{\text{ion}} n^4 T_i^{-1/2} \ln \Lambda, \quad (2.9)$$

with ν in $1/\text{s}$, n_{ion} the proton density ($1/\text{cm}^3$), T_i the *proton* temperature (eV) and $\ln \Lambda$ the Coulomb logarithm. The n^4 comes from a dipole matrix element scaling as n^2 and entering squared; the $T_i^{-1/2}$ from slower protons interacting for longer. Equation (2.9) tells us in advance that ℓ -mixing will be fast, will get faster with density, and will grow explosively up the ladder — which is the quantitative content of the bundling argument in Section 2.2.

2.7.2 The implemented formula

The rates actually used come from the Debye-cutoff form of the Pengelly–Seaton treatment [37] given by Badnell *et al.* [38] — called PSM20 in the code and in this thesis — coded in `src/rates/compute_lm.py`. For the downward transition within shell n , writing $\ell_{>} = \max(\ell, \ell')$,

$$q_{ji} = \frac{a_0^3}{\tau_0} \left(\frac{\pi \mu I_H}{k_B T_e} \right)^{1/2} \frac{D_{ji}}{\omega_\ell} F(U_m), \quad D_{ji} = 6n^2 \ell_{>} (n^2 - \ell_{>}^2), \quad (2.10)$$

with a_0 the Bohr radius, $\tau_0 = a_0/\alpha c$ the atomic unit of time, $\mu = m_H/m_e = 918.08$, $\omega_\ell = 2\ell_{>} - 1$ the weight of the lower sublevel, and D_{ji} the dimensionless dipole coupling. The prefactor a_0^3/τ_0 carries the units and everything else is dimensionless, so q_{ji} is a rate coefficient. Note $D_{ji} \sim 6n^4$ for $\ell_{>} \ll n$, recovering the n^4 of Eq. (2.9), and that temperature enters as $T_e^{-1/2}$, recovering the $T_i^{-1/2}$ under $T_i = T_e$. The upward rate follows by reciprocity, $q_{\text{up}}/q_{\text{down}} = (2\ell_{\text{hi}} + 1)/(2\ell_{\text{lo}} + 1)$, which is Eq. (2.4) for degenerate states, where the Boltzmann factor is unity and only the weights survive.

$F(U_m)$ is the Coulomb logarithm of the physical picture, in the form Badnell's Eq. 9 gives it:

$$F(U_m) = \frac{\sqrt{\pi}}{2} U_m^{-3/2} \text{erf}(\sqrt{U_m}) - \frac{e^{-U_m}}{U_m} + E_1(U_m), \quad U_m = \frac{E_{\min}}{k_B T_e}, \quad (2.11)$$

with $E_{\min}$ the minimum energy transfer set by the cut-off radius $R_c = \lambda_D$. F is dimensionless and *diverges* logarithmically as $U_m \rightarrow 0$: a smaller minimum energy transfer means a larger cut-off radius means more decades of impact parameter contributing. That behaviour is the subject of Section 2.9, because an earlier version of the code had the limit backwards.

Three approximations enter and are carried through the thesis: $T_i = T_e$, $n_{\text{ion}} = n_e$ by quasi-neutrality, and $\Delta\ell = \pm 1$ only. A fourth is easy to miss on reading the code. The Debye length inside U_m is evaluated at a fixed reference density $10^{14}/\text{cm}^3$ rather than at each grid density, on the stated grounds that F depends on density only logarithmically and varies by about 30% across the grid. The stored table has no density axis at all, so the linear $\propto n_e$ scaling applied at assembly is exact while the Coulomb-logarithm factor inside it is frozen at mid-grid. One item is left open rather than settled: Badnell’s formulation offers competing cut-offs to the Debye radius, one set by the level lifetime and one by any residual splitting of the ℓ degeneracy, and the correct R_c is the smallest of the three. Only the Debye cut-off is implemented. At these densities it is expected to be the binding one, but that has not been tested across the grid.

2.7.3 Without it, $n = 2$ is not a physical manifold

We can now close the loop that Sections 2.2, 2.4 and 2.6 each opened, and the result is the strongest single statement in this chapter.

Ask what an atom in $2s$ can do. It cannot radiate: $2s \rightarrow 1s$ is E1-forbidden and $\gamma_{2s} = 0$. It cannot be moved to $2p$ by an electron: $\Delta n = 0$ transitions were excluded from the CCC dataset on the provider’s instruction. What is left is electron-impact excitation to $n \geq 3$, electron-impact ionisation, and proton ℓ -mixing. Decomposing the total loss from $2s$ into those channels, from the stored coefficient arrays at the benchmark point: proton ℓ -mixing $1.4745 \times 10^{11}/\text{s}$; electron-impact collisions to $n \geq 3$ $3.4030 \times 10^7/\text{s}$; electron-impact ionisation $6.9216 \times 10^6/\text{s}$; radiative decay zero. ℓ -mixing carries 99.972% of everything that happens to a $2s$ atom.

Because γ_{2s} is exactly zero, every remaining channel is collisional and scales with n_e , so that share is independent of density: it depends only on temperature, running from 99.994% at $T_e = 1 \text{ eV}$ to 99.921% at 10 eV . **Over the whole 400-point grid, proton ℓ -mixing carries at least 99.92% of the $2s$ loss rate**, the minimum falling at the hot edge. There the contrast is starkest: without ℓ -mixing $2s$ would empty at $6.17 \times 10^5/\text{s}$; with it, at $7.79 \times 10^8/\text{s}$, a factor of 1262.

The conclusion is not that omitting ℓ -mixing perturbs the $n = 2$ manifold. It is that omitting it changes which process controls the manifold, by three orders of magnitude, and leaves $2s$ an artificial metastable trap. Any model of hydrogen that resolves ℓ and excludes $\Delta n = 0$ electron impact *must* include proton ℓ -mixing or it is not describing the $n = 2$ shell at all.

Worked numbers. At the benchmark point $q_{2s \rightarrow 2p} = 1.0612 \times 10^{-3} \text{ cm}^3/\text{s}$ (`K_lmix.npy`, element $[2, 1, 23]$), giving $1.4745 \times 10^{11}/\text{s}$ and a $2s$ lifetime of 6.78 ps. Within $n = 5$ the ℓ -mixing loss rates run from $3.25 \times 10^{12}/\text{s}$ for $5s$ (0.31 ps) to $3.47 \times 10^{11}/\text{s}$ for $5g$ (2.88 ps). An independent hand estimate from the impact-parameter picture, with the Debye length as cut-off and $b_{\min} = n^2 a_0$, gives 0.16 ps for $n = 5$: high by roughly six, in the expected direction, because the crude estimate uses the full orbital size for the dipole matrix element and drops the angular factors that D_{ji} supplies. An order-of-magnitude

Table 2.3: The CCC excitation rate coefficients used in this model, compared against the R-matrix with pseudostates (RMPS) calculation of Anderson *et al.*, over 85 matched transitions at $T_e = 1, 3, 5$ and 10 eV (340 comparisons). Source: `data/processed/collisions/anderson_benchmark_full.csv` and the report `ccc_anderson.md`. Transitions with $n_{\text{upper}} \leq 4$ dominate collisional–radiative kinetics in this range; the disagreement is concentrated entirely in $n_{\text{upper}} = 5$.

Subset	comparisons	within 20 %	mean error
all transitions	340	42.4 %	29.0 %
$n_{\text{upper}} \leq 4$	140	82.1 %	12.7 %
excited-state, $n_{\text{upper}} \leq 4$	104	86.5 %	10.7 %
$n_{\text{upper}} = 5$ only	200	14.5 %	40.4 %

derivation agreeing to within a factor of six for an understood reason is doing its job — it confirms that the large number in the table is physics and not a coding error. Finally, the largest single entry anywhere in the assembled rate matrix at the benchmark point is the $8s$ diagonal, $-9.5235 \times 10^{12}/\text{s}$ (`L_grid.npy`, element `[23, 5, 28, 28]`): again ℓ -mixing, at the highest resolved shell, exactly where the n^4 of Eq. (2.9) puts it. The stiffest direction of the whole problem is a process most treatments omit.

2.8 How well do we know these rates?

Internal consistency checks cannot establish that a rate is the right size. Detailed balance holds to 2.5×10^{-9} because it was imposed; column-sum conservation, which Chapter 3 uses, holds to 3.6×10^{-11} because every transition entered as a loss also entered as a gain. Both would hold for a dataset uniformly wrong by a factor of three. The only test with real severity is comparison against an independent calculation.

That comparison is against Anderson *et al.* [39, 40], who computed effective collision strengths for all $n \leq 5$, $\Delta n \neq 0$ transitions in hydrogen using the R-matrix with pseudostates (RMPS) method — a different formulation, basis and code from CCC. Their collision strengths are converted to rate coefficients and compared point by point in `src/validation/anderson_benchmark_qc.py`, with the 340-row comparison stored in `collisions/anderson_benchmark_full.csv`. Table 2.3 gives the outcome. For the two transitions this thesis leans on hardest the agreement is better than the aggregate: $1s \rightarrow 2p$ differs by -14.5% at 1 eV and -2.2% at 10 eV; $2p \rightarrow 3d$ by -6.7% and -0.7% .

Taken at face value the first row is a failure: fewer than half the comparisons agree within 20 %. The disagreement is not spread out. It sits entirely in transitions reaching $n = 5$, which supply 200 of the 340 comparisons, and has two modes with opposite signs. Where the transition is dipole-allowed between nearly degenerate shells — $4s \rightarrow 5p$, gap 0.31 eV — CCC gives rates 2 to 3 times larger than RMPS. Where it is a high-multipole transition with no dipole matrix element — $1s \rightarrow 5g$, in which ℓ changes by four — CCC gives rates 3 to 5 times smaller.

The temptation is to declare one of them right. The honest position is that we cannot. Both are approximate, and both face the same difficulty at $n = 5$: the orbitals are large and diffuse. Anderson *et al.* state that $n = 6$ orbitals would have needed an R-matrix box of $140 a_0$, too large to include, and that constraint affects how their $n = 5$ states couple

to the continuum. **The direction of the error is not established**, and this thesis does not claim it is. What can be said is where the disagreement lands: kinetics in this range is driven by $n \leq 4$ processes — ground-state excitation to $2p$ and $3p$, and stepwise transitions such as $2p \rightarrow 3d$ and $2s \rightarrow 3p$ — while $n = 5$ contributes to cascades but not to the supply of the Balmer-emitting levels. A factor of three in a weak cascade channel is not a factor of three in the observable. That is an argument, not a measurement; it would be strengthened by scaling the $n = 5$ block and recomputing the observable, and that test was not performed.

Three further soft spots are worth naming rather than leaving to be discovered. *Detailed balance in the raw cross sections*: separately from the enforced consistency above, the CCC cross sections were checked for microscopic reversibility as supplied (`src/parsers/qc_ccc.py`) and satisfy it at the 0.05 % level — a genuine physics check on the provider’s calculation, and it passes comfortably. *Lotz ionisation for the top six shells, and its echo in recombination*: the metadata records that Lotz “overestimates CCC by factor ~ 4 – 8 ” for $n = 10$ – 15 , and because Eq. (2.6) builds α_{3BR} from Q , that overestimate propagates straight into the three-body coefficients of the same shells — the shells whose feed is essentially all three-body. Chapter 6 reports, from an independent comparison against published population coefficients, that the terminal shell is over-populated by 4.9–6.2 times while $n = 10$ agrees to 4 to 20 %. The two observations agree in size and location, but the test that would establish the link has not been done, so the connection is a hypothesis, not a finding. *Vriens–Smeets for the bundled block*: 201 of the 819 excitation pairs come from a semi-empirical fit of quoted accuracy ~ 20 % rather than a close-coupling calculation, which is the price of the ladder reaching $n = 15$.

2.9 Two errors found by audit

Two implementation errors were found in this pipeline by auditing it against its own published sources. Neither was found by a test failing; both were found by reading the code against the paper it claimed to implement, and by recomputing a quantity unconditionally and comparing with the stored value. They belong in the body of the thesis for two reasons: a reader assessing the numbers is entitled to know what the audit turned up, and the *size* of each error, once bounded, is a physical result in its own right.

2.9.1 The ℓ -mixing Coulomb logarithm

An earlier version of `src/rates/compute_lmix.py` set the factor $F(U_m)$ of Eq. (2.11) to 1, justified in a comment as “the low-density limit $U_m \rightarrow 0$, $F \rightarrow 1$ ”.

The justification inverts the limit. Reading Badnell’s Eq. 9 [38] directly from the source, $F(U_m)$ is the Coulomb logarithm of the impact-parameter picture, and $F(U_m \rightarrow 0)$ diverges logarithmically rather than tending to one — as it must, since a vanishing minimum energy transfer means an unbounded cut-off radius. The values of U_m at divertor conditions are indeed small, 10^{-3} to 10^{-1} , which is precisely where F is *large*. At the benchmark temperature the correct factor for $2p \rightarrow 2s$ is $F = 6.95$ at $U_m = 1.05 \times 10^{-3}$; across transitions it runs from about 7 at low n to about 3 at $n = 8$. The old code under-counted every ℓ -mixing rate by 3 to 7.

That the correction is present in the data this thesis uses can be checked rather than taken on trust. The canonical matrix is `cr_matrix/L_grid.npy`, written 21 July 2026, SHA-256 beginning `2d92b58e1693`; the ℓ -mixing coefficient it was built from is

$q_{2p \rightarrow 2s} = 3.53722 \times 10^{-4} \text{ cm}^3/\text{s}$ at temperature index 23, while evaluating Eq. (2.10) independently with F set to 1 gives $5.08977 \times 10^{-5} \text{ cm}^3/\text{s}$. The ratio is 6.9497 and the independently computed $F(U_m)$ at the same point is 6.94961, so the stored coefficient equals $q F$ to 8 parts in 10^6 .

What the correction changed is the interesting part. Scaling every intra-shell off-diagonal by the correct F , restoring the exact column sums and recomputing the spectrum moves the excited-state relaxation time by -0.12% and the ionisation-balance time by 0.07% at the benchmark point; across the grid the effect on the relaxation time is bounded by 0.85% (register entry A12). A factor of 3 to 7 in the fastest rates in the matrix moves the slow eigenvalues by less than one percent.

That insensitivity is not luck and it is not an excuse. It is the timescale separation of Chapter 3 doing what timescale separation does. ℓ -mixing runs at picoseconds; the relaxation being measured runs at nanoseconds, and on that timescale the ℓ sublevels are fully equilibrated whether the mixing is three times faster or seven, so the slow eigenvalue cannot see the difference. Finding a factor-of-five error in an input and then proving the headline moves by less than one percent is a stronger statement about the model than never having looked. It licenses nothing elsewhere: absolute ℓ -resolved populations and any ℓ -resolved line ratio depend on the correction directly, which is why it was fixed rather than merely bounded.

2.9.2 The eigenvalue magnitude filter

Two analysis scripts extracted timescales by sorting the eigenvalues of the rate matrix and then discarding every eigenvalue of magnitude below $1/\text{s}$, with the literal `eigs = eigs[eigs < -1.0]` and a comment calling the discarded values “near-zero numerical noise”.

The slowest genuine eigenvalue of this system is not noise. It is the ionisation-balance rate, and in the cold, thin corner of the grid its magnitude falls below $1/\text{s}$ — the corresponding timescale reaches tens of seconds. Wherever that happened the filter deleted the physical slow mode and promoted the whole eigenvalue ladder by one rung, so the reported ionisation-balance time was really the relaxation time of the level above it.

19 of the 400 grid points were affected, all at $T_e \leq 1.389 \text{ eV}$ and $n_e \in [10^{12}, 1.93 \times 10^{13}] \text{ 1/cm}^3$. At every one the corrupted relaxation time equalled the corrupted ionisation-balance time of the unfiltered calculation exactly — the signature of a one-rung shift, and how the mechanism was confirmed rather than guessed. The reported range of the timescale-separation ratio changed from $1.34\text{--}1.01 \times 10^8$ to $86.77\text{--}1.72928 \times 10^9$. The benchmark point was bit-identical before and after, and so were the step-error maps at all 400 points: the corruption was real, confined, and did not touch the central result. It was not found by a check failing — every check the filtered code ran continued to pass — but by recomputing the spectrum unconditionally, storing it separately, and comparing bit-for-bit. The filtered outputs were kept rather than deleted, under names ending `_FILTERED_20260721`, so the comparison remains reproducible.

Honesty requires the last statement to be precise rather than tidy. The filter was removed from the analysis scripts that write the timescale grids, which now select on `eigs < 0.0`. The literal `eigs[eigs < -1.0]` is still present in two other files: `src/rates/solve_cr.py` line 269 and `src/rates/check_mz.py` line 10. Neither produces any number quoted in this thesis — the first is a superseded solver, the second a diagnostic print — but the repository has not been swept clean, and a reader who searches for the

literal will find it. Saying so is cheaper than being caught.

2.9.3 What both errors have in common

Both were invisible to every consistency test the pipeline ran. The ℓ -mixing error preserved column sums exactly, because ℓ -mixing is a conservative redistribution and its magnitude cancels in the sum. The eigenvalue filter produced eigenvalues that were all negative, all finite and correctly ordered. A structurally consistent, physically wrong result is the characteristic failure mode of this kind of model, and it is why Chapter 4 separates checks a wrong result would also have passed from checks it would not. The structural response is the one described in Section 2.2: grids, state ordering and matrices are loaded from the pipeline’s own files through a single module that raises on any inconsistency, so a stale array cannot be silently combined with a fresh one.

2.10 What this model assumes

Every item here is a deliberate choice, gathered so that they can be found in one place and quantified where the thesis is able to quantify them.

Maxwellian electrons. Every rate coefficient is an average over a Maxwell–Boltzmann distribution, and Eq. (2.4) — on which every de-excitation and every three-body coefficient rests — holds exactly for a Maxwellian and not otherwise. A suprathermal tail, which a real transient can produce, would break it.

$T_i = T_e$ **and** $n_{\text{ion}} = n_e$. The proton temperature in the ℓ -mixing rate is set to the electron temperature; since $\nu_{\ell\text{-mix}} \propto T_i^{-1/2}$, a factor of two error in T_i is 40% in the rate — inside the insensitivity of Section 2.9, but stated rather than assumed away. Quasi-neutrality gives $n_{\text{ion}} = n_e$; no impurities are modelled.

A fixed reference density in the Debye cut-off. U_m in Eq. (2.11) is evaluated at $10^{14}/\text{cm}^3$ rather than at each grid density; the dependence is logarithmic, and the stated variation across the grid is about 30% in F .

Optically thin. All Einstein coefficients enter bare, $\Theta_P \equiv 1$. Chapter 3 evaluates Eq. (2.8) across the grid and finds the assumption excellent over the hot half and failing along the $T_e = 1\text{ eV}$ edge; Chapter 6 quantifies the consequence.

$n_{\text{max}} = 15$, **with** $n \geq 9$ **bundled.** Seven shells are one state each, with no ℓ -mixing inside the bundled block: they are *assumed* in statistical ℓ -equilibrium rather than driven to it, and that has not been tested directly. Chapter 6 reports the measured excess in the terminal shell.

Uniform plasma, no transport, no molecules. A single well-mixed volume — in reactor terms a perfectly stirred tank, not a plug-flow reactor — of atomic hydrogen only. At the cold end of the grid the gas is largely neutral and in reality substantially molecular, and molecular-assisted recombination would feed $n = 3$ directly. That is a physics limit rather than a data limit, and it is why the grid stops at 1 eV.

An open system. The ion H^+ is a fixed external reservoir: there is no 44th state, the model does not conserve nuclei, and ionisation is a one-way leak out of the tracked manifold. This is why the ionisation-balance timescale is boundary-dependent, and why the effective coefficients this model produces cannot be compared directly with those tabulated by ADAS. No ADAS dataset enters the rate matrix at any point; ADAS appears in this thesis only as an external comparison.

2.11 Summary

This chapter has built the input side of a collisional–radiative model of atomic hydrogen: 43 levels, six elementary processes, and a rate coefficient for every pair of levels at every point of a 50×8 grid in temperature and density. Table 2.4 is the provenance record for all of it.

[TODO: Provenance item outstanding. The merged excitation arrays `K_exc_full.npy` and `K_deexc_full.npy` carry a modification date earlier than the CCC rate tables they are assembled from, which would mean `assemble_K_exc.py` has not been re-run since those tables were last regenerated. The array shapes are compatible, so nothing would have raised. Re-run the merge, byte-compare against the stored arrays, and either delete this note or record the outcome here.]

Four things carry forward. The **state space is not arbitrary**: ℓ is resolved below $n = 9$ because $2s$ and $2p$ behave completely differently, and bundled above it because ℓ -mixing makes the sublevels of a high shell indistinguishable — which was checked, not asserted. **Proton-impact ℓ -mixing is structural**: it carries at least 99.92% of the loss from $2s$ everywhere on the grid. **The data are good where it matters**: the $n_{\text{upper}} \leq 4$ transitions that drive the kinetics agree with an independent RMPS calculation at a mean absolute error of 12.7%, and the factor 2–5 disagreement is confined to $n = 5$, where neither method establishes the direction of the error. And **two implementation errors were found by audit and both were bounded**.

What none of this has done is put the atoms in motion. We have 43×43 rate coefficients and a feed term, and no idea what the system does with them. Chapter 3 assembles them into a single matrix equation, finds that the network runs on two clocks separated by four orders of magnitude, and asks the question this thesis was built to answer: whether the fast one is fast enough for the standard shortcut to be safe.

Table 2.4: Every atomic dataset entering the model, with the file it is stored in and the script that produced it. Paths are relative to `data/processed/`; the arrays are those named in the `PATHS` dictionary of `src/rates/assemble_cr_matrix.py`. All rate-coefficient arrays are tabulated on the 50-point temperature grid $T_e = 1\text{--}10\text{ eV}$. Two hydrogen energies appear in the pipeline and they are two *different physical quantities*, not two estimates of one; Section 2.3.2 says which is used where.

Quantity	Source	Stored file	Producing script
Excitation and de-excitation, $n \leq 10$	CCC (Bray, priv. comm. 2026); $\Delta n = 0$ excluded	<code>collisions/K_exc_full/</code> <code>compute_K_full.py,</code> <code>K_deexc_full.npy</code>	<code>assemble_K_exc.py</code>
Excitation, bundled shells	Vriens–Smeets Eqs. (17)–(19)	merged into the same arrays	<code>compute_K_VS.py</code>
Ionisation, $n \leq 9$	CCC total ionisation cross sections	<code>collisions/tics/K_ionization.npy</code>	<code>compute_K_TICS.py</code>
Ionisation, $n = 10\text{--}15$	Lotz semi-empirical formula	same array	<code>ionization_rates.py</code>
Radiative recombination	Johnson three-term Gaunt expansion; ℓ -split statistically	<code>recombination/alpha_Recombined.npy</code>	<code>recombination_rates.py</code>
Three-body recombination	detailed balance from ionisation, via Saha	<code>recombination/alpha_3BRecombined.npy</code>	<code>recombination_rates.py</code>
Einstein A coefficients	Hoang-Binh, ADUU v1.0	<code>Radiative/A_resolved.npy,</code> <code>A_bund_res.npy,</code> <code>A_bund_bund.npy,</code> <code>gamma_*.npy</code>	<code>radiative_rates.py</code>
Proton-impact ℓ -mixing	PSM20 Debye-cutoff formula, Badnell <i>et al.</i>	<code>lmix/K_lmix.npy</code>	<code>compute_lmix.py</code>
State ordering and grids	the pipeline’s own output	<code>collisions/K_exc_full/</code> <code>state_ordering_matrix.py,</code> <code>cr_matrix/Te_grid_L.npy</code> <code>ne_grid_L.npy</code>	<code>state_ordering_matrix.py,</code> via <code>cr_context.py</code>

Chapter 3

Two Clocks

Chapter 2 established what can happen to a single hydrogen atom in a plasma and how fast each process runs. This chapter asks the next question: if we know what happens to one atom, how do we follow 10^{12} or more of them per cubic centimetre, distributed over 43 energy levels, and what makes the answer hard?

The answer to the second half is that the system has two clocks running at once, separated by four orders of magnitude. Everything in this thesis follows from that separation — including, as Chapter 5 shows, a result that the separation itself does not predict.

What this chapter takes as given. Three things are established elsewhere and used here without re-derivation. From Chapter 1: that divertor conditions are inferred spectroscopically, that the Balmer H_α/H_β ratio is the working diagnostic, and the operating range $T_e = 1\text{--}10\text{ eV}$, $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$ over which everything is computed. From Chapter 2: the 43-level state space and the reason ℓ is resolved for $n \leq 8$ and bundled above it; the origin and accuracy of every rate coefficient; and the Maxwellian averaging that turns a cross section into a rate. This chapter assumes those rates exist and are correct, and asks only what the resulting equations do.

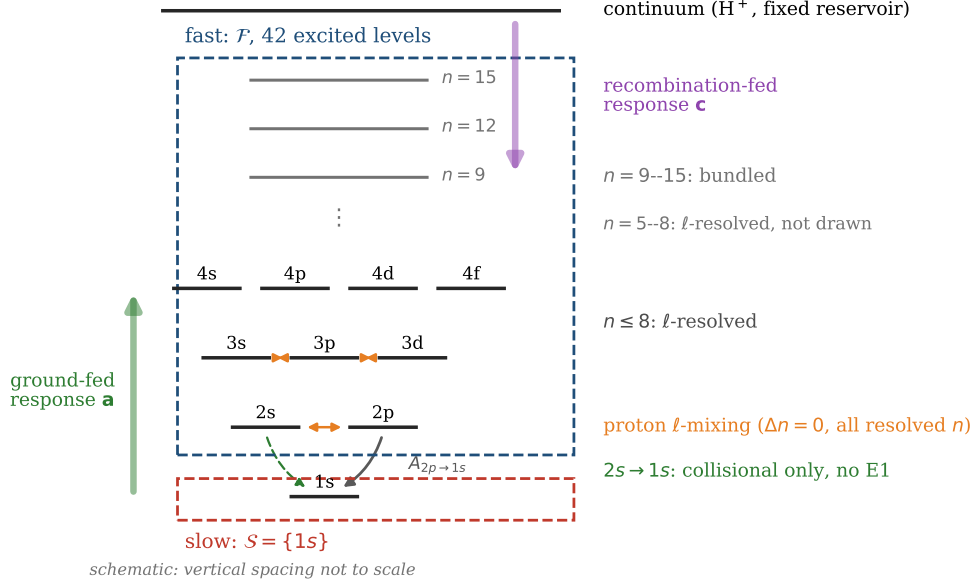
3.1 The rate equation

3.1.1 A balance on every level

Consider one energy level of the hydrogen atom — say $2p$ — and count atoms. Atoms arrive in $2p$ when an electron collision excites a ground-state atom into it, when an atom in a higher level decays into it radiatively, when a collision knocks an atom down from above, and when a proton captures an electron directly into it. Atoms leave $2p$ when they are excited upward, knocked downward, ionised, or radiate to $1s$.

The rate of change of the $2p$ population is the sum of everything arriving minus the sum of everything leaving. That is all a rate equation is.

A chemical engineer has written this equation many times. It is a species balance on a well-mixed vessel, with one species per energy level and one reaction per atomic process. Figure 3.1 shows the network and the partition that Section 3.4 will exploit. The only unusual feature is the number of species — 43 — and the fact that some of the “reactions” are radiative and therefore proceed without a collision partner.



CR data 710f4a24 · 2026-08-24

Figure 3.1: The collisional–radiative network of atomic hydrogen and the partition used in Section 3.4. Levels with $n \leq 8$ are resolved in orbital angular momentum; $n = 9\text{--}15$ are bundled with a statistical ℓ -distribution. Proton-impact ℓ -mixing (orange) couples sublevels within every resolved shell and is the dominant loss channel for the metastable $2s$ level, which has no electric-dipole route to the ground state; arrows are drawn for $n = 2$ and $n = 3$ only, the $n = 4$ sublevels being too closely spaced on this schematic for a legible arrow. The broad arrows are not elementary transitions: **a** = $-\mathbf{L}_{FF}^{-1}\mathbf{L}_{FS}$ and **c** = $-\mathbf{L}_{FF}^{-1}\mathbf{S}_F$ are *network responses*, each summing every cascade route from its reservoir into a given level (Section 3.5.3). Vertical spacing is schematic and not drawn to energy scale.

Write it out for a general level p :

$$\frac{dn_p}{dt} = \underbrace{\sum_{q \neq p} C_{qp} n_e n_q}_{\text{collisions in}} - \underbrace{\sum_{q \neq p} C_{pq} n_e n_p}_{\text{collisions out}} + \underbrace{\sum_{q > p} A_{qp} n_q}_{\text{radiation in}} - \underbrace{\gamma_p n_p}_{\text{radiation out}} - \underbrace{Q_p n_e n_p}_{\text{ionisation}} + \underbrace{S_p n_{\text{ion}}}_{\text{recombination}} \quad (3.1)$$

where n_p is the population of level p in $1/\text{cm}^3$; C_{qp} is the rate coefficient for the collisional transition $q \rightarrow p$ in cm^3/s ; A_{qp} is the Einstein coefficient for spontaneous emission $q \rightarrow p$ in $1/\text{s}$; $\gamma_p = \sum_{q < p} A_{pq}$ is the total radiative loss rate from p ; Q_p is the ionisation rate coefficient out of p in cm^3/s ; and S_p is the recombination source per unit ion density in $1/\text{s}$, given in Eq. (3.5) below.

Collisional terms carry a density factor because a collision requires a partner; radiative terms do not, because an excited atom decays whether or not anything else is nearby. Which partner matters: electron-impact terms scale with n_e , proton-impact terms with n_{ion} . For the pure hydrogen plasma considered here quasi-neutrality gives $n_{\text{ion}} = n_e$, so both scale together and the distinction is one of physics rather than of parametric dependence. This is the first hint of the two clocks: radiative rates are fixed by atomic structure, collisional rates scale with density, and the competition between them is what makes the answer depend on where in the plasma you look.

3.1.2 Matrix form

Equation (3.1) holds for each of the 43 levels. Stacking them and collecting the terms that are linear in the tracked populations gives

$$\boxed{\frac{d\mathbf{n}}{dt} = \mathbf{L} \mathbf{n} + \mathbf{S} n_{\text{ion}}} \quad (3.2)$$

with $\mathbf{n}$ the 43-component population vector ($1/\text{cm}^3$), $\mathbf{L}$ the collisional–radiative rate matrix ($1/\text{s}$), and $\mathbf{S} n_{\text{ion}}$ the recombination feed ($1/(\text{cm}^3 \text{s})$). The matrix elements are

$$L_{pq} = C_{qp}^{(e)} n_e + C_{qp}^{(i)} n_{\text{ion}} + A_{qp} \quad (p \neq q), \quad (3.3)$$

$$L_{pp} = - \left(\sum_{q \neq p} [C_{pq}^{(e)} n_e + C_{pq}^{(i)} n_{\text{ion}}] + \gamma_p + Q_p n_e \right). \quad (3.4)$$

The superscripts distinguish the two collision partners. $C^{(e)}$ denotes electron-impact transitions and is the dominant term for changes of principal quantum number. $C^{(i)}$ denotes proton-impact transitions, which in this model are non-zero only for ℓ -changing collisions within a shell ($\Delta n = 0$, $\Delta \ell = \pm 1$, $n \leq 8$). That term is easy to overlook and it is not optional. Chapter 2 shows it carries 99.92% of the total loss rate from the metastable $2s$ level. The reason is that $2s$ has no other fast exit: the transition $2s \rightarrow 1s$ is forbidden to electric-dipole radiation, and no two-photon rate is included in the present radiative dataset, so the only route out other than proton ℓ -mixing is electron-impact de-excitation, some three orders of magnitude slower. Removing proton ℓ -mixing therefore does not merely perturb the $n = 2$ manifold; it changes which process controls it. Quasi-neutrality gives $n_{\text{ion}} = n_e$ for the pure hydrogen plasma considered here, so both collisional terms scale with the same density.

Note the index order: L_{pq} with $p \neq q$ is the rate of the transition $q \rightarrow p$, a *gain* into p . Off-diagonal elements are therefore non-negative and diagonal elements are negative. This convention is the opposite of the one some texts use, and getting it backwards transposes the physics; it is stated here once and used throughout.

3.1.3 Why recombination is a feed stream, not a matrix element

There is one term in Eq. (3.1) that refuses to become a matrix element, and understanding why is worth a paragraph.

$\mathbf{L}$ multiplies $\mathbf{n}$. Every entry of $\mathbf{L}\mathbf{n}$ must therefore be proportional to the population of some *tracked* level. Excitation, de-excitation, radiative decay and ionisation all satisfy this: they take an atom that is in the manifold and move it, or remove it. Recombination does not. It is proportional to $n_e n_{\text{ion}}$ — the density of a species outside the neutral manifold entirely — and so it enters as a constant feed, independent of $\mathbf{n}$.

The chemical-engineering analogy is exact and worth holding onto for the rest of the thesis. $\mathbf{L}$ is the reaction-rate matrix of a network of 43 species in a well-mixed vessel. $\mathbf{S}n_{\text{ion}}$ is the inlet stream. And the ion population is the reservoir feeding that inlet — a quantity this model does not evolve, which becomes important in Section 3.5 and again in Chapter 6.

3.1.4 Units, and a source that is quadratic in density

The source deserves care because its units are easy to state wrongly. Writing it out,

$$\mathbf{S} = n_e (\boldsymbol{\alpha}_{\text{RR}} + n_e \boldsymbol{\alpha}_{\text{3BR}}) \quad (3.5)$$

where $\boldsymbol{\alpha}_{\text{RR}}$ is the radiative recombination coefficient (cm^3/s) and $\boldsymbol{\alpha}_{\text{3BR}}$ the three-body coefficient (cm^6/s). Three-body recombination requires a second electron to carry away the excess energy, so it carries an extra factor of n_e ; the source as a whole is therefore *quadratic* in electron density, and $\mathbf{S}$ has units of $1/\text{s}$, not cm^3/s .

This matters physically, not just dimensionally. At $n_e = 10^{12}/\text{cm}^3$ the three-body channel supplies under 10% of the feed into any level; at $n_e = 10^{15}/\text{cm}^3$ it supplies more than 99% into the high- n shells. The same equation therefore describes two physically different regimes at the two ends of the density grid, a fact used in Chapter 4 to test the two recombination datasets separately.

Dimensional check on Eq. (3.2): $[\mathbf{L}\mathbf{n}] = 1/\text{s} \cdot 1/\text{cm}^3 = 1/(\text{cm}^3\text{s})$, and $[\mathbf{S}n_{\text{ion}}] = 1/\text{s} \cdot 1/\text{cm}^3$, matching $[d\mathbf{n}/dt]$. Every term is a number of atoms per unit volume per unit time, as a balance equation must be.

3.2 Conservation, and how we know the matrix is right

A 43×43 matrix assembled from six separate atomic datasets offers a great many opportunities to make a mistake that produces plausible numbers. Before using $\mathbf{L}$ for anything, we need a test that a wrong matrix would fail.

The test comes from counting atoms. Sum Eq. (3.2) over all levels with the feed switched off. Every collisional and radiative transition moves an atom *within* the manifold: it is a loss from one level and a gain to another, so the two contributions cancel in the sum. Only ionisation genuinely removes an atom from the bound states. Therefore

$$\boxed{\sum_p L_{pq} = -Q_q n_e \quad \text{for every column } q.} \quad (3.6)$$

Each column of $\mathbf{L}$ must sum to the ionisation leak from the level that column represents — no more, no less. An error in a single off-diagonal element, a transposed index, or a missing back-reaction all break this immediately.

3.2.1 A worked example

The rule is easiest to see in a two-level system. Take $1s$ and $2p$ only, at $T_e = 3\text{ eV}$ and $n_e = 10^{14}/\text{cm}^3$, with representative coefficients $C_{12} = 1 \times 10^{-11} \text{ cm}^3/\text{s}$, $C_{21} = 1 \times 10^{-10} \text{ cm}^3/\text{s}$, $A_{21} = 6.27 \times 10^8/\text{s}$, $Q_1 = 1 \times 10^{-13} \text{ cm}^3/\text{s}$ and $Q_2 = 1 \times 10^{-10} \text{ cm}^3/\text{s}$. Then

$$\mathbf{L} = \begin{pmatrix} -1010 & 6.2701 \times 10^8 \\ 1000 & -6.2702 \times 10^8 \end{pmatrix} 1/\text{s}, \quad \sum_p L_{pq} = \begin{pmatrix} -10 \\ -10^4 \end{pmatrix} 1/\text{s} \quad (3.7)$$

and $Q_1 n_e = 10/\text{s}$, $Q_2 n_e = 10^4/\text{s}$. The rule holds.

Two features of this small matrix survive into the full one and are worth noticing now.

First, the second column is a difference of two numbers that agree to four significant figures: 6.2701×10^8 against 6.2702×10^8 , leaving 10^4 . Almost all of the $2p$ loss rate is radiative decay to the ground state, and the physically interesting part — the ionisation leak — is four orders of magnitude smaller. Conservation is therefore always quoted as a *relative* residual; an absolute one would be meaningless.

Second, that same near-cancellation is the seed of the two clocks. The $2p$ level empties in nanoseconds by radiating, but the atom that arrives back in $1s$ is still in the manifold. Only the small ionisation term actually removes it. Section 3.3 makes this quantitative.

3.2.2 The measured residual

For the full matrix, evaluated at all 400 grid points, the largest relative column-sum residual is

$$\max_{q, T_e, n_e} \left| \frac{\sum_p L_{pq} + Q_q n_e}{Q_q n_e} \right| = 3.61 \times 10^{-11}, \quad (3.8)$$

which is round-off for a matrix whose entries span eight orders of magnitude. The assembly is arithmetically sound.

It is worth being precise about what this does and does not establish. It establishes that every transition entered as a loss also entered as a gain somewhere. It says nothing whatever about whether the *magnitude* of any rate is correct: ℓ -mixing, for instance, cancels exactly in the column sum and so is invisible to this test. Checking magnitudes requires comparison against independent sources, which is the subject of Chapter 4. A test that a wrong result would also have passed is not a test.

3.3 Two clocks

3.3.1 Why the system has natural timescales at all

Equation (3.2) is linear, so with $\mathbf{L}$ and the feed held fixed its solution is a steady state plus a sum of decaying exponentials:

$$\mathbf{n}(t) = \mathbf{n}^{\text{ss}} + \sum_k c_k e^{\lambda_k t} \mathbf{v}_k, \quad \mathbf{L} \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad \mathbf{n}^{\text{ss}} = -\mathbf{L}^{-1} \mathbf{S} n_{\text{ion}}. \quad (3.9)$$

Each eigenvalue λ_k has a negative real part — the system is stable, as a system of decaying populations must be — and each contributes a mode decaying with time constant $\tau_k = 1/|\lambda_k|$. The set of τ_k is the set of clocks the system runs on.

It is worth pausing on what Eq. (3.9) has achieved, because the whole chapter turns on it. In the populations n_p the 43 balance equations are hopelessly coupled: nothing happens to $3d$ without also happening to $2p$, $4f$, and everything else. Diagonalising $\mathbf{L}$ is a change of coordinates. Instead of asking how many atoms are in each *level*, ask how much of each *eigenvector* the state contains — and in those coordinates the coupling disappears entirely. The k th coordinate obeys $\dot{c}_k = \lambda_k c_k$, which is the first-order decay $c_k(t) = c_k(0)e^{-t/\tau_k}$ familiar from single-step kinetics. Forty-three coupled equations become forty-three independent ones, each with its own clock.

The eigenvectors are therefore not abstract: each is a *pattern of populations* that decays without changing shape, and any state of the system is a superposition of those patterns fading at their own rates. Asking which physical process a mode corresponds to is then a matter of reading which levels carry its weight, which is what Section 3.3 does next. One caveat, developed in Section 3.6: this decomposition is exact, but the patterns are not orthogonal, so a small state can be the sum of two large and nearly cancelling ones.

For hydrogen at divertor conditions there are 43 of them. The striking fact, and the one this thesis is built on, is that they are not spread out evenly.

3.3.2 The spectrum

Figure 3.2 shows all 43 relaxation times, and Table 3.1 lists the five slowest at the benchmark point $T_e = 2.947$ eV, $n_e = 1.389 \times 10^{14}/\text{cm}^3$.

Table 3.1: The five slowest relaxation modes at the benchmark point. The gap between the first and second is 9982; the largest of the remaining 41 gaps is the 2.80 in this table, at $k = 2$, so the slow mode is isolated from the fast band by a factor of 3567.

k	λ_k (1/s)	τ_k (s)	τ_k/τ_{k+1}
0	-4.3999×10^4	2.2728×10^{-5}	9982
1	-4.3919×10^8	2.2769×10^{-9}	1.23
2	-5.3994×10^8	1.8521×10^{-9}	2.80
3	-1.5110×10^9	6.6181×10^{-10}	2.71
4	-4.0922×10^9	2.4436×10^{-10}	—

There is one gap and then a continuum.

The slowest mode is separated from the next by a factor of ten thousand; after that the modes crowd together, each within a factor of about two or three of its neighbour. It is not a hierarchy of three or four well-separated groups, as one might expect from the n -shell structure — it is one very slow mode sitting far above a dense quasi-continuum of fast ones.

3.3.3 What the two modes are

The eigenvectors identify them.

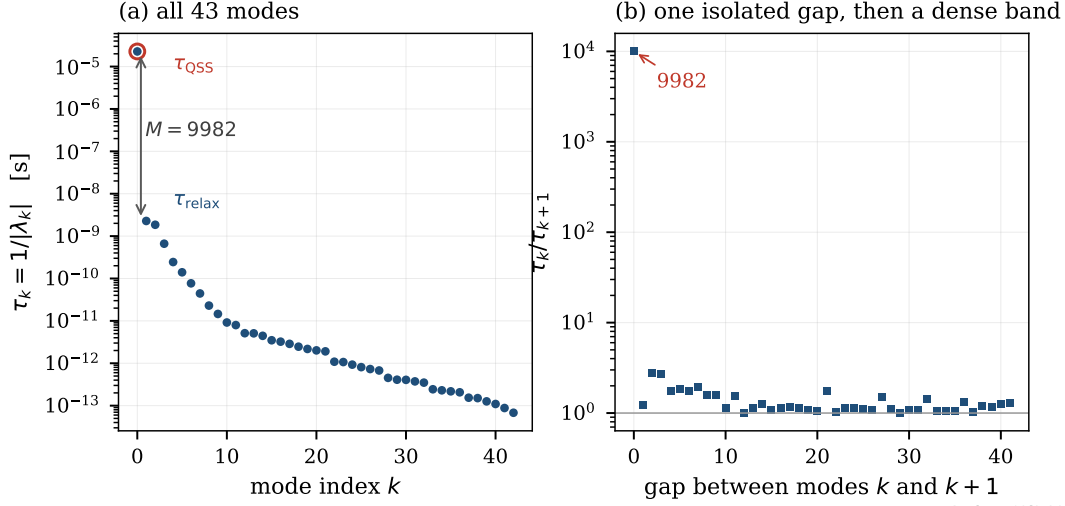


Figure 3.2: Relaxation timescales at the benchmark point. (a) All 43 modes. The slowest, τ_{QSS} , is the relaxation of the ground-state population; the next, τ_{relax} , is the slowest-decaying mode of the excited manifold — the bottleneck of the fast set — and their ratio is the separation M . (b) The ratio of consecutive timescales, which shows that the first gap is exceptional: the largest of the other 41 is 2.80, so the first exceeds every competitor by a factor of 3567 and the 42 fast modes form a dense band rather than a hierarchy.

The slow mode $\mathbf{v}_0$ is almost entirely ground state: its $1s$ component is 1.0000 and every excited component is below 10^{-3} . Its participation ratio,

$$\text{PR}(\mathbf{v}) = \frac{(\sum_i |v_i|^2)^2}{\sum_i |v_i|^4}, \quad (3.10)$$

which counts the effective number of states a mode involves — unity for a mode confined to a single level, 43 for one spread evenly over all of them — is 1.00. This mode is the relaxation of the ground-state population against the prescribed ion reservoir. It is often called the ionisation balance, and that is the right physical picture, but the term should be used with care here: this model holds n_{ion} fixed (Section 3.1), so what relaxes is n_{1s} alone and not a self-consistent neutral-ion partition. Chapter 6 assesses what that costs.

Its eigenvalue is closely related to, but not equal to, the direct ionisation rate of the ground state. Writing

$$|\lambda_0| = \eta Q_{1s} n_e, \quad (3.11)$$

the factor η ranges from 1.16 to 29.5 across the grid, and equals 3.12 at the benchmark point. It is nowhere below unity on the grid, and it is the *stepwise-ionisation enhancement*: ground-state atoms are not only ionised directly but also excited into the manifold and ionised from there. Equation (3.21) shows exactly this — η is the ratio of the effective ionisation rate to the direct one,

$$\eta = \frac{|\Omega_{\text{QSS}}|}{Q_{1s} n_e}, \quad (3.12)$$

the numerator being the full Schur complement and the denominator the direct ground-state ionisation rate alone. The denominator is emphatically *not* $|\mathbf{L}_{SS}|$: by Eq. (3.1) the ground state has no radiative loss and no ℓ -mixing partner, so $|\mathbf{L}_{SS}| = (\sum_q C_{1s \rightarrow q} + Q_{1s})n_e$, which exceeds $Q_{1s}n_e$ by the whole of the excitation flux. Since the Schur complement returns part of that flux to the ground state, $|\Omega_{\text{QSS}}| < |\mathbf{L}_{SS}|$ always, and the ratio

$\Omega_{\text{QSS}}/\mathbf{L}_{SS}$ is bounded above by unity — whereas η is bounded below by it. Note also that η is defined through $|\lambda_0|$ while Eq. (3.12) is written with $|\Omega_{\text{QSS}}|$; the two agree only to $O(M^{-1})$, which at the benchmark is one part in 10^4 and at the weakest grid point one part in 87. Consistently, η grows with density, where collisional climbing of the ladder is easier, and falls with temperature, where direct ionisation is already fast: at $T_e = 10$ eV, $n_e = 10^{12}/\text{cm}^3$ it is 1.16, and at $T_e = 1$ eV, $n_e = 10^{15}/\text{cm}^3$ it is 29.5 — almost all ionisation there proceeds stepwise. We write

$$\tau_{\text{QSS}} \equiv \frac{1}{|\lambda_0|} = 22.73 \mu\text{s}. \quad (3.13)$$

The next mode $\mathbf{v}_1$ has almost no ground-state weight and is distributed across the excited manifold. It is the excited states rearranging among themselves:

$$\tau_{\text{relax}} \equiv \frac{1}{|\lambda_1|} = 2.277 \text{ ns}. \quad (3.14)$$

Their ratio is the *separation*

$$M \equiv \frac{\tau_{\text{QSS}}}{\tau_{\text{relax}}} = 9982 \quad \text{at the benchmark point.} \quad (3.15)$$

The excited manifold settles ten thousand times faster than the ionisation balance moves. Put in units a reader can hold: if the ionisation balance took a year to equilibrate, the excited states would be done in under an hour.

Which operator M belongs to

One point of notation has to be fixed here, because it is the source of an apparent contradiction between this chapter and Chapter 5.

M is a property of an *operator*, not of an experiment. Nothing in $|\lambda_1|/|\lambda_0|$ refers to a disturbance, an initial condition, or a history. Written with its argument exposed,

$$M[\mathbf{L}] \equiv \frac{\tau_{\text{QSS}}[\mathbf{L}]}{\tau_{\text{relax}}[\mathbf{L}]} = \frac{|\lambda_1[\mathbf{L}]|}{|\lambda_0[\mathbf{L}]|}, \quad M(T_e, n_e) \equiv M[\mathbf{L}(T_e, n_e)], \quad (3.16)$$

and Eq. (3.15) is $M(2.947 \text{ eV}, 1.389 \times 10^{14}/\text{cm}^3)$ while Eq. (3.18) is the range of $M(T_e, n_e)$ over the 400 grid operators. Both are statements about matrices, and no notion of a step enters either.

A temperature step brings a second operator into scope. Writing $\mathbf{L}^- = \mathbf{L}(T_e^-, n_e)$ and $\mathbf{L}^+ = \mathbf{L}(T_e^- + \Delta T_e, n_e)$, the initial populations are those of the CR equilibrium of $\mathbf{L}^-$, but every relaxation for $t > 0$ is governed by $\mathbf{L}^+$. The two separations

$$M^- \equiv M[\mathbf{L}^-], \quad M^+ \equiv M[\mathbf{L}^+] \quad (3.17)$$

are therefore both meaningful, and it is M^+ that bounds the post-step relaxation. At the benchmark point a step of one grid interval in T_e gives $M^- = 9982$ and $M^+ = 8243$. The difference is not a disagreement between chapters: M falls steeply with temperature, with a measured local slope $d \ln M / d \ln T_e \approx -4$, so a 4.8 % step moves it by 17 %.

The alternative convention — defining M to mean the post-step operator always — was considered and rejected, for a reason worth recording. The T_e grid is logarithmic with a ratio of $10^{1/49} = 1.0481$, so a step of one grid interval from the benchmark lands

exactly on the neighbouring grid point. The post-step operator is $\mathbf{L}(T_e[24], n_e[5])$, one of the 400 operators whose separation Eq. (3.18) already reports. This is not special to that step. Because $\mathbf{L}$ is tabulated only on the grid, every step used in this thesis is an integer number of grid intervals: the larger step reported elsewhere as “+0.6 eV” is four intervals, $T_e[23] \rightarrow T_e[27]$, an increment of 0.6094 eV and a factor $1.0481^4 = 1.2068$, for which $M^+ = 4856$. *Every* post-step operator in this work is therefore also a static grid operator. Under that convention the same matrix would carry one name in this chapter and another in Chapter 5. Being pre-step or post-step is a role a matrix plays in an experiment, not a property it has; the notation should not pretend otherwise.

3.3.4 The separation across the operating range

Table 3.1 is one grid point. Across all 400,

$$0.87 \text{ ns} \leq \tau_{\text{relax}} \leq 38.9 \text{ ns}, \quad 75.4 \text{ ns} \leq \tau_{\text{QSS}} \leq 67.2 \text{ s}, \quad 86.8 \leq M \leq 1.73 \times 10^9. \quad (3.18)$$

τ_{relax} varies by a factor of 45; τ_{QSS} by seven orders of magnitude. The separation is never small: even at its weakest — the hottest, densest corner of the grid — the excited states are almost ninety times faster than the ionisation balance.

The scope of that lower figure needs stating, because it is easy to quote in a way it does not support. $M = 86.8$ is the minimum over all 400 grid points, and the point carrying it is excluded from the transient analysis of Chapter 5 by the plateau-window criterion, which requires $M > 900$ for a window to exist at all. The floor over the *analysed* set is therefore $M \geq 902$, and that floor is imposed by the criterion rather than measured. The same distinction applies to τ_{QSS} , and an earlier draft of Eq. (3.18) did not observe it: the floor $\tau_{\text{QSS}} \geq 1.18 \mu\text{s}$ is the minimum over the 346 points that satisfy $M > 900$, not over the 400. Over all 400 it is 75.4 ns, at the same [49, 7] corner, and that is the value Eq. (3.18) now reports — so that all three ranges in it share one scope. Neither number licenses a statement of the form “*the separation is at least X wherever the model is used*”: the first is a minimum over a set that includes points the analysis discards, and the second is a restatement of the cut. Both are reported here as descriptions of the spectrum, and the argument of Section 3.4 does not rest on either.

The one-gap structure of Table 3.1 also survives across the grid, though with far less margin than the benchmark suggests. Define the *isolation* of the slow mode as the first gap divided by the largest of the other 41. At the benchmark it is 3567. Its minimum over all 400 points is 24.3, at $T_e = 10 \text{ eV}$, $n_e = 10^{15}/\text{cm}^3$, where the first gap is 86.8 and its nearest competitor 3.57. A factor of 24 is still an unambiguous separation — one slow mode, then a band — but it is 147 times less margin than Figure 3.2 displays, and the benchmark value must not be quoted as though it described the grid.

That corner is worth naming, because it is where every part of this argument is weakest at once: the minimum τ_{relax} , the minimum τ_{QSS} , the minimum M and the minimum isolation all occur at the same grid point, [49, 7]. It is the single stress point of the two-clock picture. It is also, by the plateau-window criterion, outside the set analysed in Chapter 5 — which is a limitation to state rather than a convenience, since it means the weakest separation on the grid is never tested against a transient. The figure script evaluates the isolation at all 400 points and halts if it falls below ten.

τ_{relax} falls with density with a fitted logarithmic slope $d \ln \tau_{\text{relax}} / d \ln n_e$ between -0.46 and -0.53 across the temperature range, rather than the -1 that a purely collisional

relaxation would give. The difference is radiative decay, which is density-independent and increasingly dominates the excited-state loss budget as the density falls.

3.3.5 Is the definition robust?

Two definitions of τ_{relax} are available and it is worth checking they agree. The first, used above, takes the second eigenvalue of the full 43×43 matrix. The second computes the slowest eigenvalue of the excited block $\mathbf{L}_{FF}$ alone — that is, the relaxation of the excited manifold with the ground state held fixed. These are different calculations and there is no guarantee they coincide.

They agree to 0.0098 % at the benchmark point and to better than 0.346 % everywhere on the grid. The excited manifold relaxes at essentially the same rate whether or not the ground state is allowed to participate.

It would be a mistake to read that agreement as independent evidence that the subsystems separate, and the arithmetic says why. Block decoupling is a perturbation in the coupling between the two blocks, and its leading correction is of relative size M^{-1} . At the benchmark $M^{-1} = 0.0100$ % against the measured 0.0098 % — the two agree to 2 %. The disagreement between the two definitions of τ_{relax} is the first-order coupling correction; it is a measurement of the spectral gap, not a second, independent demonstration of it. Quoting it as corroboration would be the error Section 3.2 warns against: a test that a wrong result would also have passed.

Read correctly it is still worth having, and is a sharper statement than the one it replaces. Across the grid the product $M \times$ (relative disagreement) runs from 0.074 to 7.94 with median 0.494 — order unity, as first-order perturbation theory requires, with a coefficient that varies by two orders across the operating range. The two definitions differ by exactly the amount the block structure predicts, which confirms the perturbative picture rather than assuming it. The formal argument for separability is made in Section 3.4 and rests on the Schur complement, not on this comparison.

3.3.6 What the separation does and does not license

A separation of 10^4 invites an obvious simplification: treat the excited states as instantaneously equilibrated, and solve only for the slow subsystem. That is the quasi-steady-state approximation, and Section 3.4 derives it properly.

Two warnings, both of which matter later.

First, M compares two *rates*. Chapter 5 measures a *distance* — how far a diagnostic reading sits from the truth. Knowing how quickly a system recovers from a displacement tells you nothing about how far it was displaced, and these are independent quantities.

Second, eigenvalues describe the long-time behaviour of each mode in isolation. The matrix $\mathbf{L}$ is strongly asymmetric — radiative decay downward runs at $\sim 6.3 \times 10^8/\text{s}$ against collisional excitation upward at $\sim 10^4/\text{s}$ at the benchmark density, some five orders of magnitude — and asymmetry of that size makes non-normality plausible. It does not by itself prove it: a non-symmetric matrix may still be normal and have orthogonal eigenvectors. The decisive test is the numerical one of Section 3.6, where the numerical abscissa is found to be positive at every grid point while every eigenvalue is negative. For a non-normal matrix the modal amplitudes can be large and opposing, and a perturbation can grow before it decays even though every eigenvalue is negative. The spectrum bounds the asymptotic decay; it does not bound the finite-time error.

Both warnings point the same way: the two clocks tell us the system *can* be simplified, and they say nothing about what the simplification costs. Working that out is the business of the rest of this chapter.

3.4 The quasi-steady-state approximation

3.4.1 The shortcut

A separation of 10^4 invites an obvious move. If the excited states settle in nanoseconds while the ionisation balance takes microseconds, then on any timescale relevant to the second, the first has already finished. So do not solve for the excited states at all: assume they have reached whatever configuration the current conditions demand, and solve only for the slow subsystem.

This is the quasi-steady-state approximation, and every practical collisional–radiative model uses some form of it. Without it, a plasma simulation that resolves ns timesteps in order to follow the excited states cannot afford to run for the ms an experiment lasts.

The chemical-engineering version of this argument is the Bodenstein steady-state approximation. For a reaction $A \rightarrow I \rightarrow P$ with a short-lived intermediate, one sets $d[I]/dt = 0$ and solves algebraically for $[I] = (\text{production})/(\text{loss})$. The intermediate stops being a differential unknown and becomes slaved to whatever $[A]$ currently is. What follows is the same move applied to 42 intermediates at once, coupled to each other.

3.4.2 Partition and elimination

Split the 43 levels into a *slow* subspace $\mathcal{S} = \{1s\}$ and a *fast* subspace $\mathcal{F}$ containing the 42 excited levels, and partition Eq. (3.2) conformably:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{n}_S \\ \mathbf{n}_F \end{pmatrix} = \begin{pmatrix} \mathbf{L}_{SS} & \mathbf{L}_{SF} \\ \mathbf{L}_{FS} & \mathbf{L}_{FF} \end{pmatrix} \begin{pmatrix} \mathbf{n}_S \\ \mathbf{n}_F \end{pmatrix} + \begin{pmatrix} \mathbf{S}_S \\ \mathbf{S}_F \end{pmatrix} n_{\text{ion}}. \quad (3.19)$$

Section 3.3 justifies this particular partition rather than some other: the slow eigenvector is 1.0000 ground state with every excited component below 10^{-3} , so the slow mode and the ground state are the same object to the precision that matters.

Setting $d\mathbf{n}_F/dt = 0$ in the lower block gives an algebraic system for the excited populations,

$$\boxed{\mathbf{n}_F^{\text{QSS}} = -\mathbf{L}_{FF}^{-1} (\mathbf{L}_{FS} \mathbf{n}_S + \mathbf{S}_F n_{\text{ion}}).} \quad (3.20)$$

Read the right-hand side as a sentence. The bracket is everything arriving in the excited manifold from outside it: $\mathbf{L}_{FS} \mathbf{n}_S$ is excitation up from the ground state, $\mathbf{S}_F n_{\text{ion}}$ is recombination down from the ions. The factor $-\mathbf{L}_{FF}^{-1}$ divides by the loss network. It is a matrix rather than a number only because a level's losses feed other levels, which have their own losses, and so on — the division is over the whole network at once. For a single Bodenstein intermediate it would collapse to $1/k_{\text{loss}}$.

Two things make Eq. (3.20) usable. $\mathbf{L}_{FF}$ is invertible because it is a strictly column-diagonally-dominant matrix with negative diagonal: every excited level loses population to the ground state or to the continuum, so each column sum is strictly negative rather than zero, and the Levy–Desplanques theorem applies. Physically, there are no perfectly trapped states within the excited manifold. Its condition number lies between 1.48×10^3 and 1.74×10^5 across the grid, so the inversion is numerically well behaved despite the eight orders of magnitude spanned by the entries.

3.4.3 The reduced dynamics

Substituting Eq. (3.20) back into the slow block eliminates $\mathbf{n}_F$ entirely:

$$\frac{d\mathbf{n}_S}{dt} = \underbrace{(\mathbf{L}_{SS} - \mathbf{L}_{SF}\mathbf{L}_{FF}^{-1}\mathbf{L}_{FS})}_{\Omega_{\text{QSS}}} \mathbf{n}_S + (\mathbf{S}_S - \mathbf{L}_{SF}\mathbf{L}_{FF}^{-1}\mathbf{S}_F) n_{\text{ion}}. \quad (3.21)$$

A 43-dimensional differential system has become a one-dimensional one. The matrix Ω_{QSS} is the *Schur complement* of $\mathbf{L}_{FF}$ in $\mathbf{L}$, and it has a direct physical reading: $\mathbf{L}_{SS}$ is the direct loss from the ground state, and $-\mathbf{L}_{SF}\mathbf{L}_{FF}^{-1}\mathbf{L}_{FS}$ is the correction for atoms that leave the ground state, wander through the excited manifold, and either come back or are ionised on the way. It is stepwise ionisation, summed over every possible path.

This is the object the rest of the thesis is about. Ω_{QSS} is precisely the effective ionisation rate that a plasma modeller using QSS would construct, and it is the algebraic form of the effective coefficients tabulated in frameworks such as ADAS [9]. When Chapter 5 asks what the approximation costs, it is asking what this quantity costs.

3.4.4 When is it valid?

The elimination assumed the excited states are always at their instantaneous steady value. The error in that assumption can be estimated directly. Write the deviation from the instantaneous target as $\boldsymbol{\delta}(t) = \mathbf{n}_F(t) - \mathbf{n}_F^{\text{QSS}}(t)$ and subtract Eq. (3.20) from the lower block of Eq. (3.19):

$$\frac{d\boldsymbol{\delta}}{dt} = \mathbf{L}_{FF}\boldsymbol{\delta} - \frac{d\mathbf{n}_F^{\text{QSS}}}{dt}. \quad (3.22)$$

The deviation relaxes at the rate set by $\mathbf{L}_{FF}$, whose slowest mode is τ_{relax} , and is forced by the motion of the target itself. If the driving conditions change on a timescale τ_{drive} , then $|d\mathbf{n}_F^{\text{QSS}}/dt| \sim \|\mathbf{n}_F^{\text{QSS}}\|/\tau_{\text{drive}}$, and balancing the two terms in Eq. (3.22) at quasi-static equilibrium gives $\|\boldsymbol{\delta}\| \sim \tau_{\text{relax}}\|\mathbf{n}_F^{\text{QSS}}\|/\tau_{\text{drive}}$, that is

$$\frac{\|\mathbf{n}_F - \mathbf{n}_F^{\text{QSS}}\|}{\|\mathbf{n}_F^{\text{QSS}}\|} \sim \frac{\tau_{\text{relax}}}{\tau_{\text{drive}}}, \quad (3.23)$$

so the approximation is accurate whenever $\tau_{\text{drive}} \gg \tau_{\text{relax}}$. For the self-consistent problem, where the only thing driving the excited states is the slow subsystem itself, $\tau_{\text{drive}} \rightarrow \tau_{\text{QSS}}$ and the condition becomes

$$M = \frac{\tau_{\text{QSS}}}{\tau_{\text{relax}}} \gg 1. \quad (3.24)$$

Section 3.3 measured $M = 9982$ at the benchmark and $M \geq 86.8$ at every one of the 400 grid points. **The condition is satisfied everywhere on the grid, comfortably.** The floor carries the qualification of Section 3.3: the point holding it is excluded from the transient analysis of Chapter 5, and the floor over the analysed set is imposed by the window criterion rather than measured.

Two qualifications must travel with that statement.

Equation (3.23) is a spectral estimate, and a bound is available. The step from Eq. (3.22) to Eq. (3.23) identified the relaxation rate of $\boldsymbol{\delta}$ with the slowest eigenvalue of $\mathbf{L}_{FF}$. That identification is exact only for a normal operator. The quasi-static balance itself gives more: $\boldsymbol{\delta} \simeq \mathbf{L}_{FF}^{-1}\dot{\mathbf{n}}_F^{\text{QSS}}$, hence

$$\|\boldsymbol{\delta}\| \leq \|\mathbf{L}_{FF}^{-1}\|_2 \|\dot{\mathbf{n}}_F^{\text{QSS}}\|, \quad \|\mathbf{L}_{FF}^{-1}\|_2 = \frac{1}{\sigma_{\min}(\mathbf{L}_{FF})}, \quad (3.25)$$

and since $\sigma_{\min} \leq \min_k |\lambda_k|$ for every matrix, the operative timescale $\|\mathbf{L}_{FF}^{-1}\|_2$ is never shorter than the spectral one. Measured across the grid, the ratio $\|\mathbf{L}_{FF}^{-1}\|_2 |\lambda_{\min}(\mathbf{L}_{FF})|$ lies between 1.118 and 1.526, with median 1.151: the spectral estimate understates the rigorous timescale by at most 53%. Defining the corresponding bounded separation $M^{\text{eff}} = \tau_{\text{QSS}} / \|\mathbf{L}_{FF}^{-1}\|_2$,

$$M^{\text{eff}} = 8659 \text{ at the benchmark,} \quad M^{\text{eff}} \geq 77.6 \text{ at all 400 grid points,} \quad (3.26)$$

so Eq. (3.24) survives non-normality with a margin reduced by 13% and not by orders of magnitude. It is worth asking why the penalty is so small when $\kappa(\mathbf{L}_{FF})$ reaches 1.74×10^5 : the answer is that the ill-conditioning lives in the fast subspace, among modes separated by many decades in rate, and barely touches the slowest mode of $\mathbf{L}_{FF}$, which is the one that sets QSS validity. The operator is strongly non-normal in a way that is almost irrelevant to this particular question.

That said, Eq. (3.25) is still a statement about the settled deviation. Section 3.3 warned that for a strongly non-normal operator the spectrum does not control finite-time behaviour, and that warning applies to the approach as opposed to the endpoint: the balance in Eq. (3.22) is a *quasi-static* argument, valid once the fast transients have decayed and the deviation has settled onto the slow manifold. It says nothing about what δ does in the first few τ_{relax} after a disturbance, when the non-orthogonality of the eigenvectors can allow transient growth. Whether that matters for the quantity actually measured is not decidable from the spectrum, and Chapter 5 settles it by integrating the full system rather than by argument.

The condition is not sharp. At the benchmark point Eq. (3.23) gives $(M^+)^{-1} = 1.21 \times 10^{-4}$ — the post-step operator being the one that governs the relaxation being compared — whereas direct integration returns a residual of order 10^{-8} : the approximation is four orders of magnitude better than the scaling argument requires. The unwritten prefactor — set by how strongly the observable projects onto the moving part of the target — is evidently small. Equation (3.23) predicts the trend correctly and the magnitude not at all, and it should be quoted only for the former.

That would appear to end the matter. It does not, and the reason is the subject of the next section.

3.5 Two reference states

3.5.1 Why ratios

A spectrometer viewing a divertor records a line-integrated brightness. Turning that into an absolute population requires the neutral density along the sightline and the path length, neither of which is known to better than a factor of a few. Practitioners therefore work with *ratios*, in which both cancel.

The ratio used throughout this thesis is

$$R \equiv \frac{n_3}{n_4} = \frac{n_{3s} + n_{3p} + n_{3d}}{n_{4s} + n_{4p} + n_{4d} + n_{4f}}, \quad (3.27)$$

the population ratio of the $n = 3$ and $n = 4$ shells. Up to Einstein coefficients it is the Balmer H_α/H_β ratio, since H_α is the $3 \rightarrow 2$ transition and H_β the $4 \rightarrow 2$. The substitution is not free, and Section 4.5 measures what it costs. The transient error of Chapter 5,

computed from the A -weighted line ratio and from the shell ratio, agrees to 0.014 % at the benchmark point; across the whole grid the line-ratio value runs between 0.978 and 0.9999 times the shell-ratio value, a worst case of 2.2 % at the low-density edge $n_e = 10^{12}/\text{cm}^3$. The residue is the $4f$ level, which the shell sum in Eq. (3.27) contains and H_β cannot see, and it is largest at the lowest density because that is where proton ℓ -mixing is weakest. It is convenient to normalise to the ground state as well, writing $r_p = n_p/n_{1s}$, since that is the form in which the QSS solution is usually tabulated.

3.5.2 The hidden variable

Dividing Eq. (3.20) through by the ground-state population gives the QSS ratios

$$r_p^{\text{QSS}} = \frac{\left[-\mathbf{L}_{FF}^{-1}(\mathbf{L}_{FS}\mathbf{n}_S + \mathbf{S}_F n_{\text{ion}})\right]_p}{n_{1s}}. \quad (3.28)$$

It is tempting — and, in the literature, common — to treat Eq. (3.28) as a function of T_e and n_e alone, tabulate it, and invert a measured ratio against the table. That step deserves more scrutiny than it usually receives.

Look at what the numerator contains. The term $\mathbf{L}_{FS}\mathbf{n}_S$ is proportional to the ground-state density; the term $\mathbf{S}_F n_{\text{ion}}$ is proportional to the ion density. What fixes r_p^{QSS} is not either supply on its own but the *ratio* of the two.

It is convenient to measure that ratio against its equilibrium value. Define the ground-state *departure coefficient*

$$b_1 \equiv \frac{n_{1s}}{Z_1 n_e n_{\text{ion}}} \quad (3.29)$$

where Z_1 is the Saha–Boltzmann coefficient of the ground state,

$$Z_p [\text{cm}^3] = 10^6 \frac{g_p}{2g_i} \left(\frac{h^2}{2\pi m_e e T_e} \right)^{3/2} \exp\left(\frac{\chi_p}{T_e}\right), \quad \chi_p = \frac{13.606}{p^2}, \quad (3.30)$$

where T_e and χ_p are both in eV, e is the elementary charge in C (converting eV to J), the bracket evaluates in m^3 , and the factor 10^6 converts to cm^3 . Here $g_p = 2p^2$ is the statistical weight of level p , $g_i = 1$ that of the bare proton, and the factor 2 in the denominator is the spin degeneracy of the free electron. Temperatures are quoted in electron-volts throughout this thesis; no Boltzmann constant appears, and T_e should everywhere be read as the electron thermal energy.

Z_p is the factor for which $n_p = Z_p n_e n_{\text{ion}}$ holds in complete thermodynamic equilibrium, so b_1 is dimensionless and equals unity there. Departures above or below unity measure over- and under-population relative to that thermodynamic reference; no more should be read into the sign than that.

It is worth attaching a number to b_1 immediately, because the name “departure coefficient” suggests a small correction and it is nothing of the kind. Evaluated on the collisional–radiative steady state, b_1 runs from 76.6 at $[0, 7]$, the coldest and densest corner of the grid, to 2.683×10^5 at $[49, 0]$, the hottest and thinnest, and equals 956 at the benchmark point. The ground state is over-populated relative to Saha–Boltzmann by two to five orders of magnitude throughout, which is the statement that a plasma at these densities is far from thermodynamic equilibrium: radiative decay returns atoms to

the ground state faster than collisions can lift them out, and where the resonance lines escape the photons are not replaced. A quantity that varies over three decades across the operating range is not a small correction, and treating it as fixed is the assumption this thesis examines.

The location of those extremes is worth pausing on, because the obvious guess is wrong and the correction matters later. One expects the largest departure where the gas is coldest and most neutral. It is the opposite: b_1 is largest in the *hottest, thinnest* corner and smallest in the coldest, densest one. The reason is that b_1 measures distance from Saha–Boltzmann, and Saha–Boltzmann at 10 eV predicts a nearly fully ionised gas, so any surviving neutral population is an enormous departure from it. At 1 eV Saha already predicts a mostly neutral gas, so a predominantly neutral plasma is only a modest departure. The neutral *fraction* therefore runs the other way: solving $\mathbf{Ln} + \mathbf{S} = 0$ directly and restoring $n_{\text{ion}} = n_e$, the ratio n_{1s}/n_{ion} falls from 40.4 at $[0, 0]$ and 20.5 at $[0, 7]$ — gases that are 95 to 98 % neutral — to 5.3×10^{-6} at $[49, 7]$. **A large departure coefficient does not mean many neutrals**, and the two quantities peak at opposite corners of the grid.

Where the optically thin assumption holds

That last observation forces a declaration that the preceding sections left implicit. The Einstein coefficients in Eqs. (3.1), (3.3) and (3.4) enter bare: every emitted photon is assumed to leave the plasma. Where the ground-state density is high that is false — a Lyman- α photon is reabsorbed, the atom returns to $2p$, and the effective $A_{2p \rightarrow 1s}$ is reduced by the population escape factor Θ_P of Holstein [34].

Whether this matters is a question about n_{1s} , and the numbers above make it answerable. Taking a homogeneous slab of thickness D and a Doppler-broadened Lyman- α line at $T_{\text{at}} = T_e$:

point	T_e (eV)	n_e (1/cm ³)	n_{1s} (1/cm ³)	τ_c	Θ_P
[49, 0]	10	10^{12}	1.1×10^7	9×10^{-7}	1.000
[49, 7]	10	10^{15}	5.3×10^9	5×10^{-4}	1.000
[23, 5]	2.95	1.4×10^{14}	1.2×10^{11}	0.020	0.986
[0, 0]	1	10^{12}	4.0×10^{13}	11.1	0.032
[0, 7]	1	10^{15}	2.0×10^{16}	5.6×10^3	3×10^{-5}

with τ_c and Θ_P quoted at $D = 10$ cm; Θ_P at $[0, 0]$ is 0.48, 0.032 and 0.0023 for $D = 1, 10$ and 100 cm respectively, so the qualitative conclusion does not depend on the choice, but no single figure should be quoted without it. The model itself contains no geometry, and D is an input.

The reading is clean. Over the hot half of the grid the plasma is optically thin to Lyman- α to better than one part in 10^3 , and at the benchmark point $\Theta_P = 0.986$: the bare- A treatment is justified where the quantitative results of this thesis are established. Along the $T_e = 1$ eV edge it is not. There Θ_P falls to 3×10^{-5} , the effective $A_{2p \rightarrow 1s}$ collapses, and with it b_1^{CRE} , the switching points c_m/a_m , and the ground-fed fractions of Section 3.5.4.

The results of this chapter are therefore stated for $T_e \gtrsim 2$ eV, and the cold edge of the grid is retained for the asymptotic behaviour it exposes rather than as a prediction. It is also the corner where the atomic-only model is weakest for

an independent reason: at 95 to 98 % neutral and 1 eV the gas is molecular, and neither H_2 nor molecular-assisted recombination appears in $\mathbf{L}$. Chapter 6 takes both limitations up quantitatively. Neither affects the structural results of this chapter, which are statements about whatever $\mathbf{L}$ is supplied: the quasi-steady-state closure, the three-argument form of R^{QSS} and the identity $d \ln R / d \ln b_1 = f_3 - f_4$ hold for a trapped-line rate matrix exactly as they do for an optically thin one. What changes is the numerical value of b_1^{CRE} that such a matrix produces.

b_1 is a **dynamical variable**, and it is worth being precise about how it evolves. Differentiating Eq. (3.29) at fixed n_{ion} ,

$$\dot{b}_1 = \frac{\dot{n}_{1s}}{Z_1 n_e n_{\text{ion}}} - b_1 \left(\frac{\dot{Z}_1}{Z_1} + \frac{\dot{n}_e}{n_e} \right). \quad (3.31)$$

Only the first term comes from Eq. (3.21), which governs n_{1s} ; the second comes from the normalisation moving. This distinction matters for the temperature-step experiment of Chapter 5. At the instant of a step, n_{1s} is continuous but T_e is not, so $Z_1(T_e)$ jumps and b_1 changes discontinuously purely because its reference has changed. Once T_e and n_e are again held fixed, $b_1 \propto n_{1s}$ and its subsequent relaxation is that of the ground-state population, on τ_{QSS} .

Either way, $b_1(t)$ is not determined by the instantaneous (T_e, n_e) — it carries the plasma's history. Nothing in the QSS derivation fixes it.

3.5.3 Does the observable actually depend on it?

The previous subsection shows that each *individual* normalised population depends on b_1 . That is not yet an argument about R . A sceptical reader is entitled to object: perhaps every level depends on b_1 and the dependence cancels in the ratio, leaving H_α/H_β a clean function of (T_e, n_e) after all. Ratios are chosen precisely because things cancel in them. The objection must be answered, not assumed away.

It can be answered exactly, because Eq. (3.20) is linear in the two supplies. Define

$$\mathbf{a} \equiv -\mathbf{L}_{FF}^{-1} \mathbf{L}_{FS}, \quad \mathbf{c} \equiv -\mathbf{L}_{FF}^{-1} \mathbf{S}_F, \quad (3.32)$$

so that Eq. (3.20) reads

$$\mathbf{n}_F^{\text{QSS}} = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}. \quad (3.33)$$

Both vectors are dimensionless and both have a plain reading. a_p is the number of atoms that end up sitting in level p per ground-state atom present: not only those excited into p directly, but those excited anywhere and arriving at p afterwards by cascade or by a further collision. The matrix inverse is what performs that sum over every possible route. Likewise c_p is the population of p per ion, from electrons captured into any level and settling into p on the way down. Neither depends on how many ground-state atoms or ions are actually present — both are properties of $\mathbf{L}_{FF}$, $\mathbf{L}_{FS}$ and $\mathbf{S}_F$ alone, and therefore functions of (T_e, n_e) only.

So Eq. (3.33) says something simple: every excited level is fed from two directions, and its population is the sum of two terms, one proportional to the size of the reservoir below it and one to the reservoir above. What distinguishes one level from another is the *proportion* in which it draws on the two.

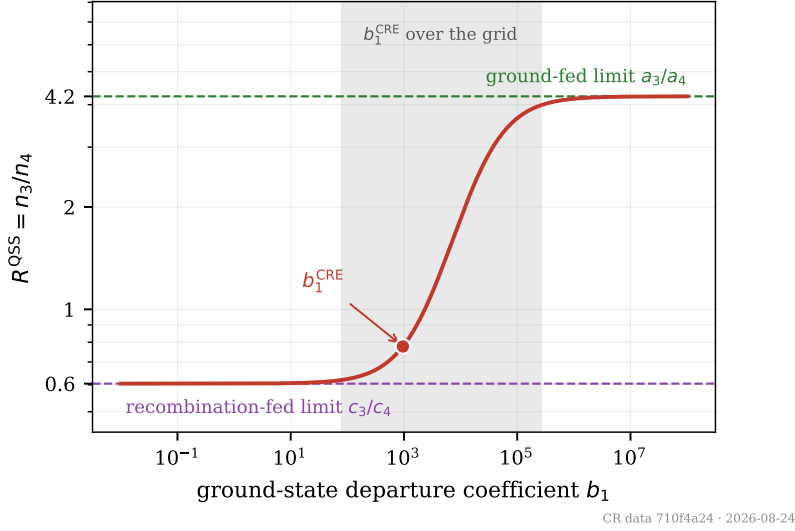


Figure 3.3: The shell ratio R^{QSS} as a function of the ground-state departure coefficient, at the benchmark point, from Eq. (3.35). The curve is a one-parameter family: it asks what R would be if b_1 took other values at fixed (T_e, n_e) . Between the two supply limits the observable changes by a factor of several; a table indexed on (T_e, n_e) alone must select one point on this curve. Both axes are logarithmic, so the visible slope is $d \ln R / d \ln b_1$ directly — the quantity plotted in Figure 3.4. The marker shows the value the CR-equilibrium closure selects, and the shaded strip the range of b_1^{CRE} over the whole grid.

Writing a_m and c_m for the sums over the ℓ -resolved states of shell m , and using $n_{1s} = b_1 Z_1 n_e n_{\text{ion}}$ from Eq. (3.29),

$$n_m^{\text{QSS}} = n_{\text{ion}} [a_m b_1 Z_1 n_e + c_m] = a_m n_{1s} + c_m n_{\text{ion}}. \quad (3.34)$$

The prefactor n_{ion} is common to both shells and cancels in the ratio, as does the emitting path length. The neutral density does *not* cancel, and it is important not to read the previous sentence that way: what an intensity ratio removes is the need to *know* the absolute neutral density and the path length in order to form it, not the dependence of the ratio on how much neutral hydrogen is present relative to the ions. That dependence is the subject of this chapter and it survives, inside the bracket, as b_1 :

$$R^{\text{QSS}}(b_1) = \frac{a_3 b_1 Z_1 n_e + c_3}{a_4 b_1 Z_1 n_e + c_4} \quad (3.35)$$

Differentiating,

$$\frac{\partial R^{\text{QSS}}}{\partial b_1} = \frac{Z_1 n_e (a_3 c_4 - a_4 c_3)}{(a_4 b_1 Z_1 n_e + c_4)^2} \quad (3.36)$$

The observable is independent of b_1 if and only if

$$a_3 c_4 = a_4 c_3 \quad \Longleftrightarrow \quad \frac{a_3}{c_3} = \frac{a_4}{c_4}, \quad (3.37)$$

that is, if and only if the $n = 3$ and $n = 4$ shells draw on the two supplies in *identical proportion*. There is no reason for them to. The $n = 4$ shell lies closer to the continuum and is more readily fed by recombination; the $n = 3$ shell is more readily reached by excitation from the ground state. Chapter 5 measures the difference and finds it substantial across most of the operating range.

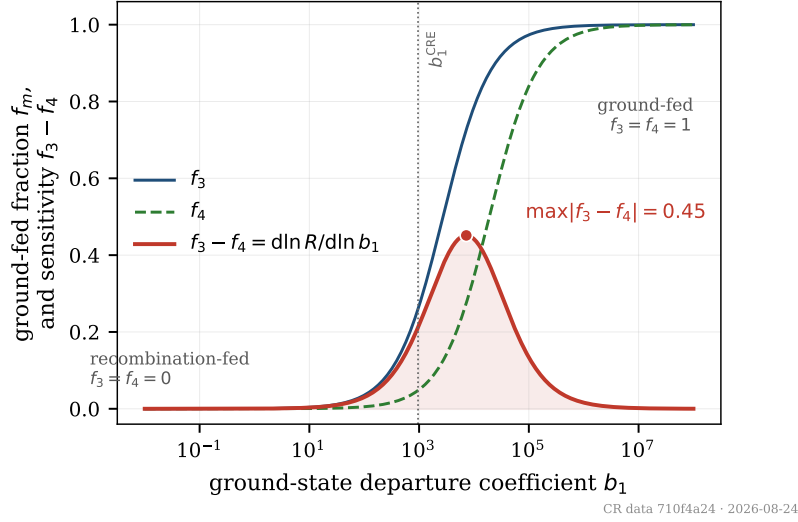


Figure 3.4: The ground-fed fractions of the two shells and their difference, at the benchmark point; the dotted vertical line marks b_1^{CRE} there. Both fractions rise from zero to unity as the ground-state reservoir grows, but they do so at different rates, and their difference — which by Eq. (3.39) is exactly $d \ln R / d \ln b_1$ — vanishes in both limits and peaks where the two supplies compete. Note that $f_3 - f_4$ is a difference of fractions, not itself a fraction.

3.5.4 The ground-fed fraction, and where the sensitivity vanishes

Equation (3.36) settles whether the dependence exists. It is not, however, the right measure of how *strong* it is, because R itself varies over orders of magnitude across the operating range and an absolute derivative is not comparable between grid points. The dimensionless measure is the logarithmic sensitivity, and it takes a form with a direct physical reading.

Define the *ground-fed fraction* of shell m ,

$$f_m \equiv \frac{a_m b_1 Z_1 n_e}{a_m b_1 Z_1 n_e + c_m} \quad (3.38)$$

the share of shell m 's population that arrived from below, by excitation out of the ground state, as opposed to from above by recombination. That f_m lies in $[0, 1]$ is not automatic: it requires $a_m \geq 0$ and $c_m \geq 0$, and both are components of $-\mathbf{L}_{FF}^{-1}$ acting on non-negative vectors. The claim therefore needs $-\mathbf{L}_{FF}^{-1}$ to be non-negative *elementwise*, which holds for the same reason $\mathbf{L}_{FF}$ is invertible (Section 3.4): $-\mathbf{L}_{FF}$ has non-positive off-diagonal entries and strictly positive column sums, making it a non-singular M-matrix, and the inverse of a non-singular M-matrix is elementwise non-negative. Physically the statement is unsurprising — adding atoms to one level cannot reduce the steady population of another — but it is worth recording, because the two limits below depend on it. Direct evaluation confirms it: at none of the 400 grid points does $-\mathbf{L}_{FF}^{-1}$ contain a negative entry.

Differentiating Eq. (3.35) logarithmically then gives

$$\frac{d \ln R}{d \ln b_1} = f_3 - f_4 \quad (3.39)$$

The Balmer ratio's sensitivity to the ground-state reservoir is exactly the difference in ground-fed fraction between the two shells. This is the mathematical core of the thesis, and Eq. (3.39) is where the physics becomes visible.

It vanishes in two limits, and only there. When $b_1 Z_1 n_e \gg c_m/a_m$ for both shells, both are fed almost entirely from the ground state, $f_3 = f_4 = 1$, and the ratio is blind to the reservoir. When $b_1 Z_1 n_e \ll c_m/a_m$, both are fed almost entirely by recombination, $f_3 = f_4 = 0$, and the ratio is blind again. *In between — where the two supplies compete — the shells are fed in different proportion and the sensitivity is largest.*

How large, and where

Equation (3.39) says the sensitivity peaks somewhere between the two limits. It does not yet say how high, or at what reservoir. Both follow in closed form, and the answer is more restrictive than one might expect.

Work in the logarithmic reservoir variable $x = \ln(b_1 Z_1 n_e)$. In that variable Eq. (3.38) is exactly a logistic,

$$f_m(x) = \frac{1}{1 + e^{-(x-x_m)}}, \quad x_m \equiv \ln\left(\frac{c_m}{a_m}\right), \quad (3.40)$$

so the two curves in Figure 3.4 are not merely similar in shape: they are the *same* curve, of the same width, displaced along the axis. Every property of the 42-state network enters only through where each sits. Their separation is

$$\Delta \equiv x_4 - x_3 = \ln\left(\frac{a_3/a_4}{c_3/c_4}\right) = \ln\left(\frac{R^{\text{grd}}}{R^{\text{rec}}}\right), \quad (3.41)$$

the logarithm of the span between the two asymptotes of Figure 3.3. Equation (3.40) also makes explicit a distinction that is easy to lose. **The switching point of shell m is $b_1 = c_m/(a_m Z_1 n_e)$, which is a property of the network, not of the Saha balance.** A plasma may lie far above the Saha–Boltzmann ground-state population, $b_1 \gg 1$, and still have both shells fed predominantly from above. The benchmark point is exactly such a case: $b_1^{\text{CRE}} = 956$, while the two shells do not become half ground-fed until $b_1 = 2.73 \times 10^3$ and 1.91×10^4 respectively, so $f_3 = 0.260$ and $f_4 = 0.048$ — three-quarters of $n = 3$ and 95 % of $n = 4$ arrive by recombination. A large departure coefficient and a predominantly ground-fed manifold are different statements, and only the second controls Eq. (3.39). Neither should be conflated with calling the plasma *ionising*, which is properly a statement about the direction of the net population flux and is not fixed by the magnitude of b_1 alone. Writing $s = x - \frac{1}{2}(x_3 + x_4)$ and using $f(-y) = 1 - f(y)$, the difference $f_3 - f_4$ is *even* in s . Its maximum therefore sits at the midpoint, and takes the value

$$\boxed{\max |f_3 - f_4| = \tanh\left(\frac{|\Delta|}{4}\right) = \tanh\left[\frac{1}{4} \left| \ln\left(\frac{R^{\text{grd}}}{R^{\text{rec}}}\right) \right| \right]} \quad (3.42)$$

where the modulus matters only for the generalisation below: for the n_3/n_4 ratio considered here $R^{\text{grd}} > R^{\text{rec}}$ and $\Delta > 0$, but for a pair of shells drawn in the opposite order Δ changes sign while the left-hand side cannot. The maximum is attained at $b_1 Z_1 n_e = \sqrt{(c_3/a_3)(c_4/a_4)}$, the geometric mean of the two switching points. The area is fixed by the same quantity: integrating Eq. (3.39) over the reservoir gives $\int (f_3 - f_4) d \ln b_1 = \ln R^{\text{grd}} - \ln R^{\text{rec}} = \Delta$. That second relation is only the fundamental theorem of calculus, and it is quoted here for what it is — a consistency check tying the two figures together, not an independent result. The figure script evaluates both and halts if either disagrees with the sampled curve.

Three things follow, and the third is the useful one.

First, the peak sensitivity is set *entirely* by how far apart the two supply limits lie. Two plasmas with the same span between R^{rec} and R^{grd} have the same worst-case sensitivity, whatever the details of their cascade networks.

Second, $\tanh$ is bounded by unity, so $|\text{d} \ln R / \text{d} \ln b_1| < 1$ everywhere and for every pair of shells. **A line ratio can never respond to the ground-state reservoir faster, in relative terms, than the reservoir itself moves.** Integrating a logarithmic sensitivity whose magnitude is everywhere below unity gives the finite-excursion statement $|\Delta \ln R| \leq |\Delta \ln b_1|$, so a stale reservoir cannot be amplified in logarithmic space, only attenuated.

It is worth being precise about what this does and does not bound. It bounds the error in the *observable*. A diagnostician does not suffer an error in R directly: they measure R and invert it for (T_e, n_e) , and the error in the inferred temperature is

$$\Delta \ln T_e = \frac{\text{d} \ln R / \text{d} \ln b_1}{\text{d} \ln R / \text{d} \ln T_e} \Delta \ln b_1. \quad (3.43)$$

The bound constrains the numerator only. Where the ratio is itself weakly sensitive to temperature — which is where Balmer diagnostics are known to be least useful — a bounded error in R can still become a large error in the inferred T_e . The inversion Jacobian in the denominator is supplied in Chapter 5; without it, no statement about diagnostic error follows from this section.

These closed forms also locate the benchmark point on the curve rather than leaving it as an anchor of unstated significance. Its sensitivity is $f_3 - f_4 = 0.212$, some 47% of the maximum 0.4514, which is attained nearly one decade higher at $b_1 = 7.2 \times 10^3$. The benchmark is therefore on the rising flank and not at the worst case, which is the honest reading of every error reported against it in Chapter 5. Two identities follow from Eq. (3.38) and fix the marker in Figure 3.3 in terms of the fractions of Figure 3.4,

$$R = R^{\text{grd}} \frac{f_4}{f_3} = R^{\text{rec}} \frac{1 - f_4}{1 - f_3}, \quad (3.44)$$

both giving $R = 0.7778$ at the benchmark. They are algebraic rearrangements rather than new physics, but they are the cheapest available cross-check between the two figures and the figure script enforces them.

Third, the attenuation is severe unless the span is large. For $\Delta \ll 1$ the peak is $\Delta/4$: a diagnostic whose two supply limits differ by only 10% has a worst-case log-sensitivity of 0.024. Sensitivity to the hidden variable is bought with span, and $R = n_3/n_4$ has span because $n = 3$ and $n = 4$ draw on the two supplies in genuinely different proportion. That is why this ratio is diagnostically useful and also why it is exposed.

Note that this is the logarithmic sensitivity, not the absolute derivative $\partial R / \partial b_1$ of Eq. (3.36). The latter does *not* vanish as $b_1 \rightarrow 0$; it tends to $Z_1 n_e (a_3 c_4 - a_4 c_3) / c_4^2$, which is finite. The two limits are statements about the *fractional* response of the observable, which is what a diagnostic error is.

The classification into ionising and recombining plasmas is due to Fujimoto [12], and the crossover between the two is exactly the region Eq. (3.39) singles out. A plasma is ionising when excitation out of the ground state supplies most of the excited population and recombining when capture from the continuum does; between those limits both supply channels of Eq. (3.33) contribute, the two shells draw on them in different proportion, and the sensitivity is largest. That crossover is a statement about the switching points c_m/a_m

of the network, and where it falls in (T_e, n_e) is measured rather than assumed; identifying it with any named divertor operating regime is a separate claim, requiring densities this section does not supply, and is not made here. Chapter 5 maps $f_3 - f_4$ over the operating range and shows the diagnostic error tracking it.

It is worth recording what Eq. (3.35) is not. It is not a fit, an expansion, or an approximation: it is an exact rewriting of Eq. (3.20), valid wherever $\mathbf{L}_{FF}$ is invertible. Nor is it new. It is the standard population-coefficient decomposition of collisional–radiative theory, introduced by Bates, Kingston and McWhirter [7] and written by Fujimoto and McWhirter [11] in the Saha-normalised form $b(p) = r_0(p) + r_1(p)b(1)$, with r_0 the recombining and r_1 the ionising population coefficient. Substituting $b(p) = n_p/[Z_p n_e n_{\text{ion}}]$ into that expression and clearing the normalisation gives the correspondence

$$a_p = r_1(p) \frac{Z_p}{Z_1}, \quad c_p = r_0(p) Z_p n_e, \quad (3.45)$$

both sides dimensionless. The two forms are the same linear solution in different coordinates; what is done here is to ask what the ratio of two such populations depends on, which Eq. (3.39) answers.

Section 4.4 tests the decomposition against Fujimoto’s published population coefficients. A second test of the same decomposition is attempted against the ADAS effective coefficients [9], and it fails. Because r_0 and r_1 are published as curves rather than tables, that test cannot be made level by level: $\mathbf{a}$ and $\mathbf{c}$ are instead summed against the ionisation rates to form the effective stage-to-stage coefficients, which *are* tabulated, and compared with SCD96 and ACD96. The comparison — Gate D of Chapter 4 — disagrees at every grid point, by a systematic factor between 9 and 65 that peaks near $T_e \approx 2$ eV, and only its ionisation half is implemented, the recombination table being read and never used. Chapter 4 reports that failure and its diagnosis rather than repairing it. Two readings are open. ADAS tabulates SCD as the *ionising* coefficient, whereas the sum formed here runs over the full steady state, so ionisation out of recombination-fed Rydberg levels enters the model’s coefficient and belongs under ACD; and this model holds the ion as a fixed reservoir with no forty-fourth state (Section 3.5), so the two quantities are not defined on the same system. Neither reading has been demonstrated, and no agreement with ADAS is claimed anywhere in this thesis.

The contraction is severe in any case — forty-two coefficients are collapsed into one, and many incorrect $\mathbf{a}$ would contract correctly — but it is not merely a rate check either: a_p and c_p enter the two coefficients with different weights, so an error in the *split* between the two supplies displaces the ionisation and recombination coefficients in opposite directions, which no single-channel comparison detects. Chapter 5 verifies the superposition $\mathbf{n}_F = \mathbf{a}n_{1s} + \mathbf{c}n_{\text{ion}}$ to a relative error of 3×10^{-14} .

3.5.5 Two reference states, and they are not the same

Equation (3.35) shows that the QSS ratio is a function of three arguments, not two. This gives two distinct things a measurement can be compared against.

The QSS reference. $R^{\text{QSS}}(T_e, n_e; b_1(t))$, evaluated at the *instantaneous* departure coefficient. This is what adiabatic elimination claims: the excited manifold is in steady state with whatever reservoir is present at that moment. Its accuracy is governed by Section 3.4’s validity condition.

The CR-equilibrium reference.

$$R^{\text{CRE}}(T_e, n_e) \equiv R^{\text{QSS}}(T_e, n_e; b_1^{\text{CRE}}(T_e, n_e)), \quad (3.46)$$

where b_1^{CRE} is obtained by additionally setting $d\mathbf{n}_S/dt = 0$ in Eq. (3.21) and solving for the steady ground-state population. This *is* a function of (T_e, n_e) alone and can therefore be tabulated.

The distinction is easy to miss because both are called “the QSS solution” in common usage, and in a steady plasma they are the same number. They separate whenever the ground-state population has not finished settling — which, in a divertor disturbed by an edge-localised mode every few milliseconds, is most of the time.

Two remarks on terminology, both of which matter for what follows. First, $b_1 = 1$ denotes complete thermodynamic equilibrium, whereas b_1^{CRE} denotes the steady state of the collisional–radiative system, which is a different and much weaker condition; in an optically thin ionising plasma b_1^{CRE} is very much larger than unity. Second, this model holds the ion density fixed (Section 3.1), so the quantity relaxing on τ_{QSS} is the ground-state population against a prescribed reservoir, not a self-consistent ionisation balance. What that restriction costs is assessed in Chapter 6.

3.5.6 Two error measures

Each reference defines an error:

$$\varepsilon^{\text{QSS}}(t) = \left| \frac{R(t)}{R^{\text{QSS}}(T_e, n_e; b_1(t))} - 1 \right|, \quad (3.47)$$

$$\varepsilon^{\text{CRE}}(t) = \left| \frac{R(t)}{R^{\text{CRE}}(T_e, n_e)} - 1 \right|. \quad (3.48)$$

ε^{QSS} asks whether the excited states have kept up with the reservoir they actually see. ε^{CRE} asks whether the tabulated answer is right. The first is the question the literature worries about; the second is the one a diagnostician suffers.

They are not the same question, and Chapter 5 shows they have very different answers. Naming them apart is the single most important definitional step in this thesis: conflating them makes it possible to measure the second and report it as the first.

3.5.7 What a two-parameter table must assume

It remains to say precisely what is being assumed when a measured ratio is inverted against a table indexed on (T_e, n_e) .

Equation (3.35) establishes that $R = R(T_e, n_e, b_1)$. A two-parameter table must therefore do one of three things: adopt a closure $b_1 = b_1(T_e, n_e)$; operate in a regime where one supply dominates both shells, so that Eq. (3.36) is negligible and the third argument does not matter; or hold some other quantity fixed that determines b_1 implicitly. Equilibrium ionisation balance is a closure, not a mathematical necessity, and it is important not to overstate the point.

It is important to be precise about which practice is being criticised, because the modern effective-coefficient databases are not themselves at fault. ADAS [9], for instance, tabulates

photon emissivity coefficients separately for the excitation-driven and recombination-driven components. Used as intended, those two components are weighted by n_{1s} and n_{ion} respectively, which is exactly the information b_1 carries — the database does not impose a closure at all. Indeed the excitation/recombination split of Eq. (3.33) is conceptually the same decomposition.

The assumption enters at the *inversion* step, when a measured ratio is mapped to (T_e, n_e) through a two-parameter lookup. Such a lookup returns a single ratio per (T_e, n_e) pair and therefore requires the two components to have been combined in a fixed proportion — that is, it requires a closure for b_1 . The usual choice is $b_1 = b_1^{\text{CRE}}$, obtained by solving the collisional–radiative system to complete steady state. A ratio inverted this way carries the assumption that the plasma’s ground-state population has settled at its steady value for the local conditions: not an assumption about how quickly excited states equilibrate, but an assumption that the plasma has *finished changing*.

The second possibility is not merely hypothetical, and it is the reason Eq. (3.36) is worth having. Where one supply dominates both shells, the table is safe regardless of what the ground state is doing. Chapter 5 maps where in the divertor operating range that is the case, and where it is not.

The remainder of this thesis quantifies what the assumption costs.

3.6 A caveat: the modes are not orthogonal

Section 3.3 presented the eigen-decomposition as a change of coordinates in which the 43 coupled equations become 43 independent decays. That is exact. But a change of coordinates is only as convenient as the coordinates are well behaved, and these are not.

The reason is a physical asymmetry running through the whole matrix. Radiative decay carries atoms downward at up to $6.3 \times 10^8/\text{s}$; electron-impact excitation carries them upward at $Cn_e \sim 10^4/\text{s}$ at the benchmark density. Downward and upward coupling therefore differ by some nine orders of magnitude, and $\mathbf{L}$ is very far from symmetric. Its eigenvectors, consequently, are very far from orthogonal.

The consequence is worth stating carefully, because it is easy to overstate. Every eigenvalue has negative real part, so every mode decays and any perturbation eventually vanishes; that much the spectrum does guarantee. What it does not guarantee is that the perturbation decays *monotonically*. When eigenvectors are nearly parallel, a state of modest size can be the sum of two much larger, nearly cancelling modal contributions. If those two modes decay at different rates the cancellation is progressively spoiled, and the state can grow — sometimes by orders of magnitude — before the decay of the individual modes finally wins. The standard diagnostic for whether this is possible at all is the numerical abscissa

$$\mu(\mathbf{L}) = \lambda_{\max}\left(\frac{1}{2} [\mathbf{L} + \mathbf{L}^{\top}]\right), \quad (3.49)$$

the largest eigenvalue of the symmetric part of $\mathbf{L}$, which gives the initial slope of $\|\mathbf{n}(t)\|$ for the worst-case perturbation. For this matrix $\mu(\mathbf{L}) > 0$ at every grid point, so transient growth is not merely possible but guaranteed for *some* disturbance.

Whether it occurs for the disturbance and the observable that matter here is a different question, and one the spectrum cannot answer. $\mu(\mathbf{L}) > 0$ is a statement about the worst case over all perturbation directions in the Euclidean norm — other norms carry their own logarithmic norms; the Balmer ratio is one particular scalar function of the state, and a temperature step is one particular disturbance. A quantity may be entirely well behaved

even when the operator that governs it admits growth in principle. This is why Section 3.4 was careful to label Eq. (3.23) an estimate rather than a bound: the eigenvalue argument describes the system after the transients have settled, and the transients themselves are outside its reach.

Chapter 5 therefore settles the question the only way it can be settled — by integrating the full 43-state system through a temperature step and measuring what the Balmer ratio does, rather than by inferring it from the spectrum.

3.7 Summary

Starting from a population balance on each of 43 levels, this chapter assembled the collisional–radiative rate equation $\dot{\mathbf{n}} = \mathbf{L}\mathbf{n} + \mathbf{S}n_{\text{ion}}$ and checked the assembly against particle conservation, which holds to 3.6×10^{-11} across the operating range.

Diagonalising $\mathbf{L}$ revealed a spectrum with one gap and then a continuum: a single slow mode at $\tau_{\text{QSS}} = 22.7 \mu\text{s}$, the relaxation of the ground-state population, sitting four orders of magnitude above a dense band of fast modes whose slowest member is $\tau_{\text{relax}} = 2.28 \text{ ns}$, the rearrangement of the excited manifold. The separation $M = \tau_{\text{QSS}}/\tau_{\text{relax}}$ is 9982 at the benchmark point and does not fall below 86.8 at any of the 400 grid points — though, as Section 3.3 records, that floor sits at a point the transient analysis excludes, and the four-orders figure quoted above is the benchmark value rather than a property of the grid, over which M spans two to nine orders.

That separation licenses the quasi-steady-state approximation: eliminate the excited populations algebraically and solve only for the slow subsystem. The elimination is the Bodenstein steady-state approximation applied to 42 coupled intermediates at once, and it produces the Schur complement $\mathbf{\Omega}_{\text{QSS}}$ — the effective ionisation rate that a plasma modeller using QSS constructs, and the algebraic form of the effective coefficients that spectroscopic inversions rest on. Its validity condition, $M \gg 1$, is satisfied everywhere on the grid and comfortably — subject to the two qualifications of Section 3.4: M compares rates where Chapter 5 measures a distance, and the spectrum of a non-normal operator bounds asymptotic decay rather than finite-time error.

The chapter’s central result is that this is not the end of the argument. The QSS ratio

$$R^{\text{QSS}}(b_1) = \frac{a_3 b_1 Z_1 n_e + c_3}{a_4 b_1 Z_1 n_e + c_4}$$

is a function of *three* arguments, not two. The third is the ground-state departure coefficient b_1 , which is dynamical: it carries the plasma’s history and takes a time τ_{QSS} to settle. Its logarithmic sensitivity is exactly

$$\frac{d \ln R}{d \ln b_1} = f_3 - f_4,$$

the difference in ground-fed fraction between the two shells, which vanishes when either supply dominates both shells and is largest where the two compete. A table indexed on (T_e, n_e) alone must therefore adopt a closure for b_1 , and the closure adopted in practice is complete collisional–radiative equilibrium.

Two error measures follow and must not be conflated. ε^{QSS} asks whether the excited states have kept up with the reservoir they actually see; ε^{CRE} asks whether the tabulated answer is right. The first is the question the literature worries about. The second is the one a diagnostician suffers.

Everything in this chapter has been an argument about equations. Whether the equations describe hydrogen is a separate question, and Chapter 4 addresses it: what would a wrong model fail, and does this one fail it?

Chapter 4

Does the Model Work?

- 4.1 What a validation gate can and cannot show
- 4.2 Internal consistency: detailed balance, coronal and Saha limits
- 4.3 Timescale hierarchy
- 4.4 External benchmark: Fujimoto's population coefficients
- 4.5 The Balmer ratio and the shell ratio
- 4.6 Errors found and corrected
- 4.7 Summary

Chapter 5

What the Model Says

Chapter 3 left the argument at a definition and a worry. The definition: a Balmer-ratio lookup table indexed on (T_e, n_e) is a table in *three* arguments with the third one hidden, and the hidden argument is the ground-state departure coefficient b_1 , which takes a time τ_{QSS} to settle after the plasma changes. The worry, which is the one the literature carries: the excited states are what a spectrometer sees, and they may not keep up with a plasma that changes on microseconds. Chapter 4 established that the rate matrix these questions are asked of reproduces independent atomic data where such data exist, and reported the two errors that an audit of this work found and corrected.

This chapter evaluates both errors over the operating range, and the two results are worth stating before the evidence rather than after it.

The first is a reversal. The quasi-steady-state (QSS) closure, the approximation that the excited populations sit in instantaneous steady state with whatever ground-state reservoir is present, is not the problem. Measured along a full time-dependent trajectory it is accurate to eight significant figures. What fails is the assumption that the reservoir itself has finished moving, which is a different assumption, made at a different step of the inversion, and never previously separated from the first.

The second is that what that failure costs is not a number. It is a function, and the function has a closed form,

$$\varepsilon_{\text{plateau}} = \left| \exp(\bar{S} G \Delta \ln T_e) - 1 \right|,$$

in which $\Delta \ln T_e$ is the size of the transient the reader cares about, G measures how far the ground-state reservoir is dragged by it, and $\bar{S}$ measures how strongly the line ratio answers to the reservoir. The last of these is bounded above by a quantity that no choice of atomic data can push past unity. So the chapter reports the structure and the bound, and lets a reader compute the error for their own transient. A single percentage would be an answer to a question about the grid this thesis happens to use, and Section 5.3 shows how badly that goes.

Both results are stated over $T_e \geq 2 \text{ eV}$. Section 5.10 shows that this is not a hedge: it is the region of the operating grid on which the model's own assumptions are mutually consistent, and two unrelated pieces of physics place the boundary in the same place.

What this chapter takes as given. From Chapter 3: the rate equation $\dot{\mathbf{n}} = \mathbf{L}\mathbf{n} + \mathbf{S}n_{\text{ion}}$ and its two timescales τ_{relax} and τ_{QSS} ; the QSS elimination and its Schur complement; the two reference states R^{QSS} and R^{CRE} and the two error measures ε^{QSS} and ε^{CRE} built on them; the two-supply decomposition $\mathbf{n}_F = \mathbf{a}n_{1s} + \mathbf{c}n_{\text{ion}}$; the ground-fed fraction

f_m , its logistic form, and the identity $d \ln R / d \ln b_1 = f_3 - f_4$. From Chapter 1: the operating range $T_e = 1\text{--}10\text{ eV}$, $n_e = 10^{12}\text{--}10^{15}/\text{cm}^3$, and the fact that an edge-localised mode (ELM) disturbs a divertor for of order $100\text{ }\mu\text{s}$. The benchmark point $T_e = 2.947\text{ eV}$, $n_e = 1.389 \times 10^{14}/\text{cm}^3$, grid index $[23, 5]$, runs through this chapter as it runs through the last.

5.1 What happens after a temperature step

Take a plasma sitting quietly at collisional–radiative equilibrium (CRE), the steady state of the full 43-level system, and change the electron temperature abruptly at $t = 0$, holding n_e fixed. Three things then happen, on three separated timescales, and the middle one is where this thesis lives.

For the first few nanoseconds the excited manifold rearranges. Every one of the 42 excited levels is a reactive intermediate in the chemical-engineering sense: it holds almost no population, it is created and destroyed thousands of times faster than the species feeding it changes, and it therefore reaches a steady state with respect to that species almost immediately. At the benchmark point this takes $\tau_{\text{relax}} = 2.28\text{ ns}$.

For the next several microseconds nothing further appears to happen, and this is the interesting part. The excited populations have settled, but they have settled onto the *old* ground-state population, because the ground state holds essentially all the atoms and cannot turn over in nanoseconds. The system occupies a state that is steady in its fast variables and stale in its slow one. Call it *partial equilibrium* (PE). At the benchmark point it lasts until the ground state has moved, on $\tau_{\text{QSS}} = 22.7\text{ }\mu\text{s}$, so the plateau is some four orders of magnitude wider than the relaxation that built it.

Only then does the plasma reach the new collisional–radiative equilibrium that a lookup table would have assigned to it the instant the temperature changed.

The ELM that motivates this thesis lasts of order $100\text{ }\mu\text{s}$. It is long against τ_{relax} by a factor of about 4×10^4 and comparable to τ_{QSS} , so a spectrometer pointed at the divertor during one spends its integration time looking at the plateau. That is the state whose error this chapter measures.

5.1.1 Two error measures, and why the step size is part of each

Section 3.5 fixed the two reference states against which an observable can be judged. Two further quantities are needed here, and both are properties of a temperature *step* rather than of a single operator, which is why neither appears in Chapter 3.

Let the plasma sit at the collisional–radiative equilibrium of $\mathbf{L}(T_e, n_e)$, and let the electron temperature step to $T_e + \Delta T_e$ at $t = 0$ with n_e held fixed. Write $\mathbf{L}^-$ and $\mathbf{L}^+$ for the operators before and after, in the notation of Section 3.3. The *step error*

$$\varepsilon_{\text{step}} = \left| \frac{R^{\text{CRE}}(T_e, n_e)}{R^{\text{CRE}}(T_e + \Delta T_e, n_e)} - 1 \right| \quad (5.1)$$

is the dimensionless distance between the two tabulated answers: what the reading would change by if the plasma re-equilibrated instantly. The *plateau error*

$$\varepsilon_{\text{plateau}} = \left| \frac{R^{\text{PE}}}{R^{\text{CRE}}(T_e + \Delta T_e, n_e)} - 1 \right|, \quad R^{\text{PE}} \equiv R^{\text{QSS}}(T_e + \Delta T_e, n_e; b_1^-), \quad (5.2)$$

is the distance between the tabulated answer and the state the plasma occupies once the excited manifold has settled but before the ground state has moved: the post-step excited response of Eq. (3.28), evaluated on the *pre-step* reservoir. Both are dimensionless, both are non-negative, and both are zero for a zero step.

Keeping them separate matters because they are not close. Across the operating grid $\varepsilon_{\text{plateau}}$ exceeds $\varepsilon_{\text{step}}$ at every one of the 680 point-and-direction pairs that possess a plateau, with a median ratio near 12 and a minimum of 1.45 (`verify_plateau_gridmap.py`). A diagnostician who pictures the reading sliding smoothly from one tabulated value to the other has not merely misjudged the size of the excursion; they have the wrong shape of transient.

That comparison carries a caution about the ratio itself. Because $\log(\varepsilon_{\text{plateau}}/\varepsilon_{\text{step}}) = \log \varepsilon_{\text{plateau}} - \log \varepsilon_{\text{step}}$ identically, any correlation between the amplification and $\varepsilon_{\text{step}}$ is partly tautological, and the amplification blows up wherever $\varepsilon_{\text{step}}$ passes through zero. The apparent grid maximum of $1271\times$ sits at the point of smallest $\varepsilon_{\text{step}}$ on the grid, where $\varepsilon_{\text{plateau}}$ is itself below the median (`verify_plateau_gridmap.py`). $\varepsilon_{\text{plateau}}$ is the primary quantity throughout this chapter and the amplification is reported only as a distribution.

Both quantities are defined by a step, so neither has a value until the step is named. $\mathbf{L}$ is tabulated only on the temperature grid, so every step used here is an integer number of grid intervals, and one interval is a factor 1.048113, that is +4.81 % heating or −4.59 % cooling (`validation/plateau_gridmap/plateau_gridmap.txt`, lines 10–11 and 18). The reader should read 4.81 % as an artifact of the choice of 50 logarithmically spaced points between 1 and 10 eV, and nothing else. Every $\varepsilon_{\text{plateau}}$ quoted in this chapter carries the step that produced it.

5.1.2 The closure everyone worries about is exact

Before the maps, the negative result that reframes them.

ε^{QSS} of Eq. (3.47) asks whether the excited states have kept up with the reservoir they actually see. Integrating the full 43-state system through the step and evaluating both errors along the same trajectory gives, at the benchmark point, $\varepsilon^{\text{QSS}} = 8.66 \times 10^{-6}$ against $\varepsilon^{\text{CRE}} = 6.34 \times 10^{-2}$; at $[0, 4]$, the grid point carrying the largest ε^{CRE} anywhere, $\varepsilon^{\text{QSS}} = 6.73 \times 10^{-9}$ against $\varepsilon^{\text{CRE}} = 3.869 \times 10^{-1}$ (`findings_09` §1, from direct integration).

The two differ by four to eight orders of magnitude, and they differ most at the point where the diagnostic error is worst. The adiabatic elimination of Section 3.4 is doing its job to a part in 10^8 precisely where the tabulated answer is off by 39 % per +4.81 % temperature step.

The reason is the spectrum. Section 3.3 found one slow mode and then a band, with nothing in between: at $\mathbf{L}[1, 4]$, the post-step operator at the worst point, the three slowest eigenvalues are -4.29 , -1.60×10^8 and -3.44×10^8 1/s, a gap of 3.7×10^7 (`findings_10` §3.2). An excited population cannot lag a reservoir that moves 10^7 times more slowly than it can respond.

The failure is therefore not in the elimination but in the closure chosen for the eliminated variable. A two-parameter table sets $b_1 = b_1^{\text{CRE}}(T_e, n_e)$, which asserts that the plasma has finished changing. The rest of this chapter is about what that assertion costs.

5.2 The error maps

Figure 5.1 maps $\varepsilon_{\text{plateau}}$ over the operating grid for a heating step of one grid interval, +4.81 % in T_e at fixed n_e . The grid is 50 temperatures by 8 densities, and each cell was evaluated by an exact linear solve rather than by a fit: the superposition $\mathbf{n}_F = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}$ on which Eq. (5.2) rests holds to a relative residual of 3.075×10^{-14} over the whole grid (`plateau_gridmap.txt`, line 19).

Three numbers anchor the map, and each is stated with the set it is an extremum over, because this project has twice been caught quoting a regime-restricted extremum globally.

- Over all 680 analysed pairs, the largest value is $\varepsilon_{\text{plateau}} = 38.69\%$ per +4.81 % step, at $[0, 4]$: $T_e = 1.000 \text{ eV}$, $n_e = 5.18 \times 10^{13} / \text{cm}^3$. It sits on the cold edge of the grid, so the temperature range is truncated there and the script that computes it says so in its own output (`plateau_gridmap.txt`, lines 41–43: `sits on a grid EDGE: True`). Section 5.10 shows that this value is not quantitative and that the point is not a divertor state.
- Over the range this thesis defends, $T_e \geq 2 \text{ eV}$, the largest value is $\varepsilon_{\text{plateau}} = 18.07\%$ per +4.81 % step, at $[15, 3]$: $T_e = 2.02 \text{ eV}$, $n_e = 1.93 \times 10^{13} / \text{cm}^3$ (`verify_eps_gridmap.py`).
- At the benchmark point $[23, 5]$, $\varepsilon_{\text{plateau}} = 6.36\%$ per +4.81 % step (`verify_eps_gridmap.py`, and `verify_plateau_gridmap.py` to five digits: 0.063612).

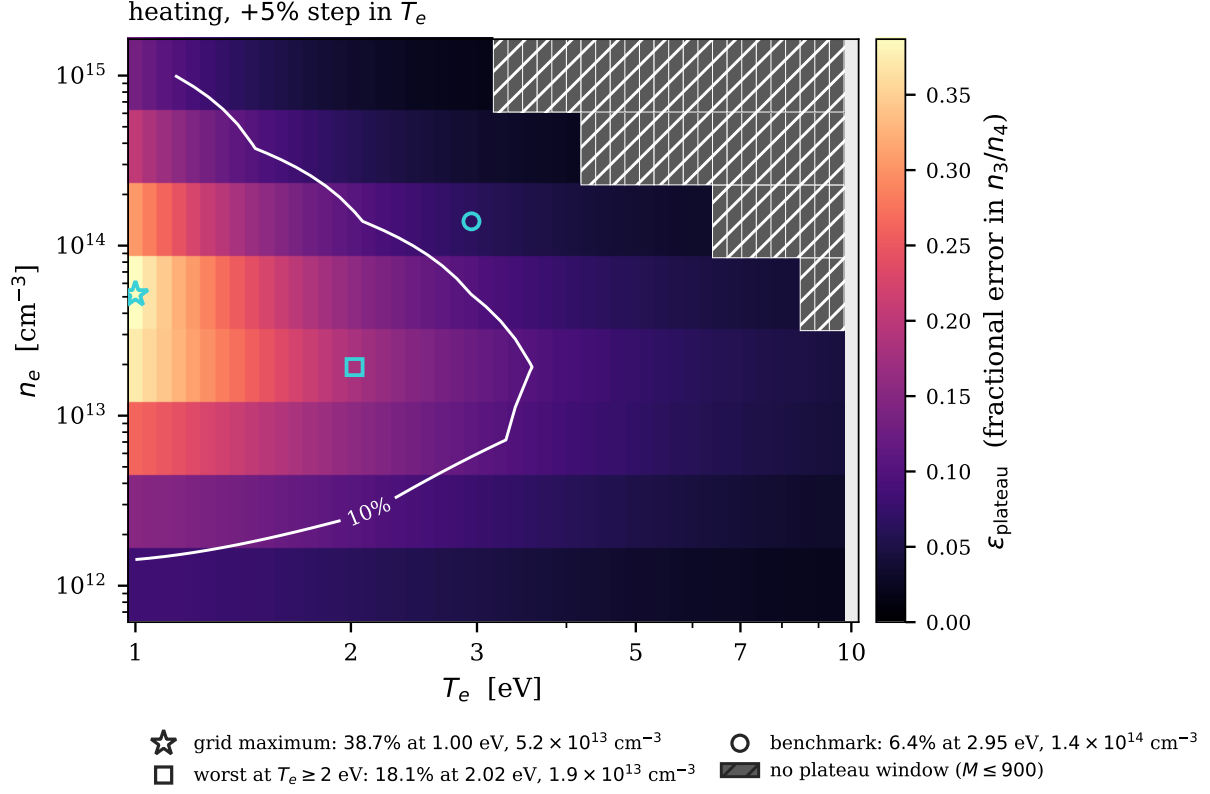
Cooling is not the mirror image of heating, and the asymmetry is mostly a step artifact rather than physics: one grid interval down is -4.59% rather than $+4.81\%$, and once the two are matched per grid point and normalised, the median heat/cool ratio is 1.036 (`thesis_ready` A8). What does not normalise away is the location of the maximum, which moves from $[0, 4]$ under heating to $[1, 3]$ under cooling, where $\varepsilon_{\text{plateau}} = 28.58\%$ and the point is no longer on a grid edge (`plateau_gridmap.txt`, lines 59–61).

5.2.1 The density structure is a broad crest, not a line

Figure 5.2 cuts the map along the density axis. Every temperature row has a single interior maximum in n_e : the error is small at both ends of the density range and largest in between. Section 5.6.1 explains why. What matters first is how sharply that maximum is located, because the temptation is to read a ridge line off a colour map and quote a density.

It is not sharply located. Across the peak column and its two neighbours, a factor 7.2 in density, $\varepsilon_{\text{plateau}}$ varies by only 6 to 22 % depending on the row, while over the full three decades the same row falls by a factor 2.2 to 6.6 (`make_ch5_figures.py`, injected into the caption of Figure 5.2). The crest is real and it is wide-topped. The grid spacing is a factor 2.683 per column, so nothing finer than that is resolved in any case.

This chapter therefore quotes the density of maximum error as the range 7×10^{12} to $5 \times 10^{13} / \text{cm}^3$, which is the pair of grid columns bracketing the modal maximum at $1.93 \times 10^{13} / \text{cm}^3$, and not as a single value. Two further facts support the range and neither supports a point value. The column at $5.18 \times 10^{13} / \text{cm}^3$ carries the row maximum in 3 of the 49 heating rows, all of them the coldest, and $1.93 \times 10^{13} / \text{cm}^3$ carries it in the rest; per-row argmax over rows 0 to 15 is $[4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3]$ (`findings_10` §1.2). And a parabolic fit in $\ln n_e$ through the three columns around the



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Figure 5.1: **Where the quasi-steady-state closure fails.** Fractional error $\epsilon_{\text{plateau}}$ in the $n = 3/n = 4$ shell population ratio committed while the excited manifold has reached partial equilibrium but the ground state has not yet responded, after a +5% step in T_e at fixed n_e — the *heating* direction. Cooling is not the mirror image: the same transition between the two coldest grid temperatures, at $n_e = 1.9 \times 10^{13} \text{ cm}^{-3}$, gives 37.1% heated and 28.6% cooled, and the grid maximum moves from $[0, 4]$ (1.00 eV, $5.2 \times 10^{13} \text{ cm}^{-3}$) to $[1, 3]$ (1.05 eV, $1.9 \times 10^{13} \text{ cm}^{-3}$), where it is 28.6%. Hatched cells have no timescale-separated plateau window ($M \leq 900$, 54 of the heated cells): a plateau value is not defined there and none is plotted. The plain grey strip at the right-hand edge is the hottest temperature row, from which a +5% step cannot reach a new grid index. The three marked cells, keyed below the axes, are the grid maximum (38.7% at $T_e = 1.00 \text{ eV}$, $n_e = 5.2 \times 10^{13} \text{ cm}^{-3}$), the largest value at $T_e \geq 2 \text{ eV}$ (18.1% at $T_e = 2.02 \text{ eV}$, $n_e = 1.9 \times 10^{13} \text{ cm}^{-3}$), and the thesis benchmark point ($T_e = 2.947 \text{ eV}$, $n_e = 1.39 \times 10^{14} \text{ cm}^{-3}$, 6.36%). Below 2 eV the values shown are upper estimates: switching on Lyman trapping divides the grid maximum by 2.5–3.3 depending on the assumed slab thickness, and below 1.15 eV it also moves which density column carries the maximum (Figure 5.4). Nothing on the cold edge of this map is quantitative.

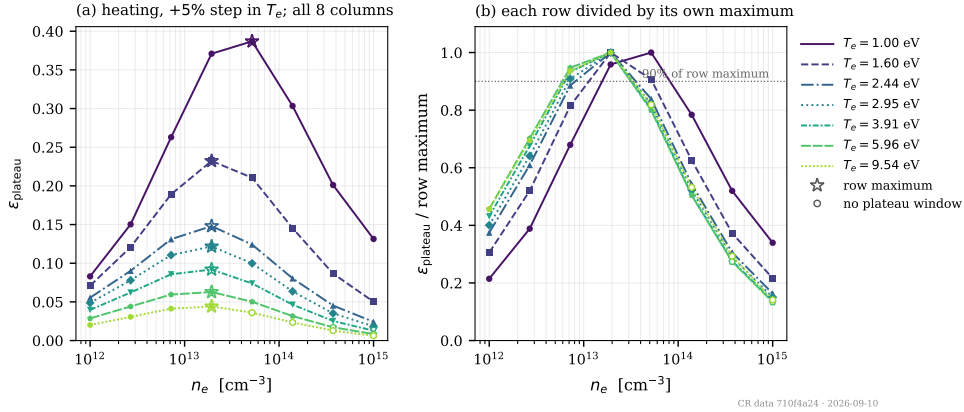


Figure 5.2: **The density dependence, all 8 columns.** $\varepsilon_{\text{plateau}}$ against n_e for 7 temperatures, heating, +5% step. Every density column computed is shown; stars mark each row's maximum, open symbols mark points with no plateau window. Each row has a single interior maximum, but it is *broad and shallow in $\log n_e$* : across the peak column and its two neighbours — a factor 7.2 in density — ε varies by only 6–22%, so the location of the maximum is not resolved to better than about one grid interval, a factor 2.68. Over the full three decades the same row falls by a factor 2.2–6.6: the structure is a real crest in density, but a wide-topped one, and the word ridge should not be read as implying a sharp or well-localised line. The column at $n_e = 5.2 \times 10^{13}$ cm $^{-3}$ carries the row maximum in 3 of 49 heating rows, all of them the coldest; at every warmer temperature the maximum sits at 1.9×10^{13} cm $^{-3}$. Both columns must be reported: quoting either alone misstates where the closure is worst, and below 1.15 eV the position moves across three columns once Lyman trapping is included (Figure 5.4), so no maximum location is claimed there at all.

maximum places its vertex at $1.65 \times 10^{13}/\text{cm}^3$, a 15 % shift off the grid node [*UNVERIFIED: the parabolic vertex 1.65×10^{13} is recorded in `findings_10` §B.3 as prose only. No script in the repository computes it and no artifact under `validation/` stores it. What would close this: a sub-grid fit written into `verify_eps_gridmap.py` that stamps its vertex and residual to `validation/`.*]

Reporting a single density would also overstate what this result is about. The crest belongs to the $n = 3$ and $n = 4$ pair specifically. An H_α/H_γ diagnostic, which uses $n = 3$ against $n = 5$, has its worst density at $7.2 \times 10^{12}/\text{cm}^3$, a factor 2.68 lower (`findings_10` §7.3). The correct form of words is that the H_α/H_β ratio is least reliable near a certain density, never that the Balmer diagnostic is.

There is no interior maximum in temperature anywhere in the defended range. The error falls monotonically from 18.07 % at 2.02 eV to 4.55 % at 8.29 eV along the crest column, both per +4.81 % step (`verify_ridge_mechanism.py`). The apparent peak near 2 eV is the floor of the scope restriction of Section 5.10, not a feature of the plasma.

5.3 Why a percentage is not a result

Every number in the previous section is attached to a +4.81 % temperature step, and 4.81 % is not a physical quantity. It is what one gets from 50 logarithmically spaced grid points between 1 and 10 eV. An examiner is entitled to ask what happens for an ELM that changes the temperature by 10 %, and a chapter whose headline is a percentage has no answer.

The answer, over the grid steps available, is that the error grows fast:

Table 5.1: Grid maximum of $\varepsilon_{\text{plateau}}$ against the size of the imposed temperature step, in grid intervals. Values from `verify_eps_gridmap.py` via `CH5_EVIDENCE.md` §1; the one-interval and four-interval maxima sit at $[0, 4]$ and $[0, 5]$ respectively, both on the cold edge, and neither is quantitative (Section 5.10). The middle row is recorded in `pivot_decision.md` §2 without a location.

step (grid intervals)	step in T_e	grid maximum of $\varepsilon_{\text{plateau}}$
1	+4.81 %	38.7 %
2	+9.85 %	87.4 %
4	+20.68 %	190.4 %

A factor 4.3 in step size moves the reported error by a factor 4.9. Quoted bare, 38.7 % is a statement about a grid, and the sentence “*the quasi-steady-state error reaches 39 %*” is not falsifiable because it does not say what it is 39 % of.

The way out is not a better choice of step. It is to report the response function instead of one of its values, which is what Section 5.4 does. That the error more than doubles between the first and second rows of Table 5.1, while the quantity introduced there moves by 2 %, is the reason for preferring the second object to the first.

5.4 The exact response to a finite step

5.4.1 The split has exactly two channels

Everything in this section and the two that follow rests on a partition made in Chapter 3: one slow state, the ground state $1s$, and 42 fast ones. That partition deserves an argument rather than an assertion, because there is an obvious candidate for a second slow state.

The $2s$ level cannot decay to the ground state by emitting one dipole photon, because both states have zero orbital angular momentum and a single dipole photon must carry one unit of it away. It decays instead by emitting two photons at once, at 8.229 1/s , a lifetime of 0.12 s against roughly a nanosecond for the $2p$ level beside it. If $2s$ held population on the slow timescale it would be a second reservoir: the supply sum $\mathbf{n}_F = \mathbf{a} n_{1s} + \mathbf{c} n_{\text{ion}}$ of Eq. (3.33) would gain a third term, the logistic of Eq. (3.40) would not merely be displaced but would change shape, and the bound of Section 5.5 would not follow.

In a plasma $2s$ is not depopulated by radiation. Proton impact mixes it into $2p$, which then decays promptly, and that route dominates the $2s$ loss budget by 99.99% . At the thinnest and coldest point on the grid, $T_e = 1\text{ eV}$, $n_e = 10^{12}/\text{cm}^3$, the total $2s$ loss rate is $|L_{2s,2s}| = 1.2564 \times 10^9\text{ 1/s}$ against a slow eigenvalue $|\lambda_0| = 0.0149\text{ 1/s}$, a separation of 8×10^{10} .

That argument leans on a proton-impact rate coefficient, so it is worth having one that does not. Delete the ℓ -mixing entirely and leave $2s$ with nothing but two-photon emission: 8.229 1/s is still 553 times faster than $|\lambda_0|$ at the same corner. **On this grid $2s$ cannot be slow under any assumption about ℓ -mixing whatever.** The density at which it would become a second reservoir is near $7 \times 10^3/\text{cm}^3$, nine orders of magnitude below the bottom of the operating range.

The direct spectral test agrees and supplies the number to quote. Over all 400 grid points the smallest ratio of $|\text{Re } \lambda|$ over the excited block $\mathbf{L}_{FF}$ to $|\lambda_0|$ is 86.5, at $T_e = 10\text{ eV}$, $n_e = 10^{15}/\text{cm}^3$, and no slow eigenvector anywhere on the grid carries dominant $2s$ weight. There is no near-degenerate second slow mode. That point and that value are the same stress point and essentially the same quantity as the timescale-separation minimum $M = 86.8$ of Eq. (3.18), computed from the excited block rather than from the full matrix, so the two are a consistency check on each other and not two results. **The worst-case separation is 86.5, and that is the figure that describes the grid;** the benchmark value $M = 9982$ describes one point and overstates the margin by two orders of magnitude.

*[SOURCE REQUIRED: the four measurements in this subsection come from the collisional-radiative audit completed 10 September 2026. The script that produced them and its stamped artifact under **validation/** must be named here before submission; no other statement in this chapter depends on them.]*

5.4.2 A better variable than the departure coefficient

Chapter 3 worked in $b_1 = n_{1s}/(Z_1 n_e n_{\text{ion}})$, because that is the variable in which the observable's response has the closed form of Eq. (3.35). Here it is better to use the reservoir itself,

$$u \equiv \frac{n_{1s}}{n_{\text{ion}}} = b_1 Z_1(T_e) n_e, \quad (5.3)$$

a dimensionless ground-state population per unit ion density. At fixed (T_e, n_e) the two are proportional, so $d \ln u = d \ln b_1$ and the sensitivity of Section 3.5.4 carries over unchanged. The reason for the change of variable is that a temperature step moves Z_1 : comparing a

pre-step b_1 with a post-step one would mix a change in the reservoir with a change in the Saha–Boltzmann normalisation, and u has no normalisation to move. That confusion is not hypothetical. An earlier version of the analysis reported in this section made exactly it, and the resulting contamination in $\Delta \ln b_1$ was comparable to the quantity being measured.

Assumption. Equation (5.3) treats n_{ion} as a fixed reservoir, unchanged by the step, as throughout this thesis (Section 3.1). This removes the freedom for the ionisation balance to respond, and it is what makes u numerically equal to the ground-state population in the solved system. Chapter 6 returns to what a mobile ion population would change, and bounds it: the total bound population per unit n_{ion} at the benchmark is 8.925×10^{-4} , so freezing n_{ion} costs at most 0.09% (`findings_10 §B.4`).

5.4.3 The decomposition

Equation (3.35) makes the post-step observable a ratio of two affine functions of u , so the logarithmic sensitivity of Section 3.5.4 is $d \ln R / d \ln u = f_3 - f_4$, the difference of the two ground-fed fractions. Integrating between the frozen and the equilibrated reservoir gives, with no approximation,

$$\ln \frac{R^{\text{PE}}}{R^{\text{CRE}}} = \int_{\ln u^+}^{\ln u^-} (f_3 - f_4) d \ln u = \bar{S} \Delta \ln u, \quad (5.4)$$

where $\Delta \ln u = \ln(u^-/u^+)$ is the reservoir displacement and

$$\bar{S} \equiv \frac{1}{\Delta \ln u} \int_{\ln u^+}^{\ln u^-} (f_3 - f_4) d \ln u \quad (5.5)$$

is the mean sensitivity over the interval the reservoir traverses. Both factors are dimensionless. Since $f_3, f_4 \in [0, 1]$ by Section 3.5.4, and $\bar{S}$ is an average of their difference, $|\bar{S}|$ is bounded by the peak sensitivity of Eq. (3.42) and so is strictly less than unity everywhere. Exponentiating gives the plateau error:

$$\boxed{\varepsilon_{\text{plateau}} = \left| e^{\bar{S} \Delta \ln u} - 1 \right|} \quad (5.6)$$

A single scalar error has become a product of two quantities that can be measured independently: how strongly the observable responds to its reservoir, and how far the reservoir moved. Nothing in Eq. (5.4) says which of them varies across the operating map, and Section 5.6 determines that.

5.4.4 The reservoir gain, and why it is the step-independent object

The second factor still contains the step. Splitting it,

$$\Delta \ln u = G \Delta \ln T_e, \quad G \equiv \frac{d \ln u}{d \ln T_e} = \left. \frac{\partial \ln u^{\text{CRE}}}{\partial \ln T_e} \right|_{n_e}, \quad (5.7)$$

defines the *reservoir gain* G : the factor by which a fractional change in electron temperature is amplified into a fractional change in the equilibrium ground-state population. It is dimensionless, it is negative throughout the grid because heating ionises the ground

state away, and unlike $\varepsilon_{\text{plateau}}$ it is a property of the operating point rather than of the experiment done to it. Substituting into Eq. (5.6),

$$\varepsilon_{\text{plateau}} = \left| \exp(\bar{S} G \Delta \ln T_e) - 1 \right|, \quad \lim_{\Delta \ln T_e \rightarrow 0} \frac{d\varepsilon_{\text{plateau}}}{d \ln T_e} = |\bar{S} G|. \quad (5.8)$$

The small-step limit on the right is the useful form: it is the diagnostic error per unit fractional temperature excursion, with the excursion divided out. At the benchmark point it is 1.31, at [15, 3] it is 3.54 and at [0, 4] it is 6.96, all computed as $|\bar{S} G|$ from the values in Table 5.2. Read the middle one as: in the worst conditions this thesis defends, a 1 % temperature transient costs the Balmer inversion about 3.5 % in the line ratio.

Table 5.2: The reservoir gain G at three grid points, for steps of one, two and four grid intervals. G is stable to 5 to 6 % across a fourfold change in step size, while $\varepsilon_{\text{plateau}}$ itself changes by a factor 4.9 over the same range (Table 5.1). Values from `pivot_decision.md` §4; see the provenance note below, which is not optional reading.

point	(T_e, n_e)	$k = 1$	$k = 2$	$k = 4$	spread
benchmark [23, 5]	2.947 eV, $1.389 \times 10^{14}/\text{cm}^3$	−5.736	−5.634	−5.439	5 %
crest [15, 3]	2.02 eV, $1.93 \times 10^{13}/\text{cm}^3$	−7.766	−7.617	−7.333	6 %
cold edge [0, 4]	1.00 eV, $5.18 \times 10^{13}/\text{cm}^3$	−14.434	−14.131	−13.551	6 %

Provenance, and what is not established. The nine values in Table 5.2 are recorded in `outputs/pivot_decision.md` §4 as measured on 10 September 2026 against the canonical matrix. No script in the repository computes them and no artifact under `validation/` stores them. The three $k = 1$ entries can be checked against artifacts that do exist, and all three pass. One grid interval is $\Delta \ln T_e = 0.046984$, so $|G| \Delta \ln T_e$ predicts $|\Delta \ln u| = 0.2695$, 0.3649 and 0.6781 at the three points, against measured values of 0.2694 (recovered below from $\varepsilon_{\text{plateau}}$ and $\bar{S}$ at the benchmark), 0.3649 (`verify_ridge_mechanism.py`, quoted in `CH5_EVIDENCE.md` §4.4) and 0.6783 (`plateau_gridmap.txt`, line 41). *[UNVERIFIED: the $k = 2$ and $k = 4$ columns, which are the entire basis of the step-independence claim, have no artifact behind them. What would close this: a script that recomputes u^{CRE} at $[i, j]$ and $[i+k, j]$ for $k = 1, 2, 4$ across the grid and stamps G to `validation/`, which would also supply the grid-wide G map that Eq. (5.8) needs in order to be usable by a reader at an arbitrary operating point.]*

5.4.5 A worked number at the benchmark

The chain closes arithmetically. At [23, 5] the mean sensitivity over the step is $|\bar{S}| = 0.2288$ (`findings_10` ADDENDUM B.2, which lists it alongside $\Delta = 1.9403$ and $\tanh(\Delta/4) = 0.4503$ at that point), the gain is $G = -5.736$, and one grid interval is $\Delta \ln T_e = 0.046984$. Then

$$\bar{S} G \Delta \ln T_e = 0.2288 \times 5.736 \times 0.046984 = 0.06166, \quad e^{0.06166} - 1 = 0.0636,$$

against the 0.063612 that `verify_plateau_gridmap.py` obtains for the same point by an exact linear solve. Three quantities from three different artifacts multiply to the recorded error to four significant figures. The arithmetic is performed here in the text and is not itself a run.

5.4.6 Use the exact form, not the linearisation

The first-order form $\varepsilon_{\text{plateau}} \approx |\bar{S} \Delta \ln u|$ is what one would write down first, and it is adequate for a small step and wrong for a large one. At the four-interval step it returns 59.4% where Eq. (5.6) gives 81.1% at [15, 3] (`verify_ridge_mechanism.py`). The exponential costs nothing and this chapter uses it throughout.

It would be convenient if the linearisation erred in one direction only, so that it could be quoted as a bound. It does not, and an earlier draft of this work claimed that it did. Which way it errs depends on which linearisation is meant. For the *endpoint* form actually computed at `verify_plateau_gridmap.py`, line 212, the ratio of measured to predicted error over the heating map runs 0.8417 to 1.3891 with median 1.0310: the linearised expression understates in 219 of 392 heating steps and overstates in the rest, and at the benchmark it overstates, with ratio 0.9459 (`findings_10` §B.3). **Any statement about the linearisation must name which linearisation it is about.**

A final caution about what the decomposition can and cannot be used for. Equation (5.4) is an identity, so comparing its two sides tests nothing: a check that $\varepsilon_{\text{plateau}}$ equals $|e^{\bar{S}\Delta \ln u} - 1|$ can fail only through arithmetic error. What the identity permits is an *attribution*. The two factors are separately measurable, so one may ask which of them carries the structure the maps show. That is a weaker kind of statement than a derivation, and it is the kind made in Section 5.6.

5.5 A bound that no atomic dataset can exceed

Section 3.5.4 showed that in the logarithmic reservoir variable both ground-fed fractions are the same unit-width logistic curve, displaced by $\Delta = \ln[(a_3/a_4)/(c_3/c_4)]$, and that their difference therefore has the closed-form maximum $\max |f_3 - f_4| = \tanh(|\Delta|/4)$ of Eq. (3.42). Since $\bar{S}$ of Eq. (5.5) is an average of $f_3 - f_4$ over an interval, it inherits the bound:

$$|\bar{S}| \leq \tanh\left(\frac{|\Delta|}{4}\right) < 1. \quad (5.9)$$

Three properties make this worth stating separately from the algebra that produced it.

It does not depend on the atomic data. The $\tanh$ is a consequence of the logistic having unit width, and the width is unity because the excited populations are exactly affine in n_{1s} , which is a structural property of the linear system and not of any rate coefficient (`findings_10` ADDENDUM B.1). Every property of the 42-state network enters only through where the two curves sit, that is through Δ . Consistent with that, Δ is invariant under rescaling the n_{1s}/n_{ion} normalisation, verified exactly under factors of 10^{10} and 10^{-7} , and is insensitive to the ℓ -weighting used to bundle a shell: population sum, p -states, d , s and statistically weighted give 1.94561, 1.94474, 1.94447, 1.94471 and 1.94574 at the benchmark, a spread of 0.07% (`findings_10` §B.5). The individual switching points x_3 and x_4 and the fractions f_p are *not* invariant under that rescaling; only their separation is.

Its physical content is a statement about amplification. Because $|d \ln R / d \ln b_1| < 1$ everywhere, integrating over any excursion gives $|\Delta \ln R| \leq |\Delta \ln b_1|$. A line ratio can never respond to the ground-state reservoir faster, in relative terms, than the reservoir itself moves. A stale reservoir cannot be amplified in logarithmic space, only attenuated.

The bound is loose in general and tight at the crest. At the benchmark $|\bar{S}| = 0.2288$ against a cap of 0.4503, so the sensitivity is at 51% of what the atomic structure permits. At [23, 3], one column into the crest, $|\bar{S}| = 0.4261$ against a cap

of 0.4394, that is 97% (both from the table in `findings_10` §B.2; the percentages are arithmetic on those two columns). Where the error is largest, the diagnostic is running at essentially its structural maximum sensitivity to the reservoir, and no refinement of the rate coefficients can make that worse.

5.5.1 What the bound does not constrain

A diagnostician does not suffer an error in R . They measure R and invert it for (T_e, n_e) , and Eq. (3.43) makes the inferred temperature error $\Delta \ln T_e = (d \ln R / d \ln b_1) / (d \ln R / d \ln T_e) \Delta \ln b_1$. Equation (5.9) constrains the numerator. It says nothing about the denominator, and Chapter 3 deferred the denominator to this chapter.

The denominator vanishes. Scanning T_e at $n_e = 5.18 \times 10^{13} / \text{cm}^3$, $\varepsilon_{\text{step}}$ of Eq. (5.1) passes through a minimum of 4.9×10^{-5} near $T_e = 6.87 \text{ eV}$, with values 7.18×10^{-4} , 3.79×10^{-4} , 4.9×10^{-5} , 2.74×10^{-4} and 5.89×10^{-4} at 6.251, 6.551, 6.866, 7.197 and 7.543 eV (`verify_plateau_gridmap.py`). Since $\varepsilon_{\text{step}}$ is the fractional distance between the two tabulated answers either side of the step, a zero means R^{CRE} is stationary in temperature there: $\partial \ln R / \partial \ln T_e = 0$.

The $n = 3/n = 4$ ratio is locally non-invertible for temperature near 6.9 eV at this density. There, a bounded error in the observable produces an unbounded error in the inferred temperature, and the plateau error is not the relevant failure mode at all: the diagnostic has no temperature resolution to lose. This is a separate failure from the one the rest of the chapter maps, it is arguably the more severe of the two for a practitioner, and it is visible only because $\varepsilon_{\text{step}}$ was computed and kept rather than discarded as uninteresting.

The scan above is one density column. *[UNVERIFIED: whether the locus $\partial \ln R / \partial \ln T_e = 0$ crosses the operating range at other densities has not been mapped. What would close this: evaluate $\varepsilon_{\text{step}}$ on the full grid, which `verify_plateau_gridmap.py` already does, and contour its minimum.]*

5.6 Where the error is largest, and why

The maps of Section 5.2 show where the diagnostic error is large. They do not say why, and a map without a mechanism is a lookup table that cannot be extrapolated past the grid that produced it.

The answer has two halves and they disagree. The structure along the density axis and the structure along the temperature axis have different causes, and reading them as one effect, which the appearance of a single crested surface invites, gets both wrong.

5.6.1 Across density: the operating point, not the span

Evaluating both factors of Eq. (5.4) at every grid point settles which of them locates the crest (`verify_ridge_mechanism.py`, over $T_e = 2\text{--}9 \text{ eV}$ and all eight density columns). The column maximising $\varepsilon_{\text{plateau}}$ is the column maximising $|\bar{S}|$ in 32 of 32 temperature rows for a one-interval step and 31 of 31 for a four-interval step; the column maximising $|\Delta \ln u|$ coincides with the error maximum in 0 of 32 and 0 of 31.

Coincidence counts are weak evidence on their own, because a quantity that barely varies still has its maximum somewhere. The amplitudes settle it. Across the three decades of density $|\bar{S}|$ varies by a factor 7.11 and $|\Delta \ln u|$ by a factor 1.01. **A quantity that**

changes by one per cent cannot produce a crest in which the error changes by a factor of seven, wherever its own maximum falls.

Table 5.3: Density profiles of the plateau error and of the two factors of Eq. (5.4), each normalised to its own row maximum and then taken as the median over the 32 temperature rows with $T_e = 2\text{--}9\text{ eV}$, for a one-interval temperature step. From `verify_ridge_mechanism.py`.

n_e (1/cm ³)	10 ¹²	2.7·10 ¹²	7.2·10 ¹²	1.9·10 ¹³	5.2·10 ¹³	1.4·10 ¹⁴	3.7·10 ¹⁴	10 ¹⁵
$\varepsilon_{\text{plateau}}$	0.437	0.683	0.937	1.000	0.807	0.512	0.280	0.138
$ \bar{S} $	0.445	0.689	0.938	1.000	0.813	0.520	0.285	0.141
$ \Delta \ln u $	0.997	0.995	0.993	0.991	0.990	0.992	0.996	1.000

The first two rows of Table 5.3 agree to 2.2% at every column and the third is flat to one per cent. The density structure of the plateau error is the density structure of the observable’s sensitivity to its reservoir.

That settles *which factor*. It does not settle why that factor has an interior maximum, and here the obvious explanation is wrong. Equation (5.9) caps the sensitivity at $\tanh(|\Delta|/4)$, so the natural guess is that Δ itself peaks near $2 \times 10^{13}/\text{cm}^3$: that the two supply channels are most widely separated there, and the sensitivity follows. That guess was written down in advance by an independent re-derivation of Chapter 3, together with a structural argument for why Δ should collapse at high density, namely that in local thermodynamic equilibrium (LTE) both channels would give the same Boltzmann shape and Δ would go to zero.

It is falsified. Δ is nearly flat across the density axis: 0.9655 at $10^{12}/\text{cm}^3$, 1.9403 at $1.39 \times 10^{14}/\text{cm}^3$ and 1.9368 at $10^{15}/\text{cm}^3$, at $T_e = 2.947\text{ eV}$. Above $7 \times 10^{12}/\text{cm}^3$ the cap $\tanh(|\Delta|/4)$ spans only 0.394 to 0.450, a 14% range. The grid never reaches LTE, and because the model is open, with $\mathbf{S}_F$ an externally imposed source not tied to n_{ion} by the Saha relation, the two channels cannot merge by construction (`findings_10` ADDENDUM B.2).

What sweeps instead is the operating point. Table 5.4 puts the two side by side: while the cap moves by 14%, the position of the CRE reservoir relative to the sensitivity peak, $\ln(u^{\text{CRE}}/u^{\text{peak}})$, runs from +1.29 to −3.29, and $|\bar{S}|$ tracks it.

Table 5.4: Why the sensitivity has an interior maximum in density, at $T_e = 2.947\text{ eV}$. The cap $\tanh(|\Delta|/4)$ is nearly constant while the operating point $\ln(u^{\text{CRE}}/u^{\text{peak}})$ sweeps through zero. $\varepsilon_{\text{plateau}}$ is for a one-interval, +4.81% step. From the independent re-derivation recorded in `findings_10` ADDENDUM B.2.

j	n_e (1/cm ³)	Δ	$\tanh(\Delta /4)$	$\ln(u^{\text{CRE}}/u^{\text{peak}})$	$ \bar{S} $
0	1.00×10^{12}	0.9655	0.2368	+1.2877	0.1754
2	7.20×10^{12}	1.6662	0.3940	+0.3842	0.3884
3	1.93×10^{13}	1.8857	0.4394	−0.2487	0.4261
5	1.39×10^{14}	1.9403	0.4503	−1.7695	0.2288
7	1.00×10^{15}	1.9368	0.4496	−3.2910	0.0668

The maximum sits where $\ln(u^{\text{CRE}}/u^{\text{peak}})$ changes sign. The physical reading is direct. Both f_3 and f_4 rise from zero to unity as the reservoir grows, and they switch

from recombination-fed to ground-fed at different reservoir strengths, so their difference vanishes in both supply limits and peaks between them. The sensitivity curve is fixed. Raising the density slides the plasma’s own CRE reservoir along it, from the ground-fed side at low density, through the mixed-supply region, to the recombination-fed side at high density. The crest is where that slide crosses the peak.

Two consequences follow, and both matter. First, this is a statement about where the operating point sits, not about how sensitive the diagnostic could be, and it replaces the weaker description of the density axis as sensitivity-dominated: both are true, but only the first says why. Second, it forbids the inference that the effect disappears at high density. The cap does not close; the plasma merely walks off the peak. On a grid extending further in density it would walk back on.

5.6.2 Three independent tests of the mechanism

The account above predicts more than the shape of one curve, and each prediction can fail.

The crest must move with the shell pair, and by a predicted amount. The reservoir strength at which a shell switches from recombination-fed to ground-fed is a version of Griem’s LTE criterion [19, 20], which places the boundary between collisionally coupled and radiatively decoupled levels at a density scaling as $n^{-17/2}$ in the principal quantum number. Moving the diagnostic from the (3, 4) pair to the (4, 5) pair should therefore move the crest by $(5/4)^{8.5} = 6.7$. Measured: 7.2. The argmax of $|f_n - f_{n+1}|$ at $T_e = 2.02$ eV runs $3.73 \times 10^{14}/\text{cm}^3$ for (2, 3), $1.93 \times 10^{13}/\text{cm}^3$ for (3, 4) and $2.68 \times 10^{12}/\text{cm}^3$ for (4, 5), with (5, 6) and above falling off the bottom of the grid (`verify_ridge_mechanism.py`).

The absolute density must match, not just the ratio. Griem’s LTE criterion evaluated at $n = 4$ and $T_e = 2$ eV gives $2.05 \times 10^{13}/\text{cm}^3$ against a measured crest at $1.93 \times 10^{13}/\text{cm}^3$, a 6 % difference.

An independent diagnostic must put the boundary level in the same place. `verify_boundary_descent.py` computes, without reference to f_3 or f_4 , the effective principal quantum number at which the collisional and radiative rates balance. At the crest density it returns $\bar{n} = 3.08$ to 3.25 at every temperature on the grid (`validation/boundary_descent.csv`), that is, between $n = 3$ and $n = 4$, which is exactly the pair whose supply fractions differ most.

What would have refuted the mechanism, and did not occur: the crest failing to move with the shell pair, or moving the wrong way. A competing explanation was refuted outright. The suggestion that both shells become ground-fed at high density because collisional coupling dominates predicts $f_3, f_4 \rightarrow 1$; measured at $T_e = 2.947$ eV, $n_e = 10^{15}/\text{cm}^3$, they are 0.071 and 0.011, so both shells become *continuum*-fed instead (`findings_10 §2`).

5.6.3 Alternatives that would produce the same appearance

An interior maximum on a coarse grid can be manufactured in several ways, and the attribution above does not by itself exclude them.

Grid resolution. The crest occupies one density column and the grid is spaced by a factor 2.683, so its location is known only to that factor. What is established is that the maximum is interior, not at either end of the density range, and that it does not move

between temperature rows above 1.15 eV. Refining the density grid requires rebuilding the rate matrix and revalidating every number in Chapter 3 against the new one; future work could determine the location more precisely.

Sub-grid drift. A parabolic fit in $\ln n_e$ puts the peak at $3.66 \times 10^{13}/\text{cm}^3$ at 1 eV and $1.51 \times 10^{13}/\text{cm}^3$ at 9.5 eV, a range of $2.56\times$, which drifts *downward* with temperature while Griem’s criterion predicts an upward drift as $\sqrt{T_e}$, a factor 3.2 (`findings_10` §7.2). The whole drift is smaller than one grid interval and therefore unresolved, so the crest should be described as constant to within a factor 2.68 over 1 to 10 eV, never as fixed. The likely reason for the sign is that the neutral fraction n_{1s}/n_{ion} collapses across the grid, dragging the supply crossover down in density faster than the Griem boundary drags it up, which is consistent with Section 5.6.1 but has not been separated from it.

Step-size convention. A crest whose position depended on the size of the imposed step would be an artifact of that choice. It does not: the error maximum falls on the same density column for a one-interval and a four-interval step in 31 of the 31 temperature rows both can reach, and so does the maximum of $|\bar{S}|$ (`verify_ridge_mechanism.py`). This tests steps of fixed *fractional* size; a step of fixed absolute size in electron-volts spans a different number of grid intervals at each temperature and is a different experiment, not tested here.

Truncation. The model resolves $n \leq 8$ and bundles $n = 9\text{--}15$. The mechanism turns on the difference between the $n = 3$ and $n = 4$ supply fractions, both fully resolved, but those fractions depend on the recombination flux cascading from above, which the bundling approximates. [*UNVERIFIED: the sensitivity of the crest to n_{max} has not been tested.* `src/validation/verify_bundling_psm20.py` exists for this purpose and has never been run.]

The population set. Two of the scripts used here apply different filters: `verify_plateau_gridmap.py` excludes the 54 points with no plateau window, and `verify_ridge_mechanism.py` does not. All 54 lie at $j \geq 4$, so the ridge script’s statistics for the two highest density columns include points at which the plateau state being computed does not physically exist (`findings_10` §B.3). The attribution of Section 5.6.1 does not depend on those columns, but a reader recomputing the profile from the raw output should apply the filter.

5.6.4 Across temperature: a displacement effect

The temperature dependence has the opposite explanation, and this is where treating the two axes alike would go wrong.

Along the crest column the plateau error falls monotonically from 18.07 % at $T_e = 2.02$ eV to 4.55 % at 8.29 eV, a factor 3.72, both per +4.81 % step. Decomposed through Eq. (5.4), $|\Delta \ln u|$ contributes a factor 2.72 of that fall and $|\bar{S}|$ only 1.29; in logarithmic terms the displacement carries 76 % of the decay and the sensitivity 19 %. At the four-interval step the split is 72 % and 13 % (`verify_ridge_mechanism.py`).

The observable therefore does not become much less sensitive as the plasma heats. **The ground state simply moves less.** In the language of Section 5.4, the reservoir gain $|G|$ falls with temperature, from 14.4 at 1.00 eV to 5.7 at 2.95 eV (Table 5.2), and a step of fixed fractional size therefore displaces the reservoir progressively less.

Why the gain behaves this way is not established here. $|\Delta \ln u|$ is flat to one per cent across three decades of density while varying by a factor 2.7 across temperature, which implies that $\partial \ln n_{1s} / \partial \ln T_e$ at collisional–radiative equilibrium is controlled by temperature alone. [*MECHANISM NOT ESTABLISHED: this is an observation from*

verify_ridge_mechanism.py, not a derived result. A natural candidate is the Arrhenius factor in the electron-impact ionisation rate out of the ground state, whose logarithmic temperature derivative is $\chi_H/T_e = 4.6$ at the benchmark against a measured $|G| = 5.7$, but the comparison has not been carried through and the two need not be equal.]

5.6.5 Three explanations this work does not support

Three readings of the crest are natural and do not survive the numbers. Each is recorded because it is among the first a reader is likely to propose, and one of them was asserted in an earlier draft of this thesis.

It is not detachment. The crest sits at $1.93 \times 10^{19}/\text{m}^3$: a factor 5.2 below the density band that published edge modelling associates with a detached ITER divertor [41], and a factor 1.6 below the modelled upstream separatrix density [2]. It falls between the two, in a region neither source characterises. Two further facts make the identification untenable rather than merely unsupported. At the citable detached band, $n_e = 10^{14}$ to $10^{15}/\text{cm}^3$ with $T_e \geq 2\text{ eV}$, the plateau error is at most 10.4 % per +4.81 % step and the ELM-averaged bound of Section 5.8 never reaches 10 %, peaking at 7.2 %. And the crest location is proportional to the assumed neutral density, which in this model is set by a local ionisation balance with no transport and no recycling, whereas in a divertor it is set by recycling; Chapter 6 measures that proportionality and finds one decade in n_{1s} moves the crest by roughly one decade in n_e . The crest is reproducible. Its identification with a named divertor operating regime is not supported by anything in this work.

It is not a peak in temperature. The error decreases monotonically with temperature throughout the defended range, as Section 5.6.4 shows. The apparent ridge line at 2 eV in Figure 5.1 is the edge of the scope restriction.

It is not simply where b_1^{CRE} meets the sensitivity maximum. That reading accounts for the density structure but fails on the temperature axis: at the four-interval step the ratio $b_1^{\text{CRE}}/b_1^{\text{peak}}$ passes through unity at $T_e = 4.7\text{ eV}$, where the error is 33 % and falling monotonically (*verify_plateau_gridmap.py*). The endpoint of the reservoir trajectory is the wrong object: Eq. (5.5) shows that what enters is the mean sensitivity over the whole interval traversed.

5.7 Does the transient visit the plateau?

Everything so far is algebraic. R^{PE} was built by holding b_1 at its pre-step value and solving the post-step excited block, which is a construction, not an observation. If the real trajectory never lingers near that state, the map measures a fiction.

It does linger. Propagating the full 43-state system through the step with `scipy.linalg.expm` and tracking $\varepsilon^{\text{CRE}}(t)$ (*verify_plateau_bridge.py*) gives, at the benchmark point and at [15, 3]:

- The excited manifold reaches its instantaneous QSS state after $1.94 \tau_{\text{relax}}$ at the benchmark and $3.72 \tau_{\text{relax}}$ at [15, 3].
- Once settled, the state deviation from QSS is 1.3×10^{-6} and 8.1×10^{-8} respectively. These are the settled values, not the maxima over the plateau interval, which are 471 and 1721 times larger because the interval begins in the tail of the fast transient. An external review of this work quoted the maxima; the settled column is the one that supports the claim that the manifold has equilibrated.

- Over the plateau, $\varepsilon^{\text{CRE}}(t)$ departs from the analytic $\varepsilon_{\text{plateau}}$ by 3.33 % at the benchmark and 1.28 % at [15, 3]. That flatness is what licenses quoting a single $\varepsilon_{\text{plateau}}$ instead of a time-dependent error, and it is measured rather than assumed.

The propagator itself was checked by running `expm` against `expm_multiply`, two independent algorithms, which agree to 1.13×10^{-10} and 3.69×10^{-12} . Two other checks that were run are *not* evidence and are recorded here so that nobody re-runs them: the ODE residual is identically zero by construction, since $d\mathbf{n}/dt$ and $\mathbf{L}^+\mathbf{n} + \mathbf{S}^+$ are the same expression evaluated twice, and the semigroup check is near-tautological for a scaling-and-squaring implementation of `expm`, which is why it returned exactly zero at both points.

Scope. This establishes the dynamic chain full transient $\rightarrow R^{\text{QSS}}(u(t)) \rightarrow R^{\text{PE}}$ at two grid points, not grid-wide. The density structure of Section 5.6.1 comes from a grid-wide algebraic calculation. A grid-wide dynamic claim would need the bridge test run at every point, which has not been done.

One honest record of a failed preregistration belongs here. The first version of the plateau criterion required $\varepsilon^{\text{QSS}} < 2 \times 10^{-3}$ throughout $20 \tau_{\text{relax}}$ to $0.02 \tau_{\text{QSS}}$, and it *failed* at [15, 3], reaching 2.778×10^{-3} at the very end of the window. The criterion was retained unchanged and the window was not shortened. It failed because the ground state had begun its expected motion while the tracking error stayed at 6.8×10^{-7} : the system had not left the QSS manifold, it had moved *along* it. The replacement interval definition, used above, was not preregistered and is not described as such.

5.8 How long the error lasts, and what an ELM averages

A spectrometer integrates. If the error decayed on nanoseconds it would be invisible in a 100 μs exposure, so the decay law matters as much as the amplitude.

The error decays as a single exponential with time constant τ_{QSS} . Fitting $\varepsilon^{\text{CRE}}(t)$ over the decay gives $\tau_{\text{eff}}/\tau_{\text{QSS}} = 1.0102$ at the benchmark and 1.0656 at [15, 3], with half-decay at 0.7002 and 0.7386 τ_{QSS} against $\ln 2 = 0.6931$ for a pure exponential (`verify_plateau_bridge.py`). In physical units the half-life is 12.95 μs at the benchmark and 1.10 ms at [15, 3]. Unlike the plateau duration, which is set by whatever tolerance defines the end of the plateau, the half-life carries no threshold dependence.

That gives a time-average over an event of duration τ_{drive} :

$$\left\langle \varepsilon^{\text{CRE}} \right\rangle_{\tau_{\text{drive}}} \geq \varepsilon_{\text{plateau}} \frac{\tau_{\text{QSS}}}{\tau_{\text{drive}}} \left(1 - e^{-\tau_{\text{drive}}/\tau_{\text{QSS}}} \right), \quad (5.10)$$

which tends to $\varepsilon_{\text{plateau}}$ as $\tau_{\text{drive}} \rightarrow 0$ and to zero as $\tau_{\text{drive}} \rightarrow \infty$, both as they should. Equation (5.10) was constructed as a conservative lower bound, on the reasoning that the observable cannot decay faster than the slowest mode. Direct 43-state eigen-propagation shows it is much better than that: the ratio of the true 100 μs average to the bound is between 0.9999 and 1.0066 at seven points including the worst on the grid (`findings_10` §3.2). The reason is the spectral gap of Section 5.1.2. With nothing between λ_0 and λ_1 , there is no intermediate mode for the observable to decay on, so the bound is essentially the answer. It should be described as a near-exact estimate whose error is under 0.7 % at every point tested, not as a loose bound.

5.8.1 The census, inside the defensible scope

Counting the pairs whose ELM-averaged error exceeds 10% at $\tau_{\text{drive}} = 100 \mu\text{s}$, with a one-interval step (`validation/divertor_map/divertor_map.csv`):

Table 5.5: ELM-averaged breakdown census at $\tau_{\text{drive}} = 100 \mu\text{s}$, one-interval step, counted on the lower bound of Eq. (5.10). Each row states the set it counts over.

scope	pairs	bound > 10 %	worst
all analysed pairs	680	202	0.3868 at [0, 4]
$T_e \geq 2 \text{ eV}$	448	45	0.1748 at [15, 3]
$T_e \geq 2 \text{ eV}$ and $n_e \geq 10^{14}/\text{cm}^3$	108	0	0.0717 at [15, 5]

The middle row is the result. Of the 202 pairs in the first row, 157 lie below 2 eV, inside the region Section 5.10 excludes, and 105 of them have a Lyman- α optical depth above unity over a 5 cm slab. The defensible statement is 45 of 448 pairs above 2 eV, worst 17.5% at $T_e = 2.02 \text{ eV}$, $n_e = 1.93 \times 10^{13}/\text{cm}^3$; and at citable divertor densities, zero breakdowns, with the worst case 7.2%. The benchmark point does not break down: its ELM-averaged error is 1.2%.

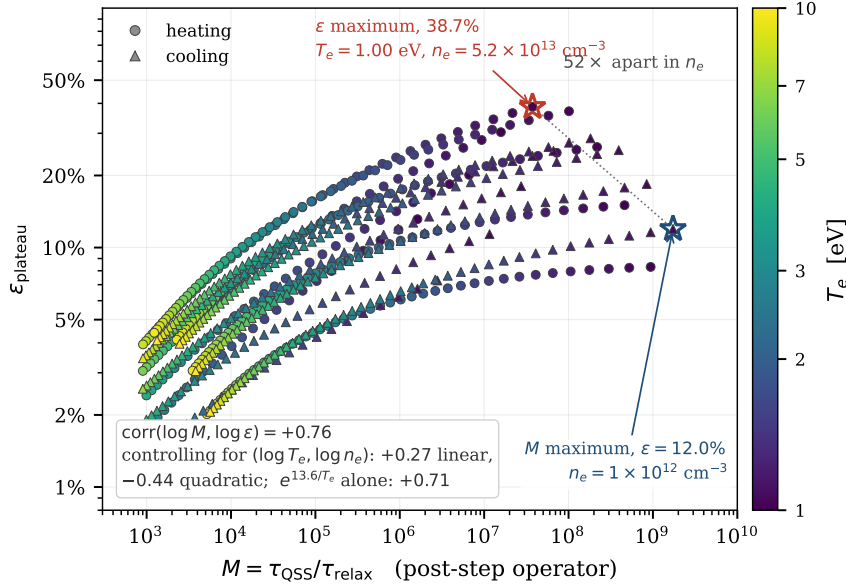
*[UNVERIFIED: the count in the first row is recorded as 202 in **findings_10** §4.1 and the figure captions, and as 201 in **thesis_ready** A11, which in the same sentence decomposes it as 105 heating plus 97 cooling pairs. Those two statements are inconsistent with each other. What would close this: recount directly from **divertor_map.csv**. Nothing in this chapter’s defensible scope depends on which is right.]*

Two sensitivities belong with the census. The count depends strongly on the threshold: at 5, 10 and 20% it is 348, 202 and 61 over the same 680 pairs, so the count should always be quoted with the threshold that produced it. The worst case does not reference the threshold at all and is unaffected by it. And the 104 points excluded for having no plateau window contain no breakdowns whatever, all sitting at $T_e \geq 3.24 \text{ eV}$ with a maximum bound of 0.0011, so the exclusion does not hide anything; including them changes 202/680 to 202/784.

5.8.2 Two structural objections, tested

The lower bound of Eq. (5.10) uses τ_{QSS} , and τ_{QSS} in this model is the relaxation time of an *open* system: the ground state loses atoms to an ion reservoir that is held fixed and never gives them back. An artificially long τ_{QSS} would inflate every entry in Table 5.5. Rebuilding the operator with an exactly particle-conserving 44th state and recomputing the whole census changes the count from 202 to 200 and the worst case from 38.68% to 38.56%; above 2 eV it changes nothing at all, 45 of 448 (**findings_10** §3.1). The closure factor on τ_{QSS} reaches 41 only where τ_{QSS} already exceeds the event duration by three orders of magnitude, and is 1.00 wherever τ_{QSS} is comparable to τ_{drive} , which is where it could matter. Chapter 6 states the caveat that this closure adds an ion reservoir without closing the electron balance and without adding transport.

The second objection is that a step is an idealisation and a real ELM has a finite rise time. Any ramp with $\tau_{\text{relax}} \ll t_{\text{ramp}} \ll \tau_{\text{QSS}}$ reaches the same plateau, because the fast manifold tracks the ramp adiabatically while the slow reservoir cannot follow it at all. The relevant Damköhler-like ratio is $\tau_{\text{QSS}}/\tau_{\text{drive}}$, which is 2332 at [0, 4], and not



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Figure 5.3: **Timescale separation does not predict closure error.** Each point is one (grid point, step direction) pair with a timescale-separated plateau window (680 of 784); M is $\tau_{\text{QSS}}/\tau_{\text{relax}}$ of the post-step operator, colour is T_e , circles heating and triangles cooling. The raw association is strong, $\text{corr}(\log M, \log \varepsilon) = +0.76$, and it is a temperature proxy rather than a dynamical statement: $\log M$ is 93% explained by $(\log T_e, \log n_e)$ alone, a bare Arrhenius factor $e^{13.6/T_e}$ containing no dynamics correlates at +0.71, and controlling for $(\log T_e, \log n_e)$ drives the association to +0.27 linearly and -0.44 — the opposite sign — quadratically. The two starred extrema make the same point without statistics: the largest M on the grid (1.7×10^9 , blue) carries $\varepsilon = 12.0\%$, while the largest error (38.7%, red) occurs $52\times$ away in density, at an M smaller by a factor 46. A large timescale separation is not evidence that the quasi-steady-state closure is accurate.

$\tau_{\text{relax}}/\tau_{\text{drive}} \approx 10^{-5}$, which describes the suppression of the fast excited-manifold error and has nothing to say about the plateau. The step idealisation is safe, and it is safe for the second reason and not the first.

[UNVERIFIED: the map steps T_e at fixed n_e , whereas an ELM raises both. Joint steps in (T_e, n_e) give comparable $\varepsilon_{\text{plateau}}$ but a time-averaged bound that can fall by a factor 4, because a higher n_e shortens τ_{QSS} ; at the benchmark the bound falls from 0.0117 to 0.0030. A joint-step map has not been run, and an examiner who knows ELM physics will ask for one (*findings_10* §8.3).]

5.9 Timescale separation does not predict it

A modeller deciding whether to trust a QSS-based inversion reaches for $M = \tau_{\text{QSS}}/\tau_{\text{relax}}$, the stiffness ratio that licenses eliminating the fast variables in the first place. Chapter 3 measured it from 86.8 to 1.73×10^9 across the grid. The natural expectation is that the diagnostic is worst where M is smallest.

It is not, and M carries no information about the diagnostic error once the operating point is accounted for.

The two extrema make the point without statistics. The largest M on the grid,

1.7×10^9 , sits at $[0, 0]$ and carries $\varepsilon_{\text{plateau}} = 12.0\%$, which is the 75th percentile of the distribution. The largest error, 38.7% at $[0, 4]$, sits $52\times$ away in density at an M smaller by a factor 46. **The maxima of the two quantities are $52\times$ apart in density, and the error at the timescale-separation maximum is a factor 3.2 below the worst.** Where M is largest is not where the diagnostic fails.

The raw association looks strong and is a temperature proxy. Over the 680 analysed pairs, $\text{corr}(\log M, \log \varepsilon_{\text{plateau}}) = +0.76$. But $\log M$ is 93% explained by $(\log T_e, \log n_e)$ alone, and a bare Arrhenius factor e^{χ_H/T_e} built from the hydrogen ionisation potential $\chi_H = 13.6\text{ eV}$, which contains no dynamics whatever, correlates with $\log \varepsilon_{\text{plateau}}$ at $+0.71$ (`make_ch5_figures.py`). Both quantities rise steeply toward low temperature for unrelated reasons, and that shared trend is nearly all of the $+0.76$.

Removing the shared trend reverses the sign. Residualising both $\log M$ and $\log \varepsilon_{\text{plateau}}$ on $(\log T_e, \log n_e)$ leaves a partial correlation of $+0.27$ over all pairs and $+0.33$ over the heating pairs alone; residualising on a full quadratic basis in the same two variables leaves -0.44 (`make_ch5_figures.py`, which refuses to draw Figure 5.3 at all unless the quadratic partial comes out negative).

Report the sign, not the magnitude. The magnitude of the controlled partial depends on the basis chosen for the control, which is a modelling choice and not a measurement. *[UNVERIFIED: thesis_ready A11 records that the quadratic partial ran -0.22 to -0.70 over eight scope and basis combinations. `make_ch5_figures.py` computes four partials, not eight, and neither the eight combinations nor the range appears in any script or artifact. What would close this: put the sweep in the script and stamp its output.]*

The negative result itself is not new as a general statement. Greenland [17] concluded, for arbitrary collisional–radiative systems, that validity criteria “are not related to the equilibrium time-scales” and that “the eigenvalues have secondary importance”. What this work adds is a quantification for one diagnostic and a map of where the criterion misleads, which is a smaller claim than discovery and a more useful one than assertion.

The reason the criterion fails here is visible in Eq. (5.6). M compares two *rates*; $\varepsilon_{\text{plateau}}$ measures a *distance*, the product of a sensitivity and a displacement, neither of which appears in M . A large M means the excited states equilibrate quickly with the reservoir, which Section 5.1.2 confirms to a part in 10^8 . It says nothing about whether the reservoir is in the right place.

5.10 Where the quantitative scope begins

Every quantitative statement in this chapter has been restricted to $T_e \geq 2\text{ eV}$. Two separate things go wrong below that boundary. Neither of them was chosen, both were measured, and they put the boundary in the same place.

5.10.1 Holding the electron density fixed stops being consistent

Every operator in this chapter is evaluated at fixed n_e : the temperature steps, the density does not. That is not an approximation whose cost can be made small by care, because in a hydrogen plasma the electrons and the neutrals are not independent quantities. Quasineutrality ties them: every ground-state atom that ionises delivers one electron, so if the reservoir collapses the electron density must rise by the same absolute amount.

Whether that matters depends on how much of the hydrogen is neutral. At $T_e = 2.947\text{ eV}$ the plasma is almost fully ionised and the neutral population is a small perturba-

tion on the electron budget, so the coupling is negligible. At $T_e = 1$ eV the model's own steady state is 95 to 98 % neutral. The ground state then holds an order of magnitude more particles than the electron population does, and a modest fractional change in it is an enormous fractional change in n_e .

Imposing conservation of nuclei across a single +4.81 % temperature step makes this quantitative. Writing $x = n_{1s}^{\text{new}}/n_{1s}^{\text{old}}$ for the fractional collapse of the reservoir, the electron density that quasineutrality would require is:

Table 5.6: Electron-density change required by conservation of nuclei across one one-interval temperature step, against the fixed n_e the model actually uses. x is the ratio of post-step to pre-step ground-state population at collisional–radiative equilibrium.

step in T_e	n_e (1/cm ³)	x	required $\Delta n_e/n_e$
1.000 $\rightarrow$ 1.048 eV	1.00×10^{12}	0.506	+20.0
1.000 $\rightarrow$ 1.048 eV	5.18×10^{13}	0.508	+13.9
2.947 $\rightarrow$ 3.089 eV	1.39×10^{14}	0.764	+0.0002

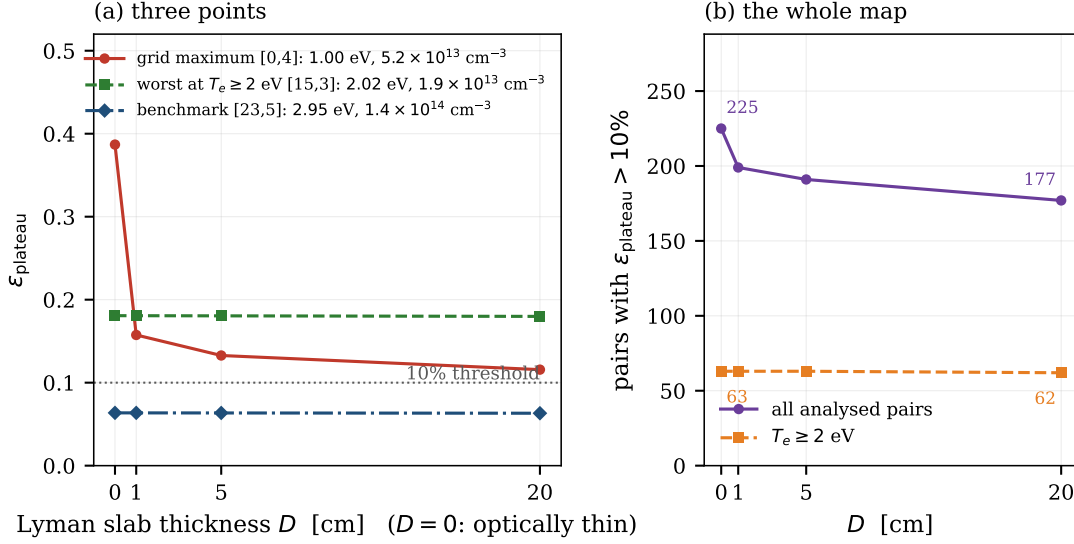
The first two rows are not a separate piece of physics from the error itself. $x = 0.506$ is $|\ln x| = 0.68$, and 0.68 is the reservoir displacement that `plateau_gridmap.txt` records at $[0, 4]$ and that Section 5.4 identifies as the source of the 38.7 %. **The displacement that produces the cold-edge error is exactly the displacement that makes the fixed-density assumption inconsistent.** Over the 392 one-step operators, 68 of them, 17 %, require $|\Delta n_e/n_e| > 10$ %, and 36, 9 %, require more than 100 %. The model evaluates the post-step operator at the old n_e at every one of them. Above 2 eV the required correction falls below 10^{-3} and the question does not arise.

This closes a gap between two objections that this thesis has until now presented as separate and separately survivable. If the plasma parcel is closed, quasineutrality forces n_e up by a factor of 10 to 20 at the cold end, and every rate coefficient in the post-step operator is then evaluated at the wrong density. If instead the parcel is fed by transport, so that n_e genuinely does stay fixed while the neutrals are consumed, then the ground state is being resupplied from outside and is not stale, which removes the mechanism that produces the error. **The two pictures cannot both be held, and the cold edge needs one of them.**

The cold corner also fails a cruder test: it is not a divertor state. The model's own steady solution at $[0, 4]$ gives $n_{1s} = 1.47 \times 10^{15}/\text{cm}^3$, which at a neutral temperature of 1 eV is a neutral pressure of 235 Pa, twelve times the upper end of the ITER divertor design range of roughly 1 to 20 Pa; at $[0, 7]$ it is 3280 Pa, 164 times over. These are fixed points of a zero-dimensional box with no transport and no pumping, and they should be read as such. *[SOURCE REQUIRED: the 1 to 20 Pa ITER divertor neutral-pressure range needs a citation to the ITER divertor physics basis; the two computed pressures and the nuclei-balance table above come from the collisional–radiative audit completed 10 September 2026 and need their script and stamped artifact named.]*

5.10.2 The plasma is not optically thin at the cold edge

The rate matrix of Chapter 3 treats the plasma as optically thin: every photon emitted escapes. For the Balmer lines that is a good assumption over the whole grid. For the Lyman series, which terminates on the ground state where essentially all the atoms



CR data 710f4a24 · 2026-09-10

Figure 5.4: **Lyman trapping, and why the quantitative scope starts at 2 eV.** (a) $\varepsilon_{\text{plateau}}$ against the assumed homogeneous-slab thickness D used for the Lyman escape factors, computed self-consistently (Θ_P depends on $n(1s)$, which depends on Θ_P) for all 14 Lyman channels; $D = 0$ is the optically thin matrix used everywhere else in this chapter. The grid maximum at $T_e = 1.00$ eV falls from 38.7% to 11.6–15.7% over $D = 1$ –20 cm: the cold-edge number is set by a slab thickness this zero-dimensional model does not contain, and must never be quoted without it. The worst point at $T_e \geq 2$ eV moves by 0.5% and the benchmark by 0.7% over the same twentyfold range. (b) Number of analysed pairs exceeding the 10% threshold, counted on the plateau value itself and not on the time-averaged ELM bound: 225 falls to 177 over the whole map, while above 2 eV it is 63 against 62 — unchanged to within one pair. (At $D = 20$ cm the analysed set is 678 pairs rather than 680: two lose their plateau window.) The $T_e \geq 2$ eV restriction is therefore not a hedge; it is the measured boundary beyond which the one neglected process most likely to invalidate the result stops mattering.

are, it fails badly at the cold edge. Computing the Doppler line-centre optical depth from the model’s own CRE solution gives a Lyman- α depth of 114 per centimetre at $T_e = 1.00$ eV, $n_e = 5.18 \times 10^{13}/\text{cm}^3$, falling to 0.036 at 2.02 eV and 0.0023 at 2.95 eV (`verify_lyman_trapping.py`). At the cold edge the Einstein coefficients in the matrix are wrong, as effective decay rates, by four to five orders of magnitude.

Rebuilding the matrix with self-consistent Lyman escape factors for all 14 Lyman channels, over an assumed homogeneous slab of thickness D , and recomputing the whole map gives Figure 5.4.

Above 2 eV nothing moves. The ELM breakdown count above 2 eV stays at 45 of 448 for $D = 1, 5$ and 20 cm, and the worst case runs 0.1748, 0.1746 and 0.1740 against 0.1748 untrapped, a change of 0.5% across a twentyfold range in a parameter this zero-dimensional model does not contain. At the benchmark the plateau error moves by 0.7% over the same range. **The $T_e \geq 2$ eV restriction is therefore not a hedge. It is the measured boundary beyond which the neglected process most likely to invalidate the result stops mattering,** which is a stronger statement than a disclaimer and was not available before the calculation was run.

Below it, the cold-edge numbers are inflated. At $[0, 4]$ the plateau error falls from

38.7% to 15.7% at $D = 1$ cm and 11.6% at $D = 20$ cm, a factor 2.5 to 3.3, and which value inside that range it takes is set by the assumed slab thickness rather than by anything the model computes. The point still exceeds the 10% threshold at every thickness, so it still breaks down; the *magnitude* is not quotable. Below 1.15 eV the density column carrying the maximum wanders across three columns, a factor 7, as a function of D , so the two coldest rows cannot support a claim about where the maximum is at all.

Chapter 6 takes up the mechanism of the fall, which is not a loss of sensitivity, and the assumptions the slab geometry imports.

5.10.3 The two boundaries coincide

Radiation trapping and quasineutrality are unrelated pieces of physics, entering through the Einstein coefficients and through the particle balance respectively. They fail together. Above 2 eV the trapped and untrapped breakdown censuses are identical and the required density correction is a part in a thousand; below it both grow quickly, and by 1 eV the first has divided the headline error by up to 3.3 while the second has invalidated the operator it was computed from.

The $T_e \geq 2$ eV restriction is therefore not a hedge and not a choice. It is the region on which this model's own assumptions are mutually consistent, and it is where the neglected process most likely to overturn the result has been shown not to. Every headline number in this chapter sits inside it. The cold-edge values are reported because they exist and because suppressing them would be worse, and they are reported as what they are: the asymptotic behaviour of a model outside its own domain of validity.

5.11 What this chapter establishes

The approximation this thesis set out to test is not the one that fails. Measured along a full 43-state trajectory, the quasi-steady-state closure is accurate to 6.73×10^{-9} at the very grid point carrying the largest diagnostic error. The excited states keep up. What does not keep up is the ground state, and the assumption a two-parameter lookup table makes about it is not that the fast states equilibrate but that the plasma has finished changing.

What that assumption costs has a closed form. The plateau error factorises exactly into a sensitivity and a reservoir displacement, Eq. (5.6), and the displacement factorises again into a reservoir gain and the size of the transient, Eq. (5.8). The gain moves by 5 to 6% across a fourfold change in step size while the error itself moves by a factor 4.9, so the gain is the object worth tabulating and the percentage is not. In the small-step limit the diagnostic error per unit fractional temperature excursion is $|\overline{SG}|$: about 1.3 at the benchmark point and about 3.5 at the worst point in the defended range.

The sensitivity factor obeys a bound, $|\overline{S}| \leq \tanh(|\Delta|/4) < 1$, that follows from the structure of the linear system and not from any rate coefficient. A line ratio cannot respond to its ground-state reservoir faster, in relative terms, than the reservoir moves. At the crest the measured sensitivity is at 97% of that bound.

Across the operating range the error has an interior maximum in density, between 7×10^{12} and $5 \times 10^{13}/\text{cm}^3$, and no interior maximum in temperature. The density maximum is a position effect: the separation between the two supply channels is nearly flat across density, and what sweeps is the plasma's own reservoir sliding along a fixed sensitivity

curve, with the maximum where $\ln(u^{\text{CRE}}/u^{\text{peak}})$ changes sign. The mechanism survived three independent tests, including a quantitative prediction of how far the maximum moves when the shell pair changes, and it is a property of the atomic system, needing no divertor geometry to produce it. It is not detachment, and it is the (3, 4) pair's maximum rather than the Balmer series'.

Inside the defensible scope, $T_e \geq 2 \text{ eV}$, 45 of 448 point-and-direction pairs commit an ELM-averaged error above 10 % at $\tau_{\text{drive}} = 100 \mu\text{s}$, the worst being 17.5 %; at citable divertor densities none do, the worst being 7.2 %. That census is unchanged across a twentyfold range in the one neglected process, Lyman trapping, most likely to have invalidated it.

That scope boundary is the chapter's least expected result. It was adopted as a precaution against radiation trapping and it turns out to be the boundary of self-consistency: below 2 eV the plasma is mostly neutral, and conservation of nuclei then demands an electron-density change of 10 to 20 times across the same temperature step whose effect is being measured at fixed n_e . The cold-edge numbers are asymptotic values of a model outside its own domain, and this chapter reports them as such and builds nothing on them.

Finally, the criterion a modeller would reach for does not see any of this coming. Timescale separation and diagnostic error have their maxima $52\times$ apart in density, and their apparent correlation reverses sign once the operating point is controlled for. M compares rates; the diagnostic error is a distance.

What the model cannot say is the subject of Chapter 6: the plasma is treated as optically thin where it is not, the atoms are held where they are put, and the gas is treated as atomic where it is mostly molecular. Each of those is quantified there rather than hedged.

Chapter 6

What This Model Cannot Say

Chapter 5 handed the reader two things: a map of how wrong a tabulated H_α/H_β inversion can be while the ground state is still catching up, and a mechanism that explains the map’s shape. The map has a ridge in it, the ridge has a closed-form explanation, and the explanation was tested three separate ways. At this point in the argument the reader believes a result.

This chapter’s job is to bound that belief without destroying it. Every number in Chapter 5 came out of one 43-state matrix describing one kind of atom under one set of assumptions, and some of those assumptions are badly violated in exactly the corner of the map where the numbers are largest. An examiner will find that corner. It is better to arrive there first, with arithmetic.

The standard this chapter holds itself to. A limitation that has been *computed* is worth more than a limitation that has been hedged. Where an objection could be turned into a calculation, it was, and the calculation is reported here with the number that would have refuted the result. Two objections were expected to be fatal and were not; they are given the same space as the ones that bite. Where an objection cannot be computed inside a model with no spatial dimension, it is stated as a *conditional* — “*if* this holds, the error is as computed” — and not softened into a caveat sentence. Where a check has simply not been run, this chapter says so.

The order is by severity. Radiation trapping and neutral transport bear directly on the headline number. Molecular chemistry is the objection a referee raises first. The remainder bound how the result may be *described* rather than how large it is.

Three abbreviations recur below and are expanded here for this chapter: *CR* for collisional–radiative, the class of model built in Chapter 3; *QSS* for quasi-steady-state, the Bodenstein-style approximation that eliminates the fast excited levels; and *CRE* for collisional–radiative equilibrium, the fully settled steady state that the diagnostic lookup table assumes and that this thesis measures the cost of assuming. An *ELM* — edge-localised mode — is the repetitive burst of heat and particles that a tokamak edge throws at the divertor every few milliseconds, and it is the transient every “event duration” below refers to; $\tau_{\text{drive}} = 100\ \mu\text{s}$ throughout, the duration used in Chapter 5.

6.1 Optical thickness: when the plasma cannot let its own light out

6.1.1 The question, and the picture behind it

Chapter 2 gave every downward transition an Einstein A coefficient: the probability per second that an excited atom drops to a lower level and emits a photon. Those coefficients entered the CR matrix $\mathbf{L}$ bare. Entering bare means one specific physical assumption — that every photon emitted anywhere in the plasma leaves it and never comes back. A plasma for which this is true is called *optically thin*.

A chemical engineer has met the failure of this assumption in another costume. Consider a reaction whose product is removed by desorption from a catalyst surface. If the product leaves the reactor, the desorption step runs at its intrinsic rate. If the product cannot leave — if it re-adsorbs somewhere else in the bed and is put back onto a site — then the *effective* desorption rate is smaller than the intrinsic one, and the whole network downstream of that step shifts. Radiation trapping is that, with photons.

Here is the specific chain. The strongest transition in hydrogen is Lyman- α : an atom in the $2p$ level drops to the ground state $1s$ and emits an ultraviolet photon. If the gas around it contains many ground-state atoms, that photon has a good chance of being absorbed by one of them, which puts *that* atom back into $2p$. Nothing has left the plasma; a $2p$ population has simply been moved from one place to another. Averaged over the volume, the $2p$ level decays more slowly than its A coefficient says. The standard way to write this down is to multiply the coefficient by an *escape factor* Θ_P between 0 and 1: $\Theta_P = 1$ is a transparent plasma, $\Theta_P = 10^{-5}$ is one in which the transition is effectively switched off. The formulation used here is Holstein’s population escape factor [34].

Why this reaches the populations, and therefore the result. If $2p$ lives longer, atoms pile up in $n = 2$. Atoms in $n = 2$ are only 10.2 eV from the ground state but only 3.4 eV from the continuum, so they are far easier to ionise than ground-state atoms. A longer $n = 2$ lifetime therefore means more *stepwise* ionisation — ionisation via an excited level rather than directly out of the ground state — which means fewer neutrals and a faster turnover of the ground-state reservoir. The ground-state reservoir is the whole mechanism of this thesis. Trapping is not a cosmetic correction to a line intensity; it acts on the slow clock.

6.1.2 Where the assumption fails, in numbers

The formalism needed is one line. A photon travelling a distance D through a gas of absorbers is attenuated by $e^{-\tau}$, where the *optical depth* $\tau = \sigma_0 n_{1s} D$ is dimensionless: n_{1s} is the ground-state density in $1/\text{cm}^3$, D the path length in cm, and σ_0 the absorption cross section at line centre in cm^2 . For a Doppler-broadened line, $\sigma_0 = 2.654 \times 10^{-2} f / (\sqrt{\pi} \Delta\nu_D)$, with f the oscillator strength (a dimensionless measure of how strongly the transition couples to light) and $\Delta\nu_D$ the Doppler width set by the temperature of the absorbing atoms. $\tau \ll 1$ is transparent; $\tau \gg 1$ is opaque.

The number that matters is n_{1s} , not n_e , and Chapter 3 already established the awkward fact that these two run in opposite directions: the plasma is most neutral where it is coldest. At the worst point of the error map — $T_e = 1.0 \text{ eV}$, $n_e = 5.18 \times 10^{13} 1/\text{cm}^3$, which is the maximum of $\varepsilon_{\text{plateau}}$ over all 680 window-resolvable points of the map, cold edge included — the model’s own CR solution gives $n_{1s}/n_{\text{ion}} = 28.3$, so quasi-neutrality

fixes $n_{1s} = 1.47 \times 10^{15} \text{ 1/cm}^3$. At 1 eV that gives $\sigma_0 = 7.74 \times 10^{-14} \text{ cm}^2$, and the Lyman- α optical depth per centimetre of path is

T_e (eV)	$\tau_{\text{Ly}\alpha}$ per cm at $n_e = 5.18 \times 10^{13} \text{ 1/cm}^3$
1.00	114
1.15	15.0
1.60	0.32
2.02	0.036
2.95	0.0023

An optical depth of 114 per centimetre is worth translating, because the figure is otherwise abstract: it means a Lyman- α photon at that point travels under a tenth of a millimetre before it is reabsorbed. It is not travelling anywhere. Over 5 cm of plasma the accumulated depth is 7900 and the escape factor is 2.7×10^{-5} — roughly one photon in forty thousand gets out. That figure is an independent reproduction of the 3×10^{-5} quoted in Chapter 3: it was recomputed here from the model’s own solution rather than copied from that chapter, and the two agree. The blunt reading is that the effective $A_{2p \rightarrow 1s}$ belonging in **L** at that point is wrong by four to five orders of magnitude.

Two rows down the same table the problem has evaporated. At 2.02 eV the depth per centimetre is 0.036 and at the benchmark point (2.947 eV, $1.389 \times 10^{14} \text{ 1/cm}^3$) it is 0.0023; a photon there crosses metres of plasma before anything happens to it. **The measured quantity is not “how opaque is this plasma” but “how fast does opacity switch on as the plasma cools”,** and the answer is a factor of 5×10^4 between 1 eV and 2.95 eV.

That is why the results of Chapter 5 were declared for $T_e \gtrsim 2 \text{ eV}$ in the first place (Section 3.5). It is also, uncomfortably, where the headline was living. Recounting the ELM breakdown census against optical depth rather than against temperature: of the 202 points at which the ELM-averaged lower bound on the error exceeds 10% — an extremum count over the 680 window-resolvable points of the full grid — 157 lie below 2 eV, 105 have $\tau_{\text{Ly}\alpha}(5 \text{ cm}) > 1$, and 26 have it above 100.

6.1.3 What we expected trapping to do, and what it did

The expectation, stated before the calculation was run, was that trapping would demolish the map. The reasoning was straightforward and it was wrong: an effective A coefficient wrong by five orders of magnitude at the worst point ought to invalidate everything computed there and, by continuity, a good deal of what was computed nearby. The project’s own records contained a matching warning — that under strong trapping the fast relaxation time τ_{relax} rises to $2.05 \mu\text{s}$ and the timescale separation M falls by a factor 158.

So the calculation was done. The Lyman series was given a self-consistent Holstein escape factor and the entire error map rebuilt. Self-consistent matters here: Θ_P depends on n_{1s} , and n_{1s} depends on Θ_P through the CR balance, so the two were iterated to a fixed point rather than evaluated once. The fixed point converged at all 400 grid points for every slab thickness tested, and non-convergence was made to raise an error rather than silently return the last iterate. All 14 Lyman channels were trapped — the seven resolved $nP \rightarrow 1S$ transitions and the seven bundled ones — not Lyman- α alone.

Three gates were passed before any result was looked at, and the third is what makes the comparison mean anything.

Table 6.1: Effect of radiation trapping on the ELM breakdown census. Each entry counts the grid points at which the ELM-averaged lower bound on the CR-equilibrium error exceeds 10%, for an event of duration $\tau_{\text{drive}} = 100 \mu\text{s}$, out of the points at which a quasi-steady-state window is resolvable; “worst” is the largest lower bound within that scope. Rows differ only in the assumed slab thickness D used to compute the self-consistent Holstein escape factor applied to all 14 Lyman transitions. The first row is the optically thin matrix used throughout Chapter 5. Above 2 eV the count does not move at all across a twentyfold range of D . The $D = 20 \text{ cm}$ row has a slightly smaller denominator because two points lose their resolvable window under trapping. Source: `validation/lyman_trapping/lyman_trapping.txt`.

matrix	all window-resolvable	$T_e \geq 2 \text{ eV}$	$T_e \geq 2 \text{ eV}, n_e \geq 10^{14}$
optically thin (canonical)	202/680, worst 0.3868	45/448, worst 0.1748	0/108, worst 0.0717
trapped, $D = 1 \text{ cm}$	176/680, worst 0.2925	45/448, worst 0.1748	0/108, worst 0.0712
trapped, $D = 5 \text{ cm}$	170/680, worst 0.2550	45/448, worst 0.1746	0/108, worst 0.0694
trapped, $D = 20 \text{ cm}$	160/678, worst 0.2237	45/446, worst 0.1740	0/106, worst 0.0627

1. The oscillator strengths were *derived* from this repository’s own A coefficients by inverting the Einstein relation, rather than imported from a table. They come out as $f = 0.4162, 0.0791, 0.0290, 0.0139$ for Lyman- α through Lyman- δ — the accepted literature values to four figures. Nothing was assumed that the model did not already contain.
2. The derived Lyman- α σ_0 was cross-checked against an independent implementation elsewhere in the repository and agrees to 0.059%. On its first execution that gate *failed*, at 1156%, and the failure was traced to a factor 4π from pairing an SI coefficient with CGS constants. The result reported below exists because a check fired and was not adjusted.
3. Rebuilding the matrix through the same code path with trapping switched *off* reproduces the canonical $\mathbf{L}$ with $\max|\text{rebuilt} - \text{canonical}| = 0$, exactly. Every trapped number below is therefore a difference against the real matrix, not against a re-derivation of it.

One new quantity had to be invented to do this at all, and it is not in the model. The escape factor depends on how far a photon must travel to get out, so it depends on the size of the plasma — the slab thickness D . A model with no spatial dimension has no D . It was therefore swept, from 1 cm to 20 cm, and never fixed; every trapped number below carries the D it was computed at. Reporting a single trapped number without its D would be reporting an assumption as a measurement.

6.1.4 Above two electron-volts, nothing moves

Table 6.1 contains the strongest robustness statement in this thesis, and it is the opposite of what was expected.

Above 2 eV the breakdown count does not change at all — 45 of 448 points in every run — and the worst case within that set moves by 0.5%, from 0.1748 to 0.1740, across a twentyfold range in assumed slab thickness. The escape factor at the benchmark point ($2.947 \text{ eV}, 1.389 \times 10^{14} / \text{cm}^3$) runs from 0.9999 at $D = 1 \text{ cm}$ to

Table 6.2: The cold-edge worst point of the optically thin error map — $T_e = 1.0$ eV, $n_e = 5.18 \times 10^{13}$ 1/cm³, heating step; the maximum of $\varepsilon_{\text{plateau}}$ over all 680 window-resolvable grid points — traced as a function of the assumed slab thickness D . $\varepsilon_{\text{plateau}}$ is the plateau error, the fractional error a lookup table makes while the ground state is stale; the ELM bound is its average over an event of $\tau_{\text{drive}} = 100$ μ s. The plateau error falls by a factor 2.5 to 3.3, but the diagnostic’s intrinsic sensitivity $|f_3 - f_4|$ barely moves: what collapses is how far the ground state is displaced, not how strongly the line ratio responds to it. Source: `validation/lyman_trapping/lyman_trapping.txt`.

D (cm)	$\varepsilon_{\text{plateau}}$	ELM bound	τ_{QSS} (s)	M	f_3	f_4	$ f_3 - f_4 $
0 (thin)	0.38690	0.38682	2.3324×10^{-1}	3.74×10^7	0.7728	0.2862	0.4866
1	0.15748	0.15739	9.0651×10^{-2}	2.75×10^6	0.6643	0.2055	0.4589
5	0.13274	0.13260	4.6940×10^{-2}	6.13×10^5	0.5967	0.1793	0.4174
20	0.11570	0.11550	2.8500×10^{-2}	2.14×10^5	0.5139	0.1501	0.3639

0.973 at $D = 20$ cm. Even at the thickest slab tested, 97% of Lyman- α photons still leave: the plasma is optically thin there in the sense that actually matters, and assuming so costs less than the resolution of the map.

This converts a disclaimer into a measurement. The $T_e \gtrsim 2$ eV restriction was chosen on an escape-factor argument *before* any of this was recomputed, and it turns out to sit almost exactly at the temperature above which trapping stops mattering. The restriction was a guess that has now been checked, and it survived. The defensible claim of Chapter 5 — 45 of 448 points above 2 eV, worst case 17.5% at $T_e = 2.02$ eV, and no breakdown whatever at citable divertor densities, where the worst lower bound over 108 points is 7.2% — is now known to be *invariant* under the one physical process most likely to have invalidated it. That is a stronger statement than a restriction and should be read as one.

6.1.5 Below two electron-volts, the cold-edge number is inflated threefold

The same calculation is much less kind to the cold edge, and it should be, because that is where the optical depth was 114 per centimetre. Table 6.2 follows the worst point of the optically thin map through the sweep.

The 38.7% becomes 11.6 to 15.7% — a factor of 2.5 to 3.3 — and which value inside that range it takes is set by a slab thickness the model does not contain. The point still exceeds the 10% threshold at every thickness tested, so the qualitative statement that the cold corner breaks down survives intact; the *number* does not, and it must never appear without its trapped range and without the optical depth that produced it.

The route by which the number falls is itself a result, and it is not the obvious route. The obvious guess is that trapping desensitises the diagnostic — that a trapped plasma simply responds less. It does not. The sensitivity $|f_3 - f_4|$, which is the factor in the exact decomposition of Section 5.6 measuring how strongly the line ratio answers to its reservoir, moves by only 6% at $D = 1$ cm and 25% at $D = 20$ cm. What collapses is the other factor: the displacement.

The chain is the one set out at the start of this section, now with numbers on it. Trapping lengthens the $n = 2$ lifetime; the longer lifetime raises stepwise ionisation; the ground-state population falls — at $T_e = 1.0$ eV, $n_e = 1.93 \times 10^{13}$ 1/cm³, n_{1s} drops from

2.99×10^{14} to 7.26×10^{13} $1/\text{cm}^3$ between $D = 1$ and 20 cm — and the slow clock τ_{QSS} shortens by a factor 8 at the worst point. A reservoir that turns over faster is a less stale reservoir, so $|\Delta \ln x|$ roughly halves. This is exactly the two-axis attribution of Section 5.6.4 arriving from a third direction: trapping acts on the displacement, not on the sensitivity.

6.1.6 Below 1.15 electron-volts the ridge location is not determined

One more thing moves at the cold edge, and it is a location rather than a magnitude. Taking the density at which $\varepsilon_{\text{plateau}}$ is maximal, row by row: at $T_e \geq 1.6$ eV that maximum falls on $n_e = 1.93 \times 10^{13}$ $1/\text{cm}^3$ in every run, optically thin and trapped alike, and the values themselves barely register the change — at the benchmark temperature $\varepsilon_{\text{plateau}} = 0.12166$, 0.12166, 0.12165, 0.12163 for $D = 0, 1, 5$ and 20 cm, with $|f_3 - f_4|$ changing in the fifth decimal place. Below 1.15 eV the maximum wanders across three density columns, a factor of 7, as a function of an assumed parameter. **The two coldest rows of the grid cannot support a claim about where the ridge is at all.**

That result cuts in the mechanism’s favour, which is worth saying explicitly because it is easy to miss. The ridge attribution of Section 5.6 was verified over $T_e \geq 2$ eV, and $T_e \geq 2$ eV is precisely the range in which trapping has now been shown to be irrelevant. The agreement with Griem’s boundary criterion is therefore not an artifact of neglected opacity — it was established where opacity does not act.

6.1.7 A contradiction resolved, and what is still open

Two statements about trapping had been circulating in this project’s records: that τ_{relax} is insensitive to Lyman- α trapping to within 1 %, and that under strong trapping τ_{relax} rises to 2.05 μs with M falling 158-fold. They look contradictory. Measured directly, both are true and neither had stated its scope: at the benchmark point (2.947 eV, $1.389 \times 10^{14}/\text{cm}^3$) τ_{relax} moves from 2.2769×10^{-9} to 2.2947×10^{-9} s at $D = 20$ cm, a change of 0.78 %; at $T_e = 1.0$ eV, $n_e = 1.93 \times 10^{13}$ $1/\text{cm}^3$ it moves from 8.8842×10^{-9} to 2.4289×10^{-7} s, a factor of 27.3. The “within 1 %” claim is a benchmark-point claim and fails by more than an order of magnitude at the cold corner. This is the same failure mode as quoting any other extremum without the set it is an extremum over, and neither sentence may travel again without its regime attached.

What this calculation does not settle should be as clear as what it does. The geometry is assumed, not derived: a homogeneous slab, uniform emission, an isotropic radiation field, a Doppler line profile, evaluated at the plasma centre. A 0-D model cannot check any of those. One assumption inside it *was* checked and is not load-bearing — the temperature of the absorbing atoms, which sets the Doppler width. Fixing it at 3 eV, the value a molecule dissociating by the Franck–Condon process would give, instead of at T_e , changes the census at $D = 5$ cm from 170/680 and a worst case of 0.2550 to 173/680 and 0.2653. Continuum lowering, and opacity in the Balmer lines themselves — which would act on the observed ratio directly rather than through the level populations — are untouched here and remain open.

Trapping, then, is an objection that has been converted into a boundary. The next one cannot be.

6.2 Transport: the atoms do not stay where the model puts them

6.2.1 The question

This is the objection that cannot be answered inside this model, and it is a more serious threat to the cold-edge number than anything in Section 6.1. It is also the simplest to state.

6.2.2 The picture: a reactor with no walls

The model of Chapter 3 is a batch reactor. A fixed parcel of gas is given a temperature, and its composition evolves until it stops changing. The slow eigenvalue of that system, τ_{QSS} , is how long the ground-state population takes to reach its new value — and it is long, because ionising an atom out of the ground state is slow.

A divertor is not a batch reactor. It is closer to a continuous-flow reactor with an enormous throughput, in which the residence time of a molecule is far shorter than the time it needs to react. Neutral atoms are born at the target surface when ions strike it and recombine there — this is *recycling* — and they fly back into the plasma. They do not wait around to be ionised locally; the ones that do not ionise leave, and fresh ones arrive continuously.

The comparison is not close. At the worst point of the map τ_{QSS} is 233 ms: nearly a quarter of a second for a stationary atom to be ionised. A deuterium atom at 1 to 3 eV crosses a 10 cm plasma in 6 to 10 μs . The atom has left the volume roughly twenty thousand times over before the local ionisation balance would have finished with it.

In a divertor, the ground-state neutral density at a given point is not set by local ionisation balance. It is set by recycling.

6.2.3 Why this bears on the result specifically

The error of Chapter 5 exists for one reason: after a temperature step, the ground-state reservoir is *stale*. Its population has not moved while the temperature has, the excited levels equilibrate against a reservoir that belongs to the old conditions, and the lookup table — which assumes the reservoir finished settling — reads the wrong answer. Everything hangs on the reservoir not being refreshed.

Neutral transport refreshes it. Any process that replaces the ground-state population on a timescale shorter than τ_{QSS} resets the staleness, and this model contains no such process. The threshold is computable from the map itself: for the ELM-averaged lower bound at the worst point to fall below the 10 % breakdown threshold, an effective ground-state renewal time of about 26 μs would suffice. Neutral transit across the plasma is faster than that.

6.2.4 The claim, restated as the conditional it is

The honest form is therefore not a caveat but a conditional, and it should be written this way wherever the number appears:

If the ground-state neutral density is frozen for the duration of the event, the error is as computed.

That conditional is not empty, and it is worth being precise about how much it concedes. It is close to exactly true for the excited manifold, which equilibrates in nanoseconds and cannot be transported anywhere in that time. It is also the standard implicit assumption of every 0-D collisional–radiative table in use — which is, after all, the object this thesis set out to test, so testing it under its own assumption is legitimate. But it is an assumption about the plasma, not about the atoms, and no amount of eigenvalue analysis inside a 0-D matrix can validate it. Measured: the timescales differ by four orders of magnitude. Assumed: that they do not interact.

What would settle it is a two-region or transport-coupled calculation — an edge code such as SOLPS (Scrape-Off Layer Plasma Simulation) supplying n_{1s} from a recycling influx, instead of n_{1s} being solved for from $\mathbf{L}\mathbf{n} + \mathbf{S}n_{\text{ion}} = 0$. That is outside the scope of this thesis and is named in Chapter 7 as the first item of further work. [TODO: Two-region or SOLPS-coupled calculation with recycling-set n_{1s} . Not attempted; cannot be bounded from within a 0-D model.]

6.3 No molecular channels, in a gas that is mostly molecules

6.3.1 The question, and why a referee asks it first

Every state in Chapter 2’s state space is a level of a single hydrogen *atom*, plus a reservoir of bare protons. There is no H_2 (or D_2), no molecular ion D_2^+ , no negative ion D^- , and no reactions between them. A search of the rate assembly for any of these returns nothing at all.

6.3.2 The picture: a third feed stream

Chapter 3 showed that excited levels are fed from exactly two directions — up from the ground state by electron collisions, and down from the ion continuum by recombination — and that the whole error mechanism is the difference in how the $n = 3$ and $n = 4$ shells split their supply between those two feeds. In reactor terms, two feed streams into one vessel, mixed in shell-dependent proportion.

Molecules open a third stream. In a cold, dense, mostly neutral gas, hydrogen atoms form H_2 ; the molecules are ionised to H_2^+ , or exchange charge, and the resulting molecular ions recombine with electrons and break apart, leaving one of the fragments in an *excited* atomic level. The whole family of such routes is called MAR — molecular activated recombination. The important feature for this thesis is not that MAR exists but where it delivers: it populates $n = 3$ directly, without the atom ever having passed through the ground-state reservoir whose staleness is the entire mechanism here. A supply channel that bypasses the reservoir does not inherit the reservoir’s lag.

6.3.3 How much gas is involved

At the worst point of the map the model’s own equilibrium solution is 96.6% neutral: $n_{1s} = 1.47 \times 10^{15}$ against $n_e = 5.18 \times 10^{13} \text{ 1/cm}^3$. A hydrogen gas at 1 eV that is 96.6%

neutral is a gas in which molecules form readily. The atomic description is not being applied at the edge of its validity there; it is being applied outside it.

The size of what is being left out is documented experimentally. In spectroscopic inversions of divertor Balmer emission at the onset of *detachment* — the operating regime in which the plasma cools enough that the ion flux to the target collapses, and the regime this diagnostic is chiefly used to identify — molecularly-induced emission is found to account for up to 60 to 70 % of the D_α light and 10 to 20 % of D_γ [21, 22]. An H_α/H_β ratio computed from atomic processes alone is not the ratio such an instrument records.

6.3.4 What survives it, and what does not

Two consequences follow, and they point in opposite directions, so both must be stated.

The *numerical* cold-edge results are atomic-model results and nothing more. Adding a third supply channel that feeds $n = 3$ preferentially would change $f_3 - f_4$, and could plausibly change it a great deal; this work offers no bound on how much, and none can be constructed from the data in this repository.

The *structural* results are indifferent to it. The exact factorisation of the plateau error, the logistic form of the ground-fed fraction f_m , and the closed-form ceiling $\max |f_3 - f_4| = \tanh(|\Delta|/4)$ are theorems about whatever matrix $\mathbf{L}$ is supplied. They hold for a matrix containing molecular states exactly as they hold for this one. What a molecular matrix changes is the numerical value of the ingredients, not the form of the result.

This is unquantified and cannot be quantified here. It is the objection most likely to be raised first by a referee drawn from the divertor spectroscopy community, and the correct answer is the honest one: this is an atomic model, its cold-edge numbers are atomic-model numbers, and the reader should treat them as such. [TODO: Molecular channels: unquantified, and no bound available from this work. Stated as a conditional.]

6.4 Where the ridge sits depends on what the model assumed about neutrals

6.4.1 The question

Section 5.6 found an interior maximum — a ridge — in the error map at $n_e \approx 2 \times 10^{13} \text{ 1/cm}^3$, and showed that it moves with the shell pair as Griem’s criterion for local thermodynamic equilibrium (LTE) predicts. The mechanism is robust; three independent tests agreed. But a mechanism and a location are different claims, and only the first of them has been established.

6.4.2 The picture: the ridge is where the two feeds are balanced

The ridge exists because the $n = 3$ and $n = 4$ shells switch from recombination-fed to ground-fed at different reservoir strengths, so their difference vanishes when either feed dominates and peaks in between. The controlling variable is therefore not the electron density directly — it is *how much neutral gas there is to feed from*. The electron density enters only because, in this model, it determines the neutral density through the CR ionisation balance.

That is the conditionality. Change what the model assumes about neutrals and the ridge moves, whatever the electron density does.

6.4.3 The measurement

Within the two-channel split the ground-fed fraction is exactly $f_p = a_1 n_g / (a_0 + a_1 n_g)$, where n_g is the ground-state density feeding the excitation channel and a_0, a_1 are the network responses of Section 3.5.3. Scaling n_g is therefore an exact operation on the ridge, not an approximation to one. Doing it, at $T_e = 2.02$ eV:

n_{1s} scaling	density at which $ f_3 - f_4 $ peaks
$\times 0.01$	$\leq 10^{12}$ 1/cm ³ (off grid)
$\times 0.1$	2.28×10^{12} 1/cm ³
$\times 1$ (the model's own CRE value)	1.68×10^{13} 1/cm ³
$\times 10$	2.48×10^{14} 1/cm ³
$\times 100$	$\geq 10^{15}$ 1/cm ³ (off grid)

One decade in the assumed neutral density moves the ridge by roughly one decade in electron density. The response is close to proportional, and the whole three-decade span of the density grid is traversed by a two-decade change in n_{1s} .

The agreement with Griem's criterion at unit scaling — 1.93×10^{13} measured against 2.05×10^{13} 1/cm³ predicted — is therefore an agreement evaluated at *this model's* collisional-radiative ionisation balance, which contains no transport and no recycling. In a real divertor the neutral density is set by recycling, which is Section 6.2's objection arriving in a different place. The sentence “the ridge sits at $n_e \approx 2 \times 10^{13}$ 1/cm³” is a statement about an ionisation balance, not about a divertor, and should be quoted with that condition attached or not quoted.

6.4.4 And it is one line pair's ridge, not the Balmer series'

There is a second conditionality in the same result, and this one is a matter of naming rather than of physics. The ridge belongs to the (3, 4) shell pair. That is why it moves when the pair is changed — and it moves a long way: an H_α/H_γ diagnostic, built on the (3, 5) pair, has its worst density at 7.2×10^{12} 1/cm³, a factor 2.68 lower.

The correct form of words is therefore “*the H_α/H_β ratio is least reliable at...*” and never “*the Balmer diagnostic is least reliable at...*”. Every line pair has its own ridge in its own place. This thesis has mapped one of them, and mapping it does not license a statement about the others.

6.5 It is not detachment

6.5.1 The question, and the temptation

The identification is tempting, and this section exists because the temptation is strong enough that the sentence would otherwise be repeated without its arithmetic. Detachment is the regime in which the divertor plasma cools until the ion flux reaching the target collapses; it is the regime a machine like ITER must operate in to survive its own exhaust power, and it is the transition the H_α/H_β diagnostic is chiefly used to identify. The ridge

sits where ionising and recombining supplies compete. Detachment is where ionising and recombining plasma compete. The words match.

6.5.2 The arithmetic

The numbers do not. Section 5.6.5 withdrew the identification and this chapter repeats the withdrawal.

The ridge sits at $1.93 \times 10^{19} \text{ 1/m}^3$. That is a factor 5.2 *below* the density band that published edge modelling associates with a detached ITER divertor [41], and a factor 1.6 *below* the modelled upstream separatrix density [1, 2]. It falls between the two, in a range neither source characterises.

Both comparison bands deserve a word of caution, because a factor of 1.6 is not a large margin to be arguing over. The separatrix figure is quoted here from a reduced ITER divertor model [2], which itself quotes it from the ITER divertor physics basis [1]; and “upstream separatrix density” is not a single quantity, since the density at the outboard midplane and the density at the X-point differ by roughly 50 % on the same flux tube. The conclusion of this section does not turn on either subtlety — the ridge is below both — but the factor 1.6 should not be quoted more precisely than the band it is measured against. [TODO: The separatrix band: confirm which quantity (outboard midplane or X-point) the factor 1.6 was computed against, and cite [1] as the primary source. Note also that this repository records the [2] reference as “Stangeby 2022, Nucl. Fusion 62” in `src/validation/grid.py` and in three `outputs/` documents; the DOI recorded there is right and the volume and year are wrong. Reported, not repaired.]

6.5.3 What that leaves

The honest statement is uncomfortable and should be made anyway. **The structure this work found is real and reproducible, and it sits at a density this thesis cannot identify with any named region of a tokamak.** It is not detachment; it is not the upstream separatrix; and calling it either would be a claim about edge transport that a 0-D atomic model cannot support. The ridge is a property of how two shells of hydrogen divide their supply. It needs no divertor geometry to produce it, and it acquires none by being described in one.

The corresponding restriction on the error map is in Chapter 5 and bears repeating here in its scoped form. Restricted to $T_e \geq 2 \text{ eV}$ *and* to the citable detached density band $n_e \geq 10^{14} \text{ 1/cm}^3$, the number of points at which the ELM-averaged lower bound exceeds 10 % is *zero* out of 108, and the largest lower bound anywhere within that set is 7.2 %. The ridge and the detached band do not overlap. Where the diagnostic is most used, this model says the transient error is real but modest.

6.6 The open boundary: an attack that was expected to be fatal

6.6.1 The question

Section 3.1 was explicit that the ion density n_{ion} is held fixed: the recombination feed $S_{n_{\text{ion}}}$ enters as a constant source term — a feed stream into a CSTR, in the reactor

picture — rather than as a matrix element, and the 43-state system has no 44th state for ionised atoms to accumulate in. Atoms ionise and vanish from the accounting. The slow eigenvalue λ_0 is therefore the CR ionisation rate of ground-state hydrogen against a prescribed reservoir, not the equilibration rate of a closed system, and this is also why the effective coefficients of this model cannot be compared directly against a closed-system atomic database such as ADAS (the Atomic Data and Analysis Structure).

6.6.2 What we expected

The obvious attack writes itself. A reactor that lets its product disappear relaxes differently from one that does not. If the long τ_{QSS} values are an artifact of the open boundary, then so is the claim that the error outlasts an ELM, and Chapter 5 collapses. The expectation was that closing the system would shorten τ_{QSS} substantially — and there was a specific reason to fear it, because a factor of order $\tau_{\text{QSS}}/\tau_{\text{drive}} \approx 2300$ at the worst point would bring the slow clock down to the ELM duration and remove the persistence entirely.

That factor, 2300, is the falsifier. It was named before the test was run. If closure produced it, the result was wrong.

6.6.3 What happened

The manifold was closed by appending a 44th state for the ions,

$$\mathbf{L}_c = \begin{pmatrix} \mathbf{L} & \mathbf{S} \\ -\text{colsum}(\mathbf{L}) & -\sum \mathbf{S} \end{pmatrix}, \quad (6.1)$$

which sends every atom that ionises into the new state instead of deleting it. Each column of $\mathbf{L}_c$ sums to zero, so particles are exactly conserved; this was verified to machine precision, and the construction reproduces this project’s own previously recorded closed-system value at the cold corner to five digits, confirming it is the same closure and not a new one.

point	τ_{QSS} open	τ_{QSS} closed	factor
cold corner, 1.0 eV, 10^{12} 1/cm ³	67.23 s	1.6226 s	41.4
worst-case operator	0.233 24 s	0.015 173 s	15.4
ridge	2.027 ms	1.999 ms	1.01
benchmark, 2.947 eV, 1.389×10^{14} /cm ³	22.728 μ s	22.708 μ s	1.00

Over the whole grid the closure factor has minimum 1.00, median 1.00 and maximum 41.4. The shape of that distribution is the answer to the attack, and it is worth reading carefully. **The factor is large only where τ_{QSS} already exceeds the event duration by three orders of magnitude, and it is unity everywhere τ_{QSS} approaches τ_{drive} .** Where the boundary condition matters, the outcome does not depend on it, because the error has already saturated; where the outcome depends on τ_{QSS} , the boundary condition does not matter.

Recomputing the entire ELM census with closed-system τ_{QSS} moves the breakdown count from 202 to 200 out of 680, and the worst case from 0.38682 to 0.38563 — about one part in 325. Restricted to $T_e \geq 2$ eV the count is 45 before and 45 after.

The falsifier did not appear. A factor of 2300 was needed at the worst point; the measured factor there is 15. The attack was survived by two orders of magnitude, and this belongs in the defence of the result rather than in a list of caveats.

6.6.4 The caveat that must not be softened

The closure adds an ion reservoir. It does *not* close the electron balance — n_e is held fixed while the ion population moves — and it adds no transport whatever. It is a test of the boundary condition on the atomic manifold, and nothing more. Section 6.2 is the objection this test does not answer, and no amount of closing the atomic system will answer it.

[TODO: The source record labels the worst-case operator as grid index $[1, 4]$ in the closure table and $[0, 4]$ elsewhere for the same $\tau_{\text{QSS}} = 0.233\,24\text{s}$. Resolve the index label before the final draft; the value itself is not in dispute.]

6.7 An external benchmark this model fails, and has not explained

6.7.1 The question

Chapter 4 presented the comparison against Fujimoto’s published population coefficients as the strongest external check this model passes. It is — for one of the two coefficients. This section reports the other, which fails, and does not repair it.

6.7.2 The picture: two coefficients, one for each feed

Chapter 3 split every excited population into two pieces corresponding to the two supply channels. The population coefficient $r_0(p)$ measures how strongly level p is fed from the ion continuum by recombination; $r_1(p)$ measures how strongly it is fed from the ground state by excitation. The published tabulation gives both, computed independently, decades ago, by a different method. Agreement is a real test; disagreement is a real failure.

6.7.3 The failure

The recombination-fed coefficients agree: r_0 matches to 0.1 to 1 % for $n \geq 4$. The ground-fed coefficients at low principal quantum number do not. Against Fujimoto’s Table 4.1(b), at $n_e = 10^{12}\,1/\text{cm}^3$, this model’s r_1 sits a factor 8.3 low at $p = 3$ and a factor 4.4 low at $p = 4$.

Those are exactly the two shells whose difference is the mechanism of Chapter 5, and $n_e = 10^{12}\,1/\text{cm}^3$ is one grid interval from the ridge. This is not a discrepancy in a corner of the model that nothing depends on.

The obvious suspect has been eliminated. If the excitation rates feeding the ground-fed channel were wrong, the deficit would follow; but those rates were benchmarked independently against R-matrix-with-pseudostates (RMPS) calculations and the dominant $\Delta\ell = 1$ channels agree at a median of -5.0% over 120 comparisons. A factor of 8 is not hiding there. No alternative suspect has been identified.

6.7.4 What it costs, and what it does not

The consequence is bounded in one direction only, and the bound comes from Section 6.4: since the ridge location responds roughly decade-for-decade to the strength of the ground-fed supply, a factor-8 error in $r_1(3)$ is not obviously harmless to where the ridge is. The

ridge *mechanism* — that the two shells switch supply channels at different reservoir strengths, and that the switching point moves with the shell pair as Griem’s criterion predicts — depends on the *ratio* of the two shells’ responses and on how it scales, not on the absolute normalisation, and survives either way.

This is reported open because it is open. [TODO: Report $\partial(\text{ridge location})/\partial(r_1 \text{ scaling})$ explicitly. If a factor-3 change in r_1 moves the ridge by more than one grid interval, the ridge *location* must be withdrawn and only the mechanism retained. Unresolved as of this draft.]

6.8 The top of the level ladder, and a check that has not been run

6.8.1 The question

A hydrogen atom has infinitely many bound levels, crowding together towards the ionisation limit. A computation cannot have infinitely many states, so a boundary was chosen: this model tracks levels up to $n_{\text{max}} = 15$ and discards the rest. Above $n = 8$ it also stops resolving orbital angular momentum ℓ , treating each shell as a single bundled state populated statistically. Both are choices, and this section states what they cost.

6.8.2 Three consequences, in decreasing order of severity

The terminal shell is over-populated. The top of any truncated ladder is fed from below and from the continuum but has nothing above it to cascade in from — and, more importantly, nothing above it to cascade *out* to that is still being tracked. The effect is measurable rather than hypothetical: against published values, this model’s ground-fed coefficient at $n = 15$ is high by a factor of 4.9 to 6.2, while at $n = 10$ it agrees to 4 to 20 %. The excess is concentrated at the terminal shell, as that picture predicts. Any result resting on the top shell is an upper bound rather than an estimate. *No result in this thesis rests on it*: the observable is built from $n = 3$ and $n = 4$, both fully ℓ -resolved and both far from the truncation. [TODO: Provenance: the 4.9–6.2 $\times$ and 4–20 % figures are recorded with the Chapter 4 benchmark but their run artifact is not stamped in validation/. Stamp before submission.]

The bundling nonetheless enters the result indirectly. Section 5.6.3 listed truncation among the alternative explanations for the ridge that this work does not exclude, and the reason is worth restating. The mechanism turns on the difference between the $n = 3$ and $n = 4$ supply fractions, both fully resolved. But those fractions depend on the recombination flux cascading down from the high levels, and the bundling approximates precisely that flux. An error in the cascade is an error in the recombination-fed channel, which is one of the two channels the mechanism compares.

There is no ℓ -mixing inside the bundled block. The proton-impact ℓ -mixing that Section 2.7 showed to be essential for the $n = 2$ manifold is present only for $n = 2$ –8; the ℓ -mixing rate array is identically zero above that. Within the bundled shells the ℓ -distribution is assumed statistical rather than computed. At high n that assumption is far more defensible than it would be at low n , because the sublevels are nearly degenerate

and every collisional process mixes them — but it is assumed, not measured, and this thesis has not measured it.

6.8.3 The check that would close this

A script to quantify the effect of the bundling on the ℓ -distribution within the bundled shells was written and **has never been executed**. It is one command. It is reported here as not run, rather than omitted, because an examiner reading Section 5.6.3 will ask for exactly this. [TODO: Run `verify_bundling_psm20.py`. Written, never executed. It is the outstanding check named in Section 5.6.3.]

6.9 Remaining assumptions, and the sensitivities that are known

Four assumptions remain. Each is stated with what is known about it, rather than listed as a disclaimer, and two of the four have been tested.

Maxwellian electrons — untested. Every rate coefficient in Chapter 2 is an average of a cross section over a Maxwell–Boltzmann electron energy distribution. Ionisation and excitation are driven by the high-energy tail of that distribution, which holds a small minority of the electrons. A 1 eV plasma next to a recycling surface has no guarantee of a Maxwellian tail, and a depleted or enhanced tail would change the rates that set both clocks. Not tested here.

$T_i = T_e$ in the ℓ -mixing rates — untested. The proton-impact ℓ -mixing rates of Section 2.7 are evaluated at an ion temperature equal to the electron temperature. In a detaching divertor the ions are frequently colder than the electrons. Not tested here.

The step is instantaneous — tested, and safe for a reason that is easy to get wrong. The error maps impose a sudden temperature step, whereas a real ELM has a finite rise time. The tempting comparison is $\tau_{\text{relax}}/\tau_{\text{drive}} \approx 10^{-5}$, which appears to say that the step idealisation is safe by five orders of magnitude. **That ratio is about the wrong quantity.** It describes the suppression of the fast *excited-manifold* step error, which is not what the maps report. The plateau error is governed by the slow ground-state response, for which the relevant ratio at the worst point is $\tau_{\text{QSS}}/\tau_{\text{drive}} = 2332$ — an unambiguous step by a wide margin. Any finite ramp with $\tau_{\text{relax}} \ll t_{\text{ramp}} \ll \tau_{\text{QSS}}$ reaches the same plateau, so the idealisation *is* safe; it is safe because of the slow ratio, and the fast ratio must never be quoted for or against this result.

The step is in temperature only — partially tested, and it matters. A real ELM raises the electron density as well as the temperature, while the maps step T_e at fixed n_e . Joint steps in both were spot-checked. They give a comparable instantaneous $\varepsilon_{\text{plateau}}$ but a *time-averaged* bound that can fall by a factor of 4 at the benchmark point, from 0.0117 to 0.0030, because a higher n_e shortens τ_{QSS} and so shortens the window over which the error survives. A full joint-step map has not been run and would cost the same as the existing one. [TODO: Joint (T_e, n_e) ELM step map. Not run. An examiner familiar with ELM physics will ask for it.]

Two sensitivities of the reported census belong in the same place, because both are properties of a reporting choice rather than of the physics, and both would otherwise be mistaken for measurements.

The step size is one interval of the temperature grid, which displaces the reservoir by $|\ln x|$ between 0.124 and 0.682 across the grid. A four-interval step gives 0.516 to 2.563 and raises $\varepsilon_{\text{plateau}}$ at the ridge from 18.1 % to 81.1 %. On step size alone, therefore, the reported numbers *understate* the error rather than overstate it.

The 10 % breakdown threshold is doing real work. At 5 %, 10 % and 20 % the count over the same 680 points is 348, 202 and 61. **The count is not robust to the threshold; the worst case is.** Any sentence of the form “ N points break down” must carry both its threshold and its denominator, or it is not a result.

6.10 What is left standing

6.10.1 What this model cannot say

It cannot say anything quantitative about the cold edge of its own grid. Below 2 eV the optically thin plateau error is inflated by a factor of 2.5 to 3.3 by neglected Lyman trapping; the size of the correction depends on a slab thickness the model does not contain; and below 1.15 eV even the location of the ridge is undetermined.

It cannot say what a real divertor does, because the ground-state population there is renewed by recycling on a 6 to 10 μs transit time, against a local ionisation balance of 233 ms at the worst point. Four orders of magnitude separate the two, and the model contains only the slower one.

It cannot say what an instrument records at the onset of detachment, because a large fraction of that light is molecular in origin and there are no molecules in ***L***.

It cannot name the density at which its own ridge sits as a region of a tokamak. The ridge falls between the modelled upstream separatrix and the detached divertor band, and belongs to neither.

And it carries an unresolved factor-8 disagreement with the standard tabulation of the ground-fed population coefficient at $p = 3$ — the coefficient the mechanism runs on — for which no explanation has been found.

6.10.2 What this model can say, with the objections computed

Above 2 eV the breakdown census is invariant under Lyman trapping across a twentyfold range of assumed slab thickness: 45 of 448 points in every run, worst case moving by 0.5 %. The restriction that produced that scope was a guess when it was written and is a measurement now.

Closing the open boundary — the objection most naturally assumed to be fatal — moves the census by two points out of 680 and the worst case by one part in 325. A factor of 2300 would have been required to refute the result. The measured factor at that point is 15.

The ridge mechanism was verified over $T_e \geq 2$ eV, which is exactly the range in which trapping has now been shown to be irrelevant, so the agreement with Griem’s boundary criterion is not an artifact of neglected opacity.

And at citable divertor densities above 2 eV — the conditions where the diagnostic is actually used — the ELM-averaged error never reaches 10 % at all, over 108 points, with

a worst case of 7.2%.

6.10.3 The bridge

The pattern is consistent, and it is why this chapter is not an apology. Every objection that could be computed inside a 0-D atomic model was computed, and the defensible claim came through all of them with its numbers unchanged. The objections that remain — neutral transport and molecular chemistry — are the ones that cannot be computed inside a 0-D atomic model at all, and they have been stated as conditionals rather than dissolved into caveats.

That is the boundary of the claim. What is inside it is a quantitative statement about when a spectroscopist may trust a lookup table and when she may not, and Chapter 7 puts it into the form she can use.

Chapter 7

Conclusions

7.1 What was found

7.2 A criterion others can apply

7.3 What it means for divertor spectroscopy

7.4 Further work

Appendix A

State Ordering

Appendix B

Atomic Data Sources

Appendix C

Numerical Methods and Convergence

Bibliography

- [1] R. A. Pitts, X. Bonnin, F. Escourbiac, H. Frerichs, J. P. Gunn, T. Hirai, A. S. Kukushkin, E. Kaveeva, M. A. Miller, D. Moulton, V. Rozhansky, I. Senichenkov, E. Sytova, O. Schmitz, P. C. Stangeby, G. De Temmerman, I. Veselova, and S. Wiesen. Physics basis for the first ITER tungsten divertor. *Nuclear Materials and Energy*, 20: 100696, 2019. doi: 10.1016/j.nme.2019.100696.
- [2] P. C. Stangeby, J. D. Lore, R. A. Pitts, J. M. Canik, and X. Bonnin. A reduced model for the ITER divertor based on SOLPS solutions for ITER $q = 10$ baseline conditions: B. A reduced model based on reversed-direction two point modeling. *Nuclear Fusion*, 63(1):016017, 2023. doi: 10.1088/1741-4326/ac9917. Published online 13 December 2022; print issue January 2023. Companion part A: Nucl. Fusion 63, 016016 (2023), doi:10.1088/1741-4326/ac9916. Cited in this repository as “Stangeby 2022, Nucl. Fusion 62”, which is incorrect.
- [3] J. D. Lore, X. Bonnin, J.-S. Park, R. A. Pitts, and P. C. Stangeby. High gas throughput SOLPS-ITER simulations extending the ITER database to strong detachment. *Nuclear Fusion*, 62(10):106017, 2022. doi: 10.1088/1741-4326/ac8a5f.
- [4] S. I. Krasheninnikov and A. S. Kukushkin. Physics of ultimate detachment of a tokamak divertor plasma. *Journal of Plasma Physics*, 83(5):155830501, 2017. doi: 10.1017/S0022377817000654.
- [5] A. Loarte, B. Lipschultz, A. S. Kukushkin, G. F. Matthews, P. C. Stangeby, N. Asakura, G. F. Counsell, G. Federici, A. Kallenbach, K. Krieger, A. Mahdavi, V. Philipps, D. Reiter, J. Roth, J. Strachan, D. Whyte, R. Doerner, T. Eich, W. Fundamenski, A. Herrmann, et al. Chapter 4: Power and particle control. *Nuclear Fusion*, 47(6):S203–S263, 2007. doi: 10.1088/0029-5515/47/6/S04. Progress in the ITER Physics Basis.
- [6] Wei Zhao, Yali Wang, Yuzhong Jin, Li Zhao, Hongxia Zhou, Lin Nie, Guangwu Zhong, Chunjia Liu, Christopher Watts, and James Paul Gunn. Preliminary design of ITER divertor Langmuir probe system. *Fusion Science and Technology*, 76(2): 79–87, 2020. doi: 10.1080/15361055.2019.1674123.
- [7] D. R. Bates, A. E. Kingston, and R. W. P. McWhirter. Recombination between electrons and atomic ions, I. Optically thin plasmas. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 267(1330):297–312, 1962. doi: 10.1098/rspa.1962.0101.
- [8] A. Yu. Pigarov, J. L. Terry, and B. Lipschultz. Study of the discrete-to-continuum transition in a Balmer spectrum from Alcator C-Mod divertor plasmas. *Plasma*

- Physics and Controlled Fusion*, 40(12):2055–2072, 1998. doi: 10.1088/0741-3335/40/12/006.
- [9] H. P. Summers, W. J. Dickson, M. G. O’Mullane, N. R. Badnell, A. D. Whiteford, D. H. Brooks, J. Lang, S. D. Loch, and D. C. Griffin. Ionization state, excited populations and emission of impurities in dynamic finite density plasmas: I. The generalized collisional–radiative model for light elements. *Plasma Physics and Controlled Fusion*, 48(2):263–293, 2006. doi: 10.1088/0741-3335/48/2/007.
 - [10] H. P. Summers. The ADAS user manual, version 2.6. <http://www.adas.ac.uk>, 2004. Atomic Data and Analysis Structure, University of Strathclyde.
 - [11] Takashi Fujimoto and R. W. P. McWhirter. Validity criteria for local thermodynamic equilibrium in plasma spectroscopy. *Physical Review A*, 42(11):6588–6601, 1990. doi: 10.1103/PhysRevA.42.6588.
 - [12] Takashi Fujimoto. *Plasma Spectroscopy*, volume 123 of *International Series of Monographs on Physics*. Clarendon Press, Oxford, 2004. ISBN 978-0-19-853028-2. doi: 10.1093/acprof:oso/9780198530282.001.0001.
 - [13] K. Verhaegh, B. Lipschultz, B. P. Duval, A. Fil, M. Wensing, C. Bowman, and D. S. Gahle. Novel inferences of ionisation and recombination for particle/power balance during detached discharges using deuterium Balmer line spectroscopy. *Plasma Physics and Controlled Fusion*, 61(12):125018, 2019. doi: 10.1088/1361-6587/ab4fle.
 - [14] T. Eich, B. Sieglin, A. J. Thornton, M. Faitsch, A. Kirk, A. Herrmann, and W. Suttrop. ELM divertor peak energy fluence scaling to ITER with data from JET, MAST and ASDEX Upgrade. *Nuclear Materials and Energy*, 12:84–90, 2017. doi: 10.1016/j.nme.2017.04.014.
 - [15] A. Kirk, I. T. Chapman, T. E. Evans, C. Ham, J. R. Harrison, G. Huijsmans, Y. Liang, Y. Q. Liu, A. Loarte, W. Suttrop, and A. J. Thornton. Understanding the effect resonant magnetic perturbations have on ELMs. *Plasma Physics and Controlled Fusion*, 55(12):124003, 2013. doi: 10.1088/0741-3335/55/12/124003.
 - [16] Keiji Sawada and Takashi Fujimoto. Temporal relaxation of excited-level populations of atoms and ions in a plasma: Validity range of the quasi-steady-state solution of coupled rate equations. *Physical Review E*, 49(6):5565–5573, 1994. doi: 10.1103/PhysRevE.49.5565.
 - [17] P. T. Greenland. On the validity of collisional–radiative models. *Journal of Nuclear Materials*, 290–293:615–618, 2001. doi: 10.1016/S0022-3115(00)00601-2.
 - [18] P. T. Greenland. Collisional-radiative models with molecules. *Proceedings of the Royal Society of London. Series A: Mathematical, Physical and Engineering Sciences*, 457(2012):1821–1839, 2001. doi: 10.1098/rspa.2001.0788.
 - [19] Hans R. Griem. Validity of local thermal equilibrium in plasma spectroscopy. *Physical Review*, 131(3):1170–1176, 1963. doi: 10.1103/PhysRev.131.1170.
 - [20] Hans R. Griem. *Principles of Plasma Spectroscopy*, volume 2 of *Cambridge Monographs on Plasma Physics*. Cambridge University Press, Cambridge, 1997. ISBN 978-0-521-45504-6. doi: 10.1017/CBO9780511524578.

- [21] K. Verhaegh, B. Lipschultz, C. Bowman, B. P. Duval, U. Fantz, A. Fil, J. R. Harrison, D. Moulton, O. Myatra, D. Wunderlich, F. Federici, D. S. Gahle, A. Perek, and M. Wensing. A novel hydrogenic spectroscopic technique for inferring the role of plasma–molecule interaction on power and particle balance during detached conditions. *Plasma Physics and Controlled Fusion*, 63(3):035018, 2021. doi: 10.1088/1361-6587/abd4c0.
- [22] T. A. Wijkamp, J. S. Allcock, X. Feng, B. Kool, B. Lipschultz, K. Verhaegh, B. P. Duval, J. R. Harrison, L. Kogan, N. Lonigro, A. Perek, P. Ryan, R. M. Sharples, I. G. J. Classen, and R. J. E. Jaspers. Characterisation of detachment in the MAST-U Super-X divertor using multi-wavelength imaging of 2D atomic and molecular emission processes. *Nuclear Fusion*, 63(5):056003, 2023. doi: 10.1088/1741-4326/acc191.
- [23] Igor Bray and Andris T. Stelbovics. Convergent close-coupling calculations of electron-hydrogen scattering. *Physical Review A*, 46(11):6995–7011, 1992. doi: 10.1103/PhysRevA.46.6995.
- [24] I. Bray, D. V. Fursa, A. S. Kheifets, and A. T. Stelbovics. Electrons and photons colliding with atoms: development and application of the convergent close-coupling method. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 35(15):R117–R146, 2002. doi: 10.1088/0953-4075/35/15/201.
- [25] L. Vriens and A. H. M. Smeets. Cross-section and rate formulas for electron-impact ionization, excitation, deexcitation, and total depopulation of excited atoms. *Physical Review A*, 22(3):940–951, 1980. doi: 10.1103/PhysRevA.22.940.
- [26] A. Hartgers, J. van Dijk, J. Jonkers, and J. A. M. van der Mullen. CRModel: A general collisional radiative modeling code. *Computer Physics Communications*, 135(2):199–218, 2001. doi: 10.1016/S0010-4655(00)00231-9.
- [27] Wolfgang Lotz. Electron-impact ionization cross-sections and ionization rate coefficients for atoms and ions from hydrogen to calcium. *Zeitschrift für Physik*, 216(3):241–247, 1968. doi: 10.1007/BF01392963.
- [28] L. C. Johnson. Approximations for collisional and radiative transition rates in atomic hydrogen. *The Astrophysical Journal*, 174:227, 1972. doi: 10.1086/151486.
- [29] M. J. Seaton. Radiative recombination of hydrogenic ions. *Monthly Notices of the Royal Astronomical Society*, 119(2):81–89, 1959. doi: 10.1093/mnras/119.2.81.
- [30] Junjie Mao and Jelle Kaastra. Parameterization of the level-resolved radiative recombination rate coefficients for the SPEX code. *Astronomy & Astrophysics*, 587:A84, 2016. doi: 10.1051/0004-6361/201527568.
- [31] D. Hoang-Binh. A program to compute exact hydrogenic radial integrals, oscillator strengths, and Einstein coefficients, for principal quantum numbers up to $n \approx 1000$. *Computer Physics Communications*, 166(3):191–196, 2005. doi: 10.1016/j.cpc.2004.11.005. CPC Program Library ADUU_v1_0.
- [32] D. Hoang-Binh. An exact calculation of hydrogenic radial integrals and oscillator strengths, for principal quantum numbers up to N equal to about 1000. *Astronomy and Astrophysics*, 238:449–451, 1990. No DOI; ADS bibcode 1990A&A...238..449H.

- [33] W. L. Wiese and J. R. Fuhr. Accurate atomic transition probabilities for hydrogen, helium, and lithium. *Journal of Physical and Chemical Reference Data*, 38(3):565–720, 2009. doi: 10.1063/1.3077727.
- [34] T. Holstein. Imprisonment of resonance radiation in gases. *Physical Review*, 72(12):1212–1233, 1947. doi: 10.1103/PhysRev.72.1212.
- [35] K. Behringer. Escape factors for line emission and population calculations. Technical Report IPP 10/11, Max-Planck-Institut für Plasmaphysik, Garching, Germany, 1998. No DOI. Implemented as ADAS214; see <https://www.adas.ac.uk/man/chap2-14.pdf>.
- [36] Eite Tiesinga, Peter J. Mohr, David B. Newell, and Barry N. Taylor. CODATA recommended values of the fundamental physical constants: 2018. *Reviews of Modern Physics*, 93(2):025010, 2021. doi: 10.1103/RevModPhys.93.025010.
- [37] R. M. Pengelly and M. J. Seaton. Recombination spectra: II. Collisional transitions between states of degenerate energy levels. *Monthly Notices of the Royal Astronomical Society*, 127(2):165–175, 1964. doi: 10.1093/mnras/127.2.165.
- [38] N. R. Badnell, F. Guzmán, S. Brodie, R. J. R. Williams, P. A. M. van Hoof, M. Chatzikos, and G. J. Ferland. H, He-like recombination spectra – IV. Clarification and refinement of methodology for l -changing collisions. *Monthly Notices of the Royal Astronomical Society*, 507(2):2922–2929, 2021. doi: 10.1093/mnras/stab2266.
- [39] H. Anderson, C. P. Ballance, N. R. Badnell, and H. P. Summers. An R-matrix with pseudostates approach to the electron-impact excitation of H I for diagnostic applications in fusion plasmas. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 33(6):1255–1262, 2000. doi: 10.1088/0953-4075/33/6/311.
- [40] H. Anderson, C. P. Ballance, N. R. Badnell, and H. P. Summers. Corrigendum: An R-matrix with pseudo-states approach to the electron-impact excitation of H I for diagnostic applications in fusion plasmas. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 35(6):1613–1615, 2002. doi: 10.1088/0953-4075/35/6/701.
- [41] C. Guillemaut, R. A. Pitts, A. S. Kukushkin, and M. O’Mullane. Radiative power loading in the ITER divertor. *Fusion Engineering and Design*, 86(12):2954–2964, 2011. doi: 10.1016/j.fusengdes.2011.07.008.